Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform

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A capstone project report submitted to the faculty of
Computer Science and Engineering
California State University, San Marcos

May 2026
(Version 5.0)

Technical Report Series: CSU-SM-CSE-39-2026-SE-001-Team-009

Contents

1. Abstract

ALVION is an Android application that detects signs of driver drowsiness and distraction using the phone's front-facing camera and on-device computer vision. Many new vehicles include driver-monitoring features, but millions of older cars do not. Our goal is to help close that gap with a low-cost, widely accessible solution that can run on everyday Android devices and provide timely safety alerts during real driving. In addition to being a prototype safety application, ALVION serves as a practical testbed to evaluate the real-world capability of Qualcomm-enabled mobile AI for this use case. We focus on whether mobile vision models and on-device acceleration are strong and stable enough to support driver-monitoring signals under realistic conditions (lighting changes, head movement, brief tracking loss, and different mounting angles). These findings help inform what is feasible today on commodity smartphones and what improvements are needed for more robust edge-based driver safety systems. The system analyzes live video to extract visual cues related to fatigue and inattention, applies configurable thresholds and time-based logic, and triggers clear alerts through sound, on-screen prompts, and vibration when attention appears to drift. While core detection runs on-device for low latency, the application also includes standard product features such as user authentication (e.g., Firebase [3]) to support a realistic app workflow. We do not attempt vehicle integration, cloud-based video processing, large-scale deployment, or regulatory certification. This document outlines the project's objectives, assumptions, and boundaries so developers, mentors, and faculty can stay aligned as the system evolves from concept to implementation. It describes the architecture, component responsibilities, and design decisions with an emphasis on building something practical within an academic timeline and extensible for future teams.

2. Report Revision History

Version	Date	Description of Changes
1.5	Nov 25, 2025	Initial draft submitted. Revised structural and behavioral design placeholders per faculty feedback; added preliminary UML comments, ERD rationale, and UI mockups. Corrected terminology in diagrams and aligned architecture descriptions with grading criteria.
2.0	Dec 2, 2025	Completed full capstone design document. Expanded all Requirements; finalized Architecture, Structural, and Behavioral Designs (MVC diagram, class diagrams, ERD, sequence and state diagrams); added UI mockups; completed Exploratory Studies, System Implementation, and System Testing sections.
2.5	Dec 11, 2025	Addressed instructor comments on context/scope diagrams, user–use-case cross-references, non-functional requirement grouping, and algorithm complexity; polished figures and layout.
3.0	Feb 28, 2026	Revised report to reflect updated user requirements and system development changes; confirmed system designs in Section 6; added/updated test case designs and execution reports; incorporated changed/new user requirements into refined system requirements and updated system design.
3.5	Mar 20, 2026	Revised based on instructor feedback. Added Requirements Traceability (Section 4.3) mapping UF-A–UF-J to components and test cases. Introduced Sections 8.4 and 8.5 for functional/non-functional requirement evaluation. Refined scope, use-case alignment, and system consistency.
4.0	Apr 27, 2026	Change-management revision. Corrected detection thresholds (drowsiness 1200 ms, distraction 4000 ms/2000 ms). Added new requirements UF-K, SF-11–SF-15. Updated architecture, ERD, UI, and algorithm sections to document TFLite sunglasses detection, Firestore trip history, GPS speed, phone orientation monitoring, and presence check. Fixed structural gaps to conform to capstone report guidelines.
5.0	May 11, 2026	Final revision for capstone submission. Completed proofreading and language polish across all sections. Verified all chapters are present and consistently structured. Confirmed all eleven user functional requirements (UF-A through UF-K) are implemented and documented. No structural changes; this version represents the final submitted state of the report.

3. Problem Statement

3.1 Business Background

Driver monitoring is becoming a regulatory and industry standard for road safety. The European Union now mandates such systems in new vehicles beginning in 2024 [6], and other markets are expected to follow. Automakers and regulators view these systems as essential for reducing human-error-related crashes, but adoption in existing vehicles remains limited because of cost, integration complexity, and consumer concerns over data privacy. Retrofit products, while available, often require expensive hardware installations or continuous cloud connectivity for processing. These barriers restrict adoption, especially in regions where consumers rely on older cars or are reluctant to share biometric data online. A smartphone-based solution reduces those barriers: most users already own compatible Android devices with capable cameras and neural-processing hardware. ALVION's approach uses these resources to demonstrate how Qualcomm's mobile AI platforms can perform effective, real-time driver-monitoring inference entirely on-device. This provides a testbed for cost-effective safety innovations and shows potential business alignment with regulatory trends. The project also establishes a baseline for future Qualcomm-enabled automotive safety research and gives developers a model for edge-based AI deployment that respects user privacy and reduces infrastructure costs.

3.2 Needs

Improving driver safety requires early detection of distraction and drowsiness in a manner that is both reliable and user-friendly. ALVION responds to several intertwined needs:

- **Safety:** The system should protect both the driver and other road users by recognizing behavioral cues such as prolonged eyelid closure, yawning, or abnormal head pose patterns that precede dangerous inattention. Distraction-affected crashes alone caused 3,208 U.S. fatalities in 2024 [7].
- **Privacy-Aware Design:** Drivers are wary of surveillance systems that upload facial video to the cloud. ALVION is designed to keep *core detection and inference on-device* and avoid transmitting raw camera footage, while still supporting standard app functionality such as user authentication (e.g., Firebase [3]).
- **Accessibility and Cost:** Full-scale automotive integrations are expensive. By leveraging common smartphone hardware, ALVION provides a low-cost entry point that can benefit drivers in older vehicles.
- **Responsiveness:** Alerts must reach the driver when they matter most. Real-time analysis enables immediate feedback that can help prevent incidents instead of only documenting them.
- **Reusability and Scalability:** The project serves as a reference implementation that can be adapted, expanded, and optimized by future teams, supporting continuous improvement in mobile AI safety systems and evaluation of mobile AI model capability.

3.3 Objectives

ALVION's objectives focus on producing a working, reproducible prototype that demonstrates the technical and practical viability of phone-based driver monitoring:

- **Develop a proof-of-concept Android application** capable of detecting drowsiness and distraction using the front camera and a lightweight on-device model optimized for Snapdragon NPUs [13].
- **Implement configurable alert mechanisms** that combine audio, visual, and haptic feedback, allowing users to tune thresholds and notification styles for comfort and responsiveness.
- **Achieve efficient real-time performance** without dependence on cloud computing, ensuring low latency and minimal power usage consistent with mobile operation.
- **Document all design decisions, architecture, and testing methodology** in a reproducible form so that future development teams can extend or refine the system.
- **Conduct iterative testing and evaluation** under varying lighting, motion, and device conditions to measure accuracy, usability, and overall system robustness.

Through these objectives, ALVION demonstrates a concrete pathway from concept to functional prototype, aligning technical feasibility with ethical design and industry safety goals.

4. Requirements

4.1 User Requirement

4.1.1 Glossary of Relevant Domain Terminology

ALVION The Android-based driver drowsiness and distraction detection mobile application developed for this capstone project. Runs entirely on-device to ensure privacy and real-time performance.

Driver Monitoring System (DMS) A safety technology that uses sensors or cameras to assess driver attention, alertness, and potential fatigue while operating a vehicle.

Head Yaw The horizontal rotation of a driver's head relative to the camera. Excessive yaw over time indicates potential distraction or inattention.

Yaw Delta The angular difference between the driver's current head yaw and their calibrated forward baseline.

Android Studio The official integrated development environment (IDE) for Android applications. Used to develop and build the ALVION mobile app using Kotlin.

Kotlin A modern programming language fully supported by Android, known for concise syntax and null safety. Used as the primary implementation language for ALVION.

Jetpack Compose Android's declarative UI framework written in Kotlin. It simplifies the creation of reactive, adaptive user interfaces without the need for XML layout files.

CameraX [**2**] A Jetpack library that provides easy access to camera functions such as preview, capture, and frame analysis. Used in ALVION to stream frames for real-time inference.

ML Kit Face Detection [**1**] Google's on-device machine learning SDK used in ALVION to provide real-time eye-openness probabilities and head pose (Euler angles) without requiring a custom-trained model.

Firebase Authentication [**3**] A cloud-based service used to manage user identity and secure login flows within the application.

Calibration Buckets Specific data collections (e.g., "forward", "left", "right") used during the setup phase to establish baseline eye-openness and head-pose values for an individual driver.

Mirror Symmetry Logic A heuristic used during calibration to infer eye-closure thresholds for one side of the face based on the other, improving system robustness when a driver only calibrates one side mirror.

Exponential Moving Average (EMA) A statistical smoothing technique applied to eye-openness scores to reduce noise and prevent flickering alerts caused by momentary blinks or sensor jitters.

Debounce / Alert Cooldown Software mechanisms that prevent repeated, rapid-fire triggers of an alert within a short time window, ensuring that warnings remain helpful rather than distracting.

Latency The time delay between camera frame capture and alert output. Low latency is critical to ensure timely driver warnings.

FPS (Frames Per Second) A measure of how many frames are processed per second during video analysis. Maintaining $\geq$15 FPS ensures smooth real-time inference.

On-device Inference The process of running AI models locally on the phone rather than sending data to cloud servers. Essential for preserving privacy and minimizing delay.

Privacy by Design A principle ensuring that user data (especially facial video) remains on the device and is not transmitted externally without explicit consent.

4.1.2 User Groups

This section describes the primary user groups who interact with ALVION today, as well as optional stakeholders shown in the context/scope diagrams as potential future extensions.

- **End Users (Drivers):** Primary users who run the app during driving and rely on real-time alerts and feedback to help detect potential drowsiness or distraction.
- **Developers / Maintainers:** Internal users responsible for building, testing, maintaining, and improving the system (e.g., updating detection logic, calibration flow, app UX, and configuration).
- **Optional / Future Stakeholders:** Roles such as fleet managers or system administrators may be supported in later iterations (e.g., aggregated summaries or managed configuration), but are not part of the current MVP.

4.1.3 Project Context

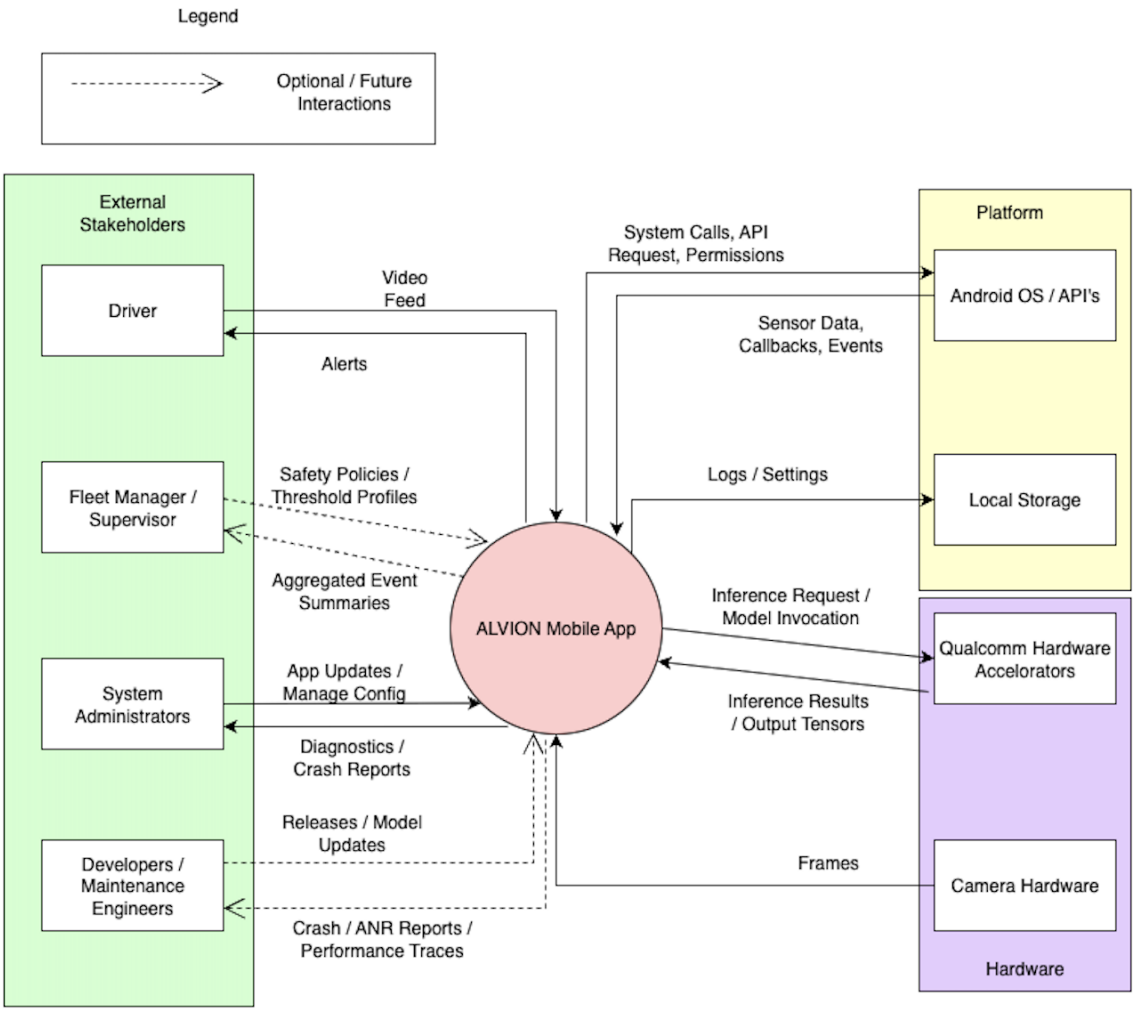

Figure 4.1: System context diagram for the ALVION driver drowsiness and distraction detection mobile application

The context diagram illustrates how the ALVION mobile application interacts with its surrounding environment. The driver provides a live camera feed through the smartphone's front-facing camera and receives real-time alerts through audio, vibration, and visual cues. The application communicates with the Android OS using standard APIs for camera access [2], permissions, audio output, vibration, and local storage. Core detection is performed on-device using a mobile face-detection pipeline (e.g., ML Kit [1]) to estimate signals such as eye openness and head pose for time-based alerting. The system may also integrate cloud-backed application features such as authentication (e.g., Firebase [3]) to support a realistic app workflow. Optional connections shown in the diagram (e.g., aggregated summaries or managed configuration) represent potential future extensions rather than required MVP functionality.

4.1.4 Functional Requirements

4.1.4.1 Project Scope

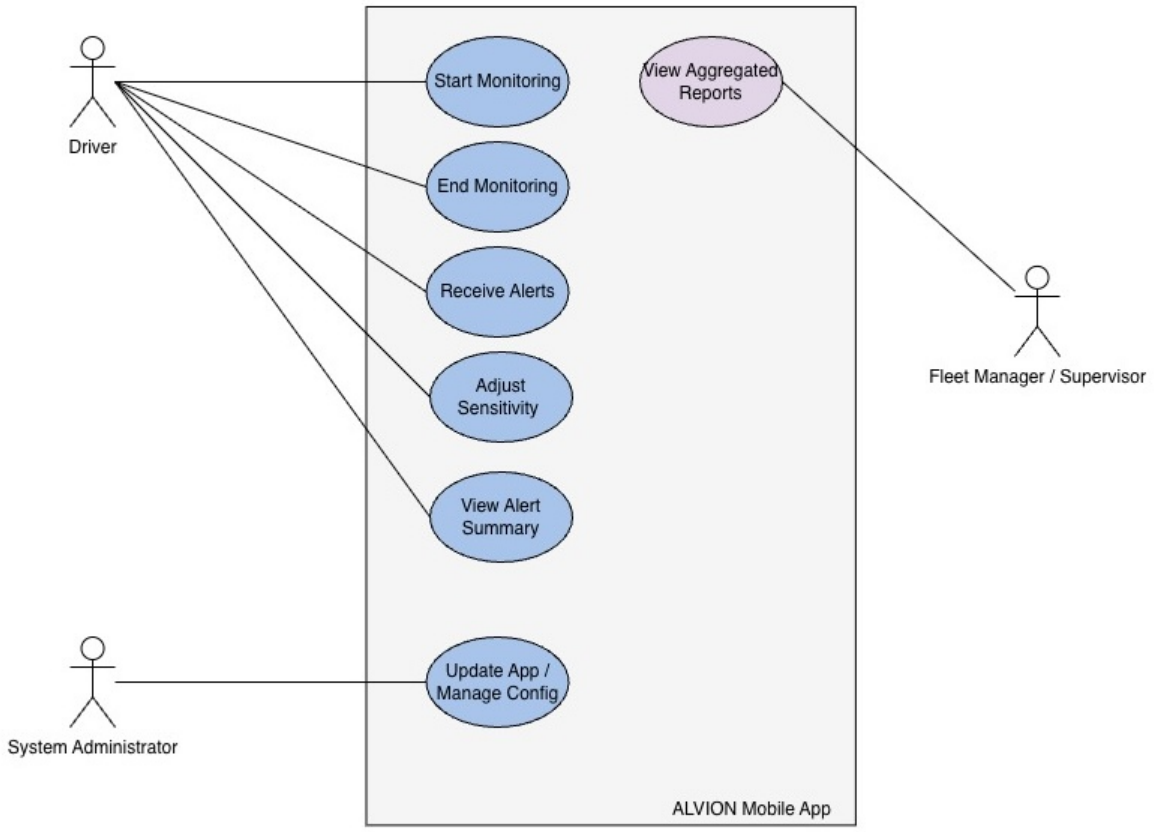

Figure 4.2: Scope diagram to illustrate major user interactions with ALVION

4.1.4.2 User Scenarios

The following scenarios describe how end users interact with the ALVION mobile application during setup, monitoring, and post-session review. Each scenario corresponds to a use case identified in the Project Scope diagram and summarized in Appendix U.

- **Start Monitoring Session:** The driver launches ALVION, completes sign-in if required, grants camera permissions, and starts a monitoring session. The app activates the front-facing camera through CameraX and initializes the face-detection pipeline to estimate eye openness and head pose in real time.
- **Receive Drowsiness Alert:** During monitoring, ALVION evaluates eye-closure behavior over time. If eye closure persists beyond a configured duration threshold, the system issues an audible and on-screen alert (and optional vibration) to prompt the driver to refocus or take a break.
- **Receive Distraction Alert:** ALVION monitors head orientation relative to the calibrated forward baseline. If the driver's head remains turned away from the forward direction beyond the configured threshold window, the system triggers a distraction alert.

- **Acknowledge or Mute Alerts:** After an alert is issued, the driver can acknowledge it and, if needed, temporarily mute alerts. The system applies a cooldown interval to prevent repeated notifications in rapid succession.
- **Adjust Settings in Profile Tab:** The driver opens the Profile/Settings screen to update preferences (e.g., alert enablement, sensitivity thresholds, vibration/sound options, and emergency contact settings). Settings persist between sessions and may be stored securely (e.g., via Firebase) depending on the current implementation.
- **Calibrate / Align Camera:** On first use, after device repositioning, or when calibration quality degrades, the driver completes a short calibration flow to align the camera and establish a baseline. The app provides live feedback to guide the driver until calibration succeeds.
- **Emergency Contact / Call for Help:** If the driver believes they are unsafe to continue (e.g., repeated drowsiness warnings), the driver can use an emergency action to place a call to **911** or a designated family member/emergency contact. This action is intentionally user-initiated to avoid accidental calls.
- **View Trip History:** After ending a session, the driver can navigate to the **History** tab to review past trips. Each trip card displays the date, duration, start time, and a categorized list of safety detections (drowsiness, distraction, presence check, phone orientation). Trip records are automatically saved to Firebase Firestore and persist across app restarts.
- **Live Speed and Duration Metrics:** While a session is active, the app displays real-time trip duration and current GPS speed (km/h) in a metrics grid below the camera view. GPS speed is obtained via the Android Fused Location Provider and requires location permission.
- **Phone Orientation Warning:** If the phone is detected as physically upside-down during monitoring (using the accelerometer and display rotation API), the system immediately issues an alert prompting the driver to flip the device. This detection is debounced to avoid false triggers from sensor noise.

4.1.4.3 User Functional Requirements

This section provides short descriptions of the user functional requirements identified for the ALVION mobile application. Each requirement defines a specific capability the system provides to meet the goals of the drivers and the technical needs of the development team.

1. **UF-A: Account Management** — The system shall allow drivers to create an account and securely log in using Firebase Authentication [3] to manage personal session history and preferences.
2. **UF-B: Drowsiness Detection and Alerting** — The system shall continuously analyze eye openness using the PERCLOS metric [4] and issue a warning when sustained eyelid closure (exceeding a defined millisecond threshold) indicates drowsiness. (Supported by Use Cases UC-001 and UC-002.)
3. **UF-C: Distraction Detection and Alerting** — The system shall detect when the driver's head yaw deviates from the calibrated forward baseline for longer than a defined interval and trigger an attention-restoring alert. (Supported by Use Case UC-010.)
4. **UF-D: Configurable Sensitivity and Alert Settings** — The system shall provide a settings interface that allows the driver to adjust detection sensitivity, alert volume, and the "cooldown" period between consecutive alerts. (Supported by Use Case UC-004.)
5. **UF-E: Start and Stop Monitoring Session** — The driver shall be able to manually start and end a monitoring session, which handles the lifecycle of the CameraX stream [2] and the

ML Kit inference pipeline [1]. (Supported by Use Case UC-001.)

6. **UF-F: Real-Time Alert Delivery** — The system shall deliver immediate visual, audio, and haptic (vibration) feedback when drowsiness or distraction is detected to ensure driver response. (Supported by Use Cases UC-002 and UC-010.)
7. **UF-G: Acknowledge or Snooze Alerts** — The system shall allow the driver to acknowledge an active alert or temporarily snooze notifications during short recovery periods. (Supported by Use Case UC-003.)
8. **UF-H: Multi-Stage Camera Calibration** — The system shall provide a calibration routine to establish the driver's forward-facing baseline. This includes "Mirror Logic" to infer eye-closure thresholds for side-mirrors based on partial calibration data. (Supported by Use Case UC-005.)
9. **UF-I: Trip History and Session Persistence** — The system shall automatically save each completed monitoring session as a `Trip` record in Firebase Firestore, including start/end timestamps, duration, and a timestamped list of alert events (drowsiness, distraction, presence check, phone orientation). The driver shall be able to view all past trips in a dedicated **History** tab and clear the history if desired. (Supported by Use Cases UC-006 and UC-007.)
10. **UF-J: Privacy and Local Data Control** — The system shall process all camera frames locally. No raw video or identifiable facial images shall be transmitted externally; only anonymized alert metadata (event type, timestamp) may be stored in the cloud via Firestore. (Emphasized in Use Cases UC-001 and UC-006.)
11. **UF-K: Sunglasses and Occlusion Detection** — The system shall use a custom TFLite binary classifier to detect when the driver is wearing sunglasses during a monitoring session. When sunglasses are detected, the drowsiness detection pipeline shall be suppressed and the driver shall be notified that eye-based monitoring is paused, while head-pose distraction monitoring continues unaffected.

4.1.5 Non-functional Requirements

The following non-functional requirements capture quality and constraint expectations from the driver's perspective. Full descriptions are provided in Appendix R.

4.1.5.1 Product: Usability Requirements

The driver must be able to start a monitoring session in under two taps. Alerts must be clear, well-timed, and non-distracting.

4.1.5.2 Product: Performance Requirements

The system must respond in real time ($\leq$200 ms alert latency) and maintain smooth frame analysis without perceptible lag.

4.1.5.3 Product: Availability/Reliability/Security

Core monitoring must remain stable for a full driving session ($\geq$60 minutes). User credentials must be securely managed and not exposed.

4.1.5.4 Organizational: Development Requirements

The system shall be built on standard Android tools (Kotlin, Jetpack Compose) to ensure maintainability by future development teams.

4.1.5.5 Organizational: Operational Requirements

Core detection must function without an active internet connection. GPS and history features may require connectivity.

4.1.5.6 Organizational: Environmental Requirements

The app must operate reliably under varied in-car lighting (day, night, overcast) and across different mounting angles.

4.1.5.7 External: Legislative Requirements on Safety/Security

The system shall comply with mobile privacy best practices by keeping facial data on-device and not storing biometric identifiers.

4.1.5.8 External: Cultural and Social Requirements

The system should function equitably across drivers of diverse skin tones and facial structures as supported by ML Kit.

4.1.5.9 External: Interoperability Requirements

The application targets Android SDK 24 minimum, ensuring broad device compatibility without requiring proprietary hardware.

4.2 System Requirements

4.2.1 Functional Requirements

4.2.1.1 System Functional Requirements

The ALVION system shall implement the following core functional features to enable real-time driver monitoring and alerting:

- **SF-01: Real-Time Drowsiness Detection** — The system shall utilize the ML Kit Face Detection API [1] to analyze eye-openness probabilities. Using the PERCLOS metric [4], it shall trigger an alert when the EMA-smoothed eye score remains below the calibrated threshold for more than 1,200 ms (default drowsiness trigger duration).
- **SF-02: Distraction Detection** — The system shall track head yaw (horizontal rotation) and issue a distraction alert using a tiered dwell-time hierarchy: 4,000 ms when the driver's head is within a calibrated mirror-check range, and 2,000 ms when the deviation exceeds that range (e.g., looking at a phone or passenger).
- **SF-03: Multi-Stage Alert Mechanisms** — The system shall deliver synchronous alerts using audio tones, haptic vibration, and visual UI overlays to ensure the driver is promptly notified of safety risks.

- **SF-04: Account and History Management** — The system shall provide a secure login interface via Firebase Authentication [3], allowing drivers to maintain personal session statistics and persistent device configurations.
- **SF-05: Dynamic Calibration and Mirror Logic** — The system shall establish a forward-facing baseline during setup. It shall include "Mirror Symmetry" logic to automatically infer detection thresholds for side-viewing angles based on partial user calibration.
- **SF-06: Configurable Thresholds** — The system shall allow the driver to adjust eye-closure sensitivity, head-pose thresholds, and the "alert cooldown" period via a dedicated Settings interface.
- **SF-07: Session Control and Lifecycle** — The application shall allow users to manually start and stop monitoring, correctly managing the Android CameraX lifecycle [2] and ML inference pipeline [1].
- **SF-08: Snooze and Acknowledge Logic** — The system shall provide interactive UI elements to acknowledge or temporarily snooze active alerts, preventing repetitive notifications once the driver has regained focus.
- **SF-09: Local Privacy and Data Protection** — All frame analysis shall occur strictly on-device. The system shall store session summaries and event logs in secure, app-scoped storage, with no transmission of raw video data to external servers.
- **SF-10: Session Summary Reporting** — Upon ending a session, the app shall generate a report summarizing the frequency and type of alerts triggered and persist this record to Firebase Firestore for the authenticated user.
- **SF-11: Real-Time GPS Speed Display** — During an active monitoring session, the system shall display the driver's current speed in km/h using the Android Fused Location Provider. Location updates shall be polled at 1-second intervals and the display shall revert to 0 km/h when the session ends or location permission is unavailable.
- **SF-12: Phone Orientation Monitoring** — The system shall detect when the device is physically upside-down during a session using the accelerometer sensor and display rotation API. Upon confirming an inverted orientation (debounced by 250 ms), it shall issue a warning alert and log a `PHONE_UPSIDE_DOWN` event to the current session record.
- **SF-13: TFLite Sunglasses Detection** — The system shall run a custom binary TFLite classifier (`sunglasses_model.tflite`) on a cropped face region every 3 frames during monitoring. When sunglasses are detected with confidence ≥ 0.70 for 2 consecutive inferences, drowsiness detection shall be suppressed and the eye-occlusion state shall be set. Detection shall be cleared when confidence falls below 0.45 for 2 consecutive inferences.
- **SF-14: Presence / Liveness Check** — When eye landmarks are occluded and the detected face position and head rotation remain statistically static for 10,000 ms (variance below defined thresholds), the system shall issue a presence-check alert prompting the driver to confirm alertness by moving their head.
- **SF-15: Face Proximity Guard** — The system shall detect when the driver's face occupies more than 60% of the camera frame area (indicating the device is too close to the face) and temporarily suspend safety monitoring for that frame, preventing unreliable landmark readings from triggering false alerts.

4.2.1.2 Data Requirements

The ALVION application manages data related to user identity, real-time facial analysis, and session-specific calibration. Since the system utilizes a pre-trained ML Kit model, data handling is focused on user authentication and the transient processing of camera frames.

Data Inputs

- **User Authentication Data:** Email and password credentials provided during login/signup, managed via Firebase Authentication [3].
- **Live Camera Feed:** RGB frames captured via CameraX [2], used exclusively for real-time inference and never stored.
- **Facial Landmark Metrics:** Raw probability scores (eye-openness) and Euler angles (head rotation) provided by the ML Kit Face Detection API [1].

Data Processing and Storage

- **User Profiles (Cloud):** Basic user identity data is stored securely in Firebase Authentication to manage accounts across sessions.
- **Trip History (Cloud):** Each completed monitoring session is saved as a `Trip` document in Firebase Firestore (`trips` collection), containing the user ID, date label, start/end timestamps, duration, and a list of `TripAlert` events (type and timestamp). This data is scoped per authenticated user and persists indefinitely until the driver clears it.
- **Calibration Baselines (Local):** Yaw offsets and eye-closure thresholds are calculated during calibration and stored locally in SharedPreferences to maintain the user's forward baseline for the duration of the app's use.
- **Session Metrics (Volatile):** Running EMA scores for eye openness and head-pose state are maintained in-memory and discarded once the monitoring session ends. Raw video frames are never written to storage.

Data Quality and Privacy

- **On-Device Processing:** All facial analysis is performed locally. No raw images, video, or biometric "face-print" data is transmitted to Firebase or any external server.
- **Threshold Clamping:** Eye-openness values are strictly clamped between 0 and 1 to ensure consistent alert triggers across different lighting conditions.
- **Secure Authentication:** Password management and token handling are delegated to Firebase, ensuring industry-standard security for user accounts.

While user identity and anonymized alert metadata are managed in the cloud, all privacy-sensitive facial data remains volatile and local to the driver's device.

4.2.2 Non-functional Requirements

This section defines the technical constraints and quality standards the ALVION system must meet. These requirements serve as the benchmarks for system testing and performance evaluation.

4.2.2.1 Product: Usability Requirements

- **SP-01-01** — The interface shall allow a driver to initiate monitoring in under two taps to ensure minimal distraction during vehicle setup.

- **SP-02-01** — The system shall maintain hands-free operation; no manual input shall be required or accepted while a monitoring session is active.
- **SP-04-01** — The application shall display a safety disclaimer upon launch that must be acknowledged before the driver can access monitoring features.

4.2.2.2 Product: Performance Requirements

- **SP-05-01** — The inference pipeline shall maintain a minimum of 15 FPS and an alert latency of $\leq$200 ms on supported Snapdragon devices.
- **SP-05-02** — Average CPU utilization shall remain below 40% by utilizing the Android Neural Networks API (NNAPI) [12] for ML Kit hardware acceleration.
- **SP-07-01** — The system shall include thermal management logic that reduces inference frequency if the device temperature exceeds safe operating limits.

4.2.2.3 Product: Availability, Reliability, and Security

- **SP-03-01** — Visual, auditory, and haptic alerts shall be synchronized within $\pm$50 ms to provide a clear, unified warning to the driver.
- **SP-08-01** — The application shall remain operational for at least 60 minutes of continuous monitoring without crash or significant performance degradation.
- **SP-09-01** — In the event of a camera obstruction or ML failure, the system shall provide a "Tracking Lost" notification within 3 seconds of continuous landmark loss.
- **SP-10-01** — All facial analysis shall be performed locally. No raw images or video data shall be transmitted externally or stored permanently.

4.2.2.4 Organizational: Development Requirements

- **SO-01-01** — The project shall be implemented using Kotlin and Jetpack Compose, targeting Android SDK 33 (Android 13) or higher.

4.2.2.5 Organizational: Operational Requirements

- **SO-09-01** — Core detection and alerting features must function without an active internet connection (strictly offline monitoring).

4.2.2.6 Organizational: Environmental Requirements

- The system shall operate across typical automotive lighting environments, including low-light night driving and high-glare daytime conditions.

4.2.2.7 External: Legislative Requirements on Safety/Security

- **SE-01-01** — The application shall comply with mobile safety standards by disabling configuration menus while the vehicle is detected to be in motion.

4.2.2.8 External: Cultural and Social Requirements

- **SE-03-01** — The system shall be validated for consistent landmark detection across varied skin tones and common eyewear (glasses).

4.2.2.9 External: Interoperability Requirements

- The system shall integrate with Firebase Authentication and Firestore using standard Google Play Services APIs, without dependency on proprietary device-specific SDKs.

4.3 Requirements Trace Table

A Requirements Traceability Table is provided in Appendix R to ensure that all user requirements are systematically mapped to system functionality, design components, and testing procedures.

This table establishes clear relationships between:

- User Functional Requirements (UF-A through UF-K)
- System Functional Requirements (SF-01 through SF-15)
- Design components such as the `FaceDetectionAnalyzer`, `FaceStateEvaluator`, `SunglassesClassifier`, `HistoryRepository`, and `SpeedTracker`
- Corresponding test cases used for validation

This traceability ensures that every requirement is:

- Properly implemented within the system
- Verified through testing
- Consistent with the overall system architecture

By maintaining this mapping, the project guarantees completeness, reduces the risk of missing functionality, and improves maintainability for future development.

5. Exploratory Studies

5.1 Relevant Development Frameworks

We are building a phone-first prototype that runs entirely on the device. To ensure real-time performance and reliability on Android, we have selected a modern, native stack that minimizes external dependencies.

5.1.1 Android Studio (Kotlin)

What it is. Android Studio is the official IDE; **Kotlin** is the primary implementation language. **Why we use it.** It provides the most robust support for modern Android development, including built-in profiling tools to monitor CPU and thermal impact during long driving sessions.

5.1.2 Front end: Jetpack Compose (Material3)

What it is. Compose is Android's modern declarative UI toolkit [8]. **Why we use it.** It allows us to build responsive, high-performance UI overlays (e.g., the camera preview and alert badges) that react instantly to changes in the driver's attention state without the overhead of XML-based layouts.

5.1.3 Camera: CameraX (Preview + Analysis)

What it is. A Jetpack library [2] that simplifies camera integration. **Why we use it.** CameraX allows us to split the camera feed into two paths: a low-latency preview for the user and a continuous "Analysis" stream of image proxies. It handles device rotation and lifecycle management automatically, ensuring the app doesn't crash when backgrounded.

5.1.4 Inference: ML Kit Face Detection

What it is. An on-device Qualcomm SDK [1] for vision tasks. **Why we use it.** ML Kit provides a pre-trained, highly optimized pipeline for facial landmarking. It returns the exact metrics needed for our safety logic—specifically eye-openness probabilities and head Euler angles (yaw)—without the need for custom model training or hosting.

5.1.5 Backend: Firebase Authentication and Firestore

What it is. A secure cloud-based identity and database platform [3]. **Why we use it.** Firebase Authentication handles secure login and account management. Firebase Firestore stores each completed trip as a structured document (alert types, timestamps, duration) scoped to the authenticated user, enabling persistent trip history across sessions without transmitting any raw facial data.

5.1.6 On-Device Custom Classifier: TFLite

What it is. The TensorFlow Lite runtime for Android (now distributed as LiteRT) [10]. **Why we use it.** ML Kit does not provide a sunglasses-detection signal. We trained and deployed a custom binary TFLite model (`sunglasses_model.tflite`) that classifies 128×128 cropped face images

as sunglasses / no-sunglasses. Running this classifier every 3 frames suppresses false drowsiness detections caused by eye occlusion from dark lenses.

5.1.7 Location: Fused Location Provider

What it is. Google Play Services' battery-efficient location API. **Why we use it.** We use the Fused Location Provider [11] to stream GPS speed updates at 1-second intervals during active sessions, displayed in the live metrics grid. This context (speed = 0 vs. highway speed) helps drivers understand their monitoring state.

5.1.8 Project Glue: Coroutines and Flow

What it is. Kotlin's asynchronous programming primitives. **Why we use it.** We use **Coroutines** [9] to keep the heavy ML analysis off the main UI thread. **Flow** allows us to stream detection events (e.g., a "Drowsy" signal) from the logic layer to the UI layer in real-time with minimal latency.

Tool roles at a glance

Tool	Role	Why it's in the stack
Android Studio + Kotlin	App development	Native toolchain; full hardware access.
Jetpack Compose	Declarative UI	Reactive alerts and polished components.
CameraX	Frame Analysis	Reliable frame capture; handles rotation.
ML Kit Face Detection	Vision Inference	Pre-trained metrics for eyes and yaw.
TFLite Interpreter	Sunglasses Detection	Custom binary classifier for eye occlusion.
Firebase Auth + Firestore	User Accounts + History	Login and persistent trip records.
Fused Location Provider	GPS Speed	Real-time speed display during sessions.
Kotlin Coroutines	Concurrency	Keeps UI responsive during ML tasks.

5.2 Relevant Solution Techniques

ALVION utilizes a heuristic-based detection engine built on top of pre-trained facial landmark metrics provided by ML Kit. Rather than training a custom classifier, our technique focuses on **subject-specific calibration** and **temporal smoothing** to ensure accuracy across different drivers and mounting angles.

5.2.1 Calibration Strategy: Multi-Angle Scan

A core innovation of the system is the multi-angle calibration routine, which establishes the driver's unique visual geometry before monitoring begins. This prevents false alerts caused by natural variations in seating position or device mounting.

- **Forward Baseline Establishment:** The driver is prompted to look straight ahead at the road. The system captures a series of frames to establish a `forwardYawBaseline`. All subsequent head movements are calculated as a *Yaw Delta* relative to this baseline.
- **Mirror Angle Bucketing:** The system provides specific "buckets" for the Left Mirror, Center/Forward, and Right Mirror. During calibration, the driver scans these positions, and the system records the specific Euler Y (yaw) angles and eye-openness ranges for each.
- **Mirror Symmetry Logic:** To minimize setup time, the system implements symmetry-based inference. If a driver only calibrates the "Left Mirror" bucket, the system automatically calculates the expected "Right Mirror" yaw and eye-thresholds by mirroring the left-side data relative to the forward baseline.

5.2.2 Detection Logic: Eye Thresholds and Head Pose

The detection engine processes the calibrated data using two parallel logic paths for Drowsiness and Distraction.

Drowsiness (Sustained Eye Closure): The system implements a **PERCLOS Proxy [4, 5]** (Percentage of Eyelid Closure) using a duration-based trigger.

1. **EMA Smoothing:** Raw eye-openness scores from ML Kit are processed using an *Exponential Moving Average* [14] ($\alpha = 0.45$) to filter out momentary blinks and sensor noise.
2. **Dynamic Thresholding:** Instead of a fixed 0.5 threshold, the system uses the eye-openness stats gathered during calibration to set a custom "closed" threshold for each driver.
3. **Temporal Trigger:** If the EMA eye-score remains below the threshold for more than 1,200 ms, a "Drowsy" event is published to the alert manager.

Distraction (Yaw Deviation): Distraction is detected by monitoring the *Yaw Delta* against a dwell-time hierarchy.

1. **Mirror-Check Tolerance:** If the driver is looking toward a calibrated mirror angle, the system allows a dwell time of 4,000 ms to account for normal traffic checks.
2. **Beyond-Mirror Deviation:** If the yaw delta exceeds the mirror range (indicating the driver is looking at a passenger or a phone), a shorter dwell time of 2,000 ms is enforced before an alert is triggered.
3. **Yaw Hysteresis:** To prevent alert "flickering" near the threshold, the system requires the head to return 8° back toward the center before the distraction state is reset.

5.2.3 Evaluation and Tuning Strategy

Since the system uses a pre-trained model for landmarks, our evaluation focus shifts from "Model Accuracy" to "Pipeline Robustness."

- **Environmental Validation:** The pipeline is tested across three primary lighting conditions: High Noon (direct sunlight), Overcast (diffuse light), and Night (artificial cabin light).

- **Occlusion Testing:** Evaluation specifically targets how well the ML Kit landmarks maintain stability when the driver wears prescription glasses or sunglasses.
- **Latency Benchmarking:** We measure the "End-to-End" latency—from the moment the eyes close to the moment the haptic motor vibrates—to ensure we stay below the 200 ms performance requirement.

5.2.4 What "Success" Looks Like

A successful implementation on the Snapdragon target device is defined by:

- **Zero-Cloud Dependency:** All calibration and detection logic executes locally via the ML Kit SDK.
- **Stable 24–30 FPS:** Smooth camera preview and concurrent analysis without thermal throttling.
- **Low False-Positive Rate:** The "Mirror Symmetry" and "EMA Smoothing" logic correctly distinguishes between a long blink and true drowsiness.

5.3 Broader Impacts

The ALVION project demonstrates the feasibility of bringing advanced safety features to a wider demographic without the need for high-cost, integrated vehicle hardware. By leveraging commodity smartphone sensors and on-device ML, the project creates meaningful impacts in road safety and mobile AI application design.

Impact on Individuals (Drivers)

- **Safety for Legacy Vehicles.** The most immediate impact is on drivers of older vehicles that lack built-in Driver Monitoring Systems (DMS). ALVION provides these users with a low-cost alternative to expensive retrofit hardware, offering real-time alerts that can prevent accidents caused by fatigue or distraction.
- **Empowerment through Awareness.** Because the system provides immediate feedback and a personal "Trip History," it encourages drivers to recognize their own fatigue patterns. This empowers individuals to make informed decisions about when to take breaks, fostering long-term safer driving habits.
- **Low-Barrier Awareness.** Because the system requires only a standard Android device, it encourages safer driving habits through immediate feedback, helping drivers recognize their own fatigue patterns before they become a critical danger.

Impact on Organizations

- **Scalable Fleet Safety.** For organizations such as delivery services or ride-sharing platforms, ALVION offers a software-defined safety solution that can be deployed across a fleet without hardware installation costs. This allows organizations to improve safety standards and reduce liability with minimal capital expenditure.
- **Technical Template for Developers.** The project serves as a proof-of-concept for running complex, real-time vision pipelines (ML Kit and CameraX) on Snapdragon hardware. It provides a template for organizations looking to build responsive, safety-critical applications that must operate reliably at the "edge" without constant cloud connectivity.

- **Resource-Efficient Safety.** The project proves that sophisticated safety signals—like yaw delta and eyelid closure—can be computed with minimal battery and thermal impact, setting a baseline for what is achievable in modern edge-based AI.

Impact on Society

- **Road Safety Democratization.** High-end safety technology is often locked behind luxury vehicle tiers. ALVION contributes to a more equitable landscape where road safety tools are accessible to anyone with a smartphone, potentially reducing the global frequency of fatigue-related collisions.
- **Privacy-First Safety Standards.** By processing all facial data locally and only storing anonymized metadata in Firestore, ALVION sets a societal benchmark for "Privacy by Design." It proves that public safety can be improved without compromising individual civil liberties or creating large-scale biometric databases.
- **Infrastructure Independence.** Unlike systems that require continuous cloud processing or dedicated vehicle sensors, ALVION operates entirely independently. This makes it a viable safety tool in areas with poor network coverage or for users who prioritize offline, autonomous operation.

6. System Design

6.1 Architectural Design

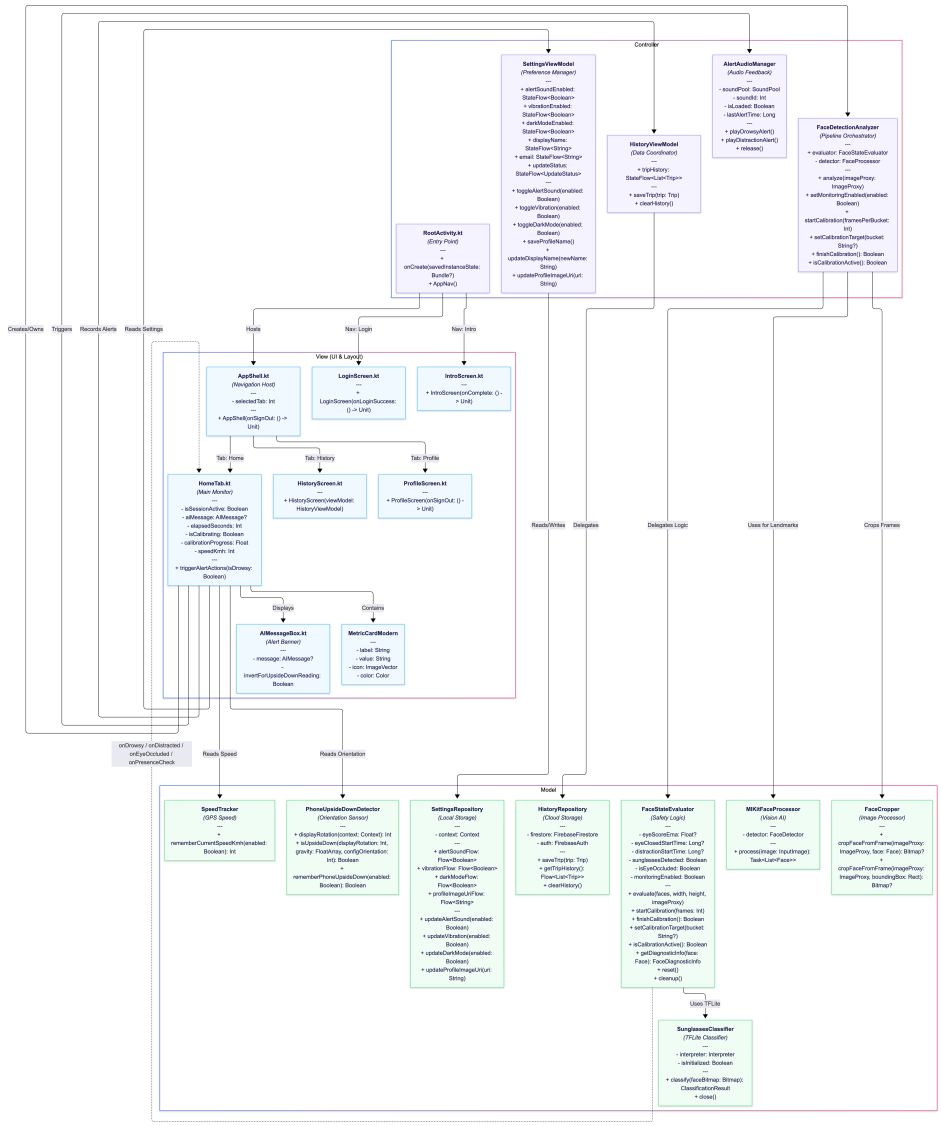

Figure 6.1: Architectural Design (MVC Pattern).

6.1.1 Architecture Overview

What the diagram shows. The ALVION system is organized using the **Model-View-Controller (MVC)** pattern to ensure a clean separation between the real-time vision pipeline and the user interface.

- **View (Jetpack Compose):** The `LoginScreen`, `IntroScreen`, `HomeTab`, `HistoryScreen`, and `ProfileScreen` are hosted via `AppShell` and routed by `RootActivity` based on Firebase

Auth state. Views are responsible for rendering the camera feed, displaying alert banners, showing trip history, and capturing calibration input.

- **Controller (Orchestration Layer):** `RootActivity` manages top-level navigation. `FaceDetectionAnalyze` coordinates the CameraX frame stream, ML Kit inference, and per-frame sunglasses detection via `FaceCropper`. `AlertAudioManager` triggers low-latency audio alerts via `SoundPool`. `HistoryViewModel` and `SettingsViewModel` manage trip data and user preference lifecycles, exposing `StateFlow` streams to the View layer.
- **Model (Safety Engine and Data):** `FaceStateEvaluator` is the core safety engine, maintaining driver state (*Normal*, *Drowsy*, *Distracted*, *Eye-Occluded*) using EMA smoothing, calibration baselines, and heuristic rules. It fires callbacks (`onDrowsy`, `onDistracted`, `onEyeOccluded`) to the View layer. `MlKitFaceProcessor` handles face landmark detection, `SunglassesClassifier` runs a TFLite binary classifier for occlusion detection, and `FaceCropper` converts YUV camera frames to RGB bitmaps for model input. `HistoryRepository` persists trips to Firebase Firestore, `SettingsRepository` manages preferences via Jetpack `DataStore`, `SpeedTracker` streams GPS speed, and `PhoneUpsideDownDetector` detects device inversion via the accelerometer.

How data moves. When the driver starts a session, **CameraX** begins streaming RGB frames to the `FaceDetectionAnalyzer`. Each frame is converted into an `InputImage` and passed to **ML Kit**. The resulting facial landmarks (eye-openness and Euler angles) are sent to the `FaceStateEvaluator`, which compares them against the stored calibration baselines. Concurrently, the face bounding box is used by `FaceCropper` to extract a cropped image for the `SunglassesClassifier`; if sunglasses are confirmed, the EMA eye pipeline is bypassed. If a threshold is exceeded for the required duration (e.g., 1,200 ms for drowsiness), the evaluator triggers a callback. The UI reacts by displaying a warning banner and initiating haptic/audio feedback. When the session ends, `HistoryRepository` writes a `Trip` document to Firestore.

Why we did it this way. This architecture prioritizes low-latency "edge" processing. By separating the **State Evaluator** from the **Inference Engine**, we can swap or update the landmark detection SDK (e.g., moving from ML Kit to another provider) without rewriting our core safety logic. Using a controller to manage the **CameraX** lifecycle ensures that hardware resources are only active when a session is in progress, preserving battery and preventing thermal issues.

6.2 Structural Design

6.2.1 UML Class Diagram(s)

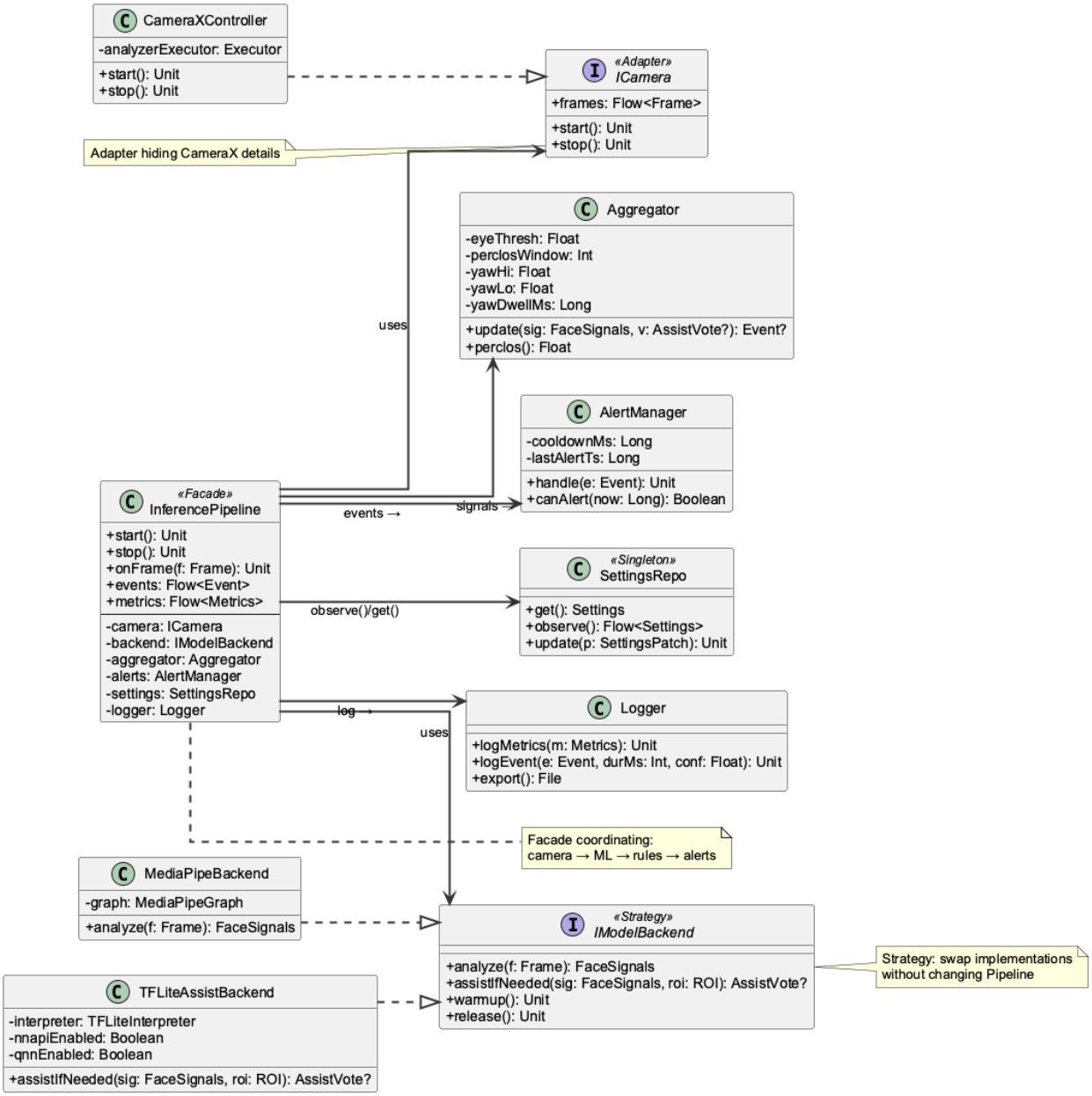

Figure 6.2: Structural overview of the ALVION system, highlighting the separation between the CameraX analyzer, the ML Kit processor, and the safety evaluation logic.

Design patterns used. To ensure the system is both testable and maintainable, the following design patterns are implemented:

- **Facade** (`FaceDetectionAnalyzer`) — Provides a simplified interface for the `HomeTab` to

interact with the complex CameraX analysis lifecycle and the ML Kit detector.

- **Strategy** (`FaceProcessor`) — An interface that abstracts the face detection implementation. While the system currently uses `MlKitFaceProcessor`, this allows the development team to swap in alternative detectors or mock processors for unit testing.
- **Observer / Callback** — The analyzer utilizes high-order Kotlin functions (callbacks) like `onDrowsy` and `onDistracted` to notify the UI layer of safety events without creating tight coupling between the logic and the view.
- **Singleton / Repository** (`SettingsRepo`) — A single source of truth for user preferences (sensitivity, alert volume) and calibration baselines, backed by Android SharedPreferences.
- **Delegation** — The `FaceDetectionAnalyzer` delegates all mathematical safety evaluations to the `FaceStateEvaluator` class, allowing the vision pipeline to remain focused on frame handling.

6.2.2 Entity–Relationship Diagram (ERD)

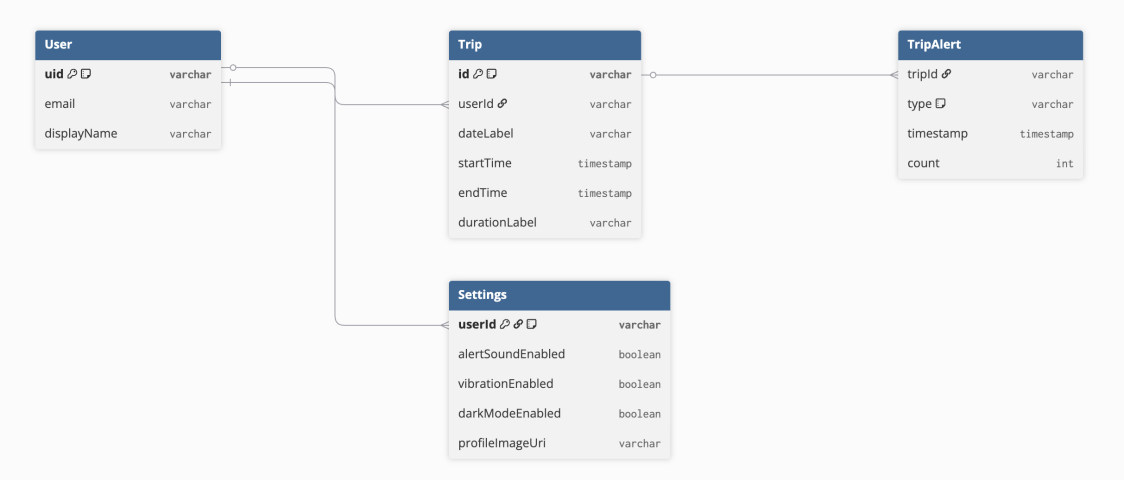

Figure 6.3: Data persistence model: User profiles are stored in the cloud (Firebase), while device-specific settings and calibration baselines are persisted locally.

Data Rationale. The ALVION system minimizes local storage to prioritize driver privacy. The data model is split into two distinct layers:

1. **Cloud Layer (Firebase):** Manages two data stores:
   - **Firebase Authentication** — User credentials and profile metadata, enabling secure login across devices.
   - **Firebase Firestore (`trips` collection)** — Completed monitoring sessions. Each `Trip` document stores the user ID, date label, start/end timestamps, duration label, and a list of `TripAlert` objects (event type: `DROWSINESS`, `DISTRACTION`, `PRESENCE_CHECK`, or `PHONE_UPSIDE_DOWN`; and a Firestore `Timestamp`). Documents are indexed by user ID and ordered by start time.
2. **Local Layer (SharedPreferences):** Stores the `CalibrationBaseline` (forward yaw, mirror angles) and `Settings` (alert toggles, vibration, dark-mode preference). These are device-specific, as the physical mounting angle of the phone may change between different vehicles.

Unlike traditional monitoring systems, ALVION does not persist raw image data or video logs, ensuring that the biometric "footprint" of the driver remains strictly volatile.

6.3 User Interface Design

The ALVION user interface is built using **Jetpack Compose** and follows **Material 3** design guidelines. The UI is designed to be high-contrast and minimalist, ensuring that the driver can interact with the app safely before a journey and receive clear, non-distracting feedback while driving.

6.3.1 Onboarding and Authentication Flow

Before monitoring begins, the driver is guided through an onboarding sequence that explains the system's purpose and safety limits. This is followed by a secure login/signup flow managed via Firebase.

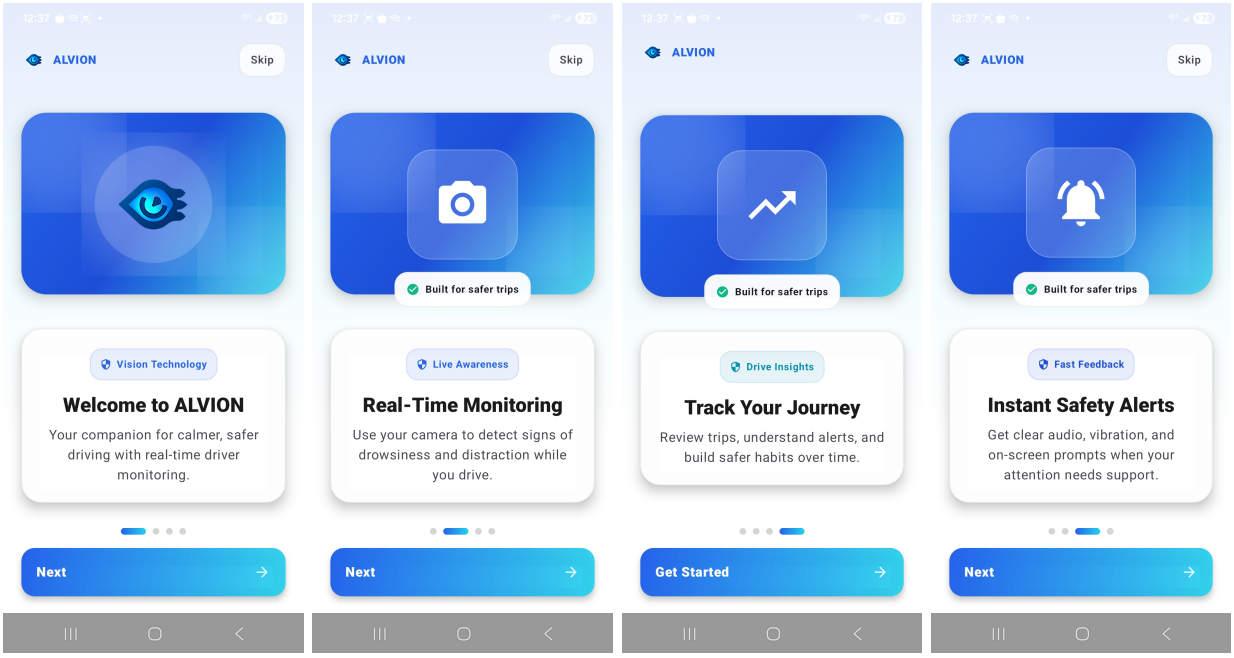

Figure 6.4: * Intro 1

Figure 6.5: * Intro 2

Figure 6.6: * Intro 3

Figure 6.7: * Intro 4

Figure 6.8: Onboarding sequence explaining the ALVION safety features.

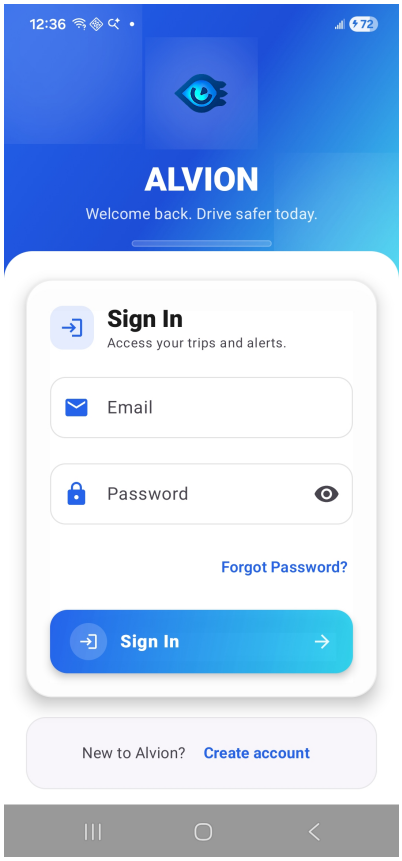

Figure 6.9: Sign In Screen

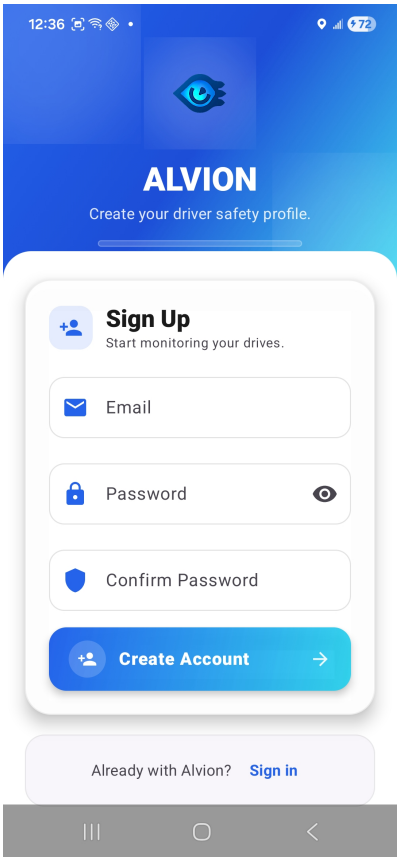

Figure 6.10: Sign Up Screen

6.3.2 Main Dashboard and Monitoring

The dashboard provides a central hub for starting sessions and managing the user profile. The monitoring screen (*Face Detection*) features a minimalist overlay that prioritizes road visibility while providing real-time status cues.

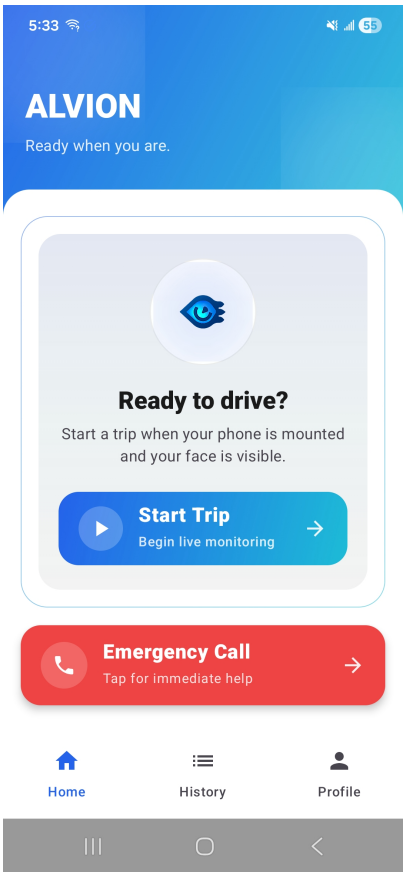

Figure 6.11: Main Dashboard

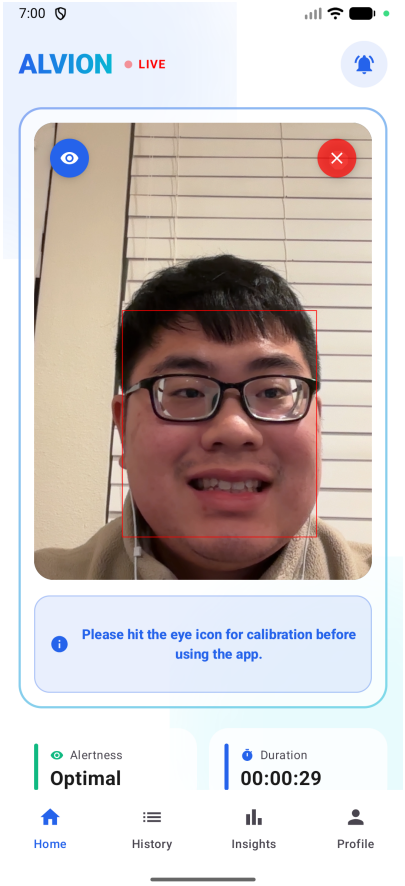

Figure 6.12: Monitoring Mode

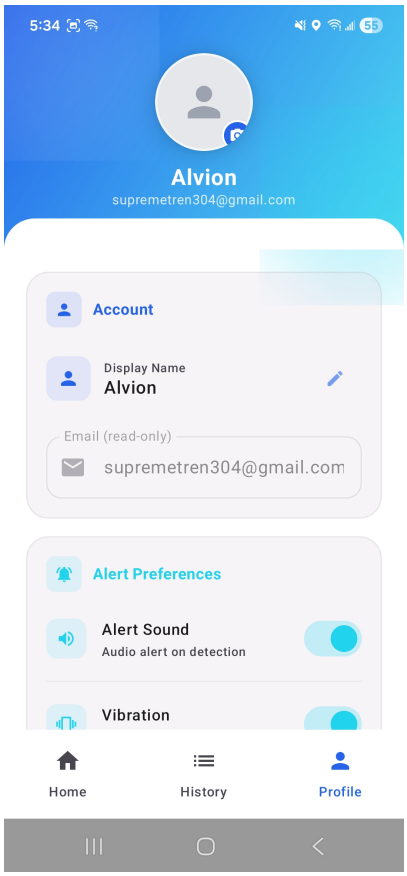

Figure 6.13: User Profile

Figure 6.14: Core application screens for monitoring and profile management.

Design Considerations for Drivers:

- **High-Contrast Accents:** Large buttons and red alert banners are used to ensure visibility in varied cabin lighting.
- **Live Overlay:** The monitoring screen uses a semi-transparent graphic overlay (see *GraphicOverlay.kt*) to show face tracking status without obscuring the camera preview.
- **Three-Tab Navigation:** The app shell (`AppShell.kt`) provides three bottom-navigation tabs: **Home** (camera + monitoring), **History** (trip records), and **Profile** (settings and account).
- **Live Metrics Grid:** When a session is active, a two-card metrics row is displayed below the camera view showing real-time **Duration** (elapsed HH:MM:SS) and **Speed** (GPS km/h). When idle, this area shows the **Emergency Call** button.
- **History Tab:** The History screen (`HistoryScreen.kt`) displays a chronological list of past

trips pulled from Firestore. Each trip card shows the date, duration, alert count badge, and expandable alert-event list with icons colour-coded by event type (amber = drowsiness, rose = distraction, blue = presence check, purple = phone upside-down). A "Clear History" button allows bulk deletion.

- **Appearance Toggle:** The Profile screen includes a Dark Mode toggle, persisted via SharedPreferences, and a profile-picture selector that loads an image from device storage.

latex

6.4 Behavioral Design

6.4.1 Sequence Diagram

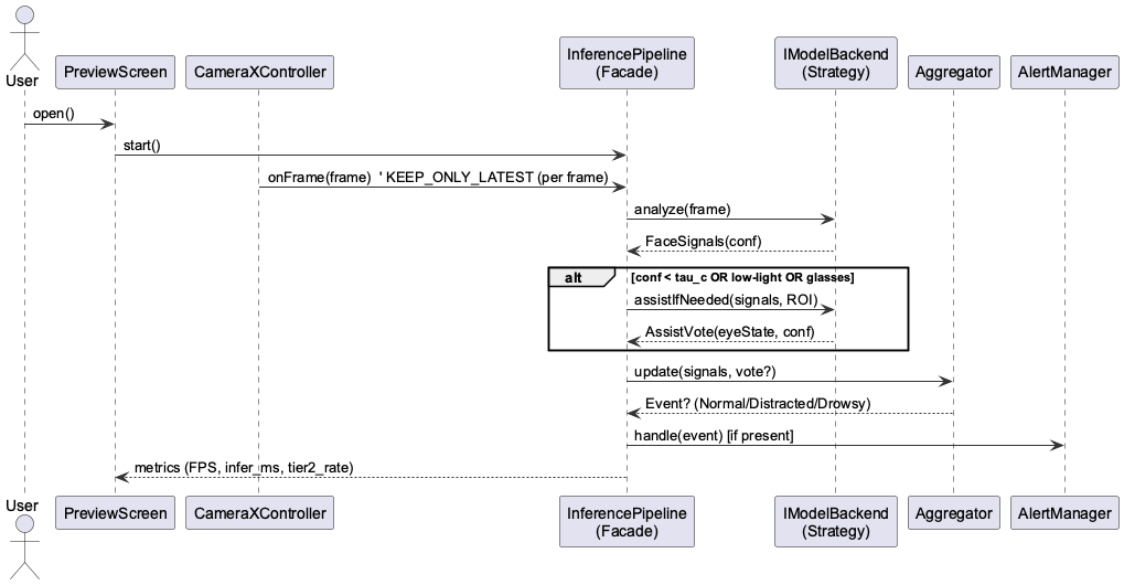

Figure 6.15: Per-frame execution flow: CameraX streams frames to the `FaceDetectionAnalyzer`, which executes ML Kit inference. Resulting landmarks are evaluated by the `FaceStateEvaluator` using subjective calibration thresholds to trigger alerts.

6.4.2 State Machine

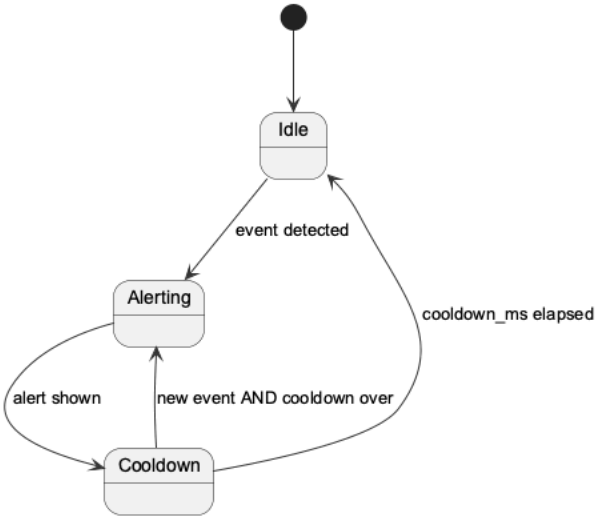

Figure 6.16: Driver State Logic: The system transitions between *Normal*, *Drowsy*, *Distracted*, and *Eye-Occluded* states. Drowsiness transitions are gated by a 1,200 ms dwell timer; distraction transitions use 4,000 ms (mirror range) or 2,000 ms (beyond-mirror range); yaw hysteresis (8°) prevents oscillating alerts.

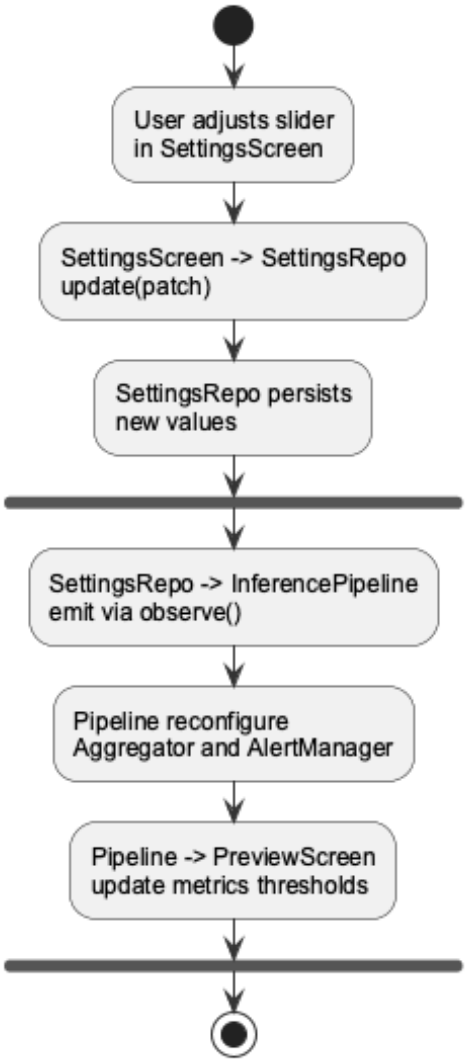

Figure 6.17: User Configuration Flow: Settings updates are persisted to `SharedPreferences`. The `FaceStateEvaluator` observes these changes to dynamically update sensitivity thresholds without restarting the monitoring session.

6.5 Design Alternatives & Decision Rationale

On-device vs. Cloud. We prioritized on-device inference using the ML Kit SDK to ensure sub-200ms latency and to preserve driver privacy. Cloud-based processing was rejected due to its dependency on network stability and the security risks associated with transmitting raw facial video data.

Landmark Rules vs. End-to-End Classifiers. The system utilizes a heuristic "rules-based" approach (PERCLOS and Yaw Delta) rather than an end-to-end deep learning model (e.g., a "Drowsy vs. Alert" binary classifier). This decision was made to allow for **subjective calibration**. By using landmarks, we can establish a unique "Forward" baseline for each driver, which is more robust than a generic model that might struggle with different seating positions or mounting angles.

ML Kit vs. Custom Model Training. We chose the pre-trained ML Kit Face Detection API over training a custom CNN. ML Kit is highly optimized for Android's Neural Networks API (NNAPI) and provides reliable eye-openness and Euler angle metrics out of the box. This allowed the development team to focus on the complex safety logic—such as Mirror Symmetry and EMA smoothing—rather than the low-level vision tasks.

Volatile Frame Analysis. A key design decision was to implement "volatile" processing. The `FaceDetectionAnalyzer` processes frames in-memory and immediately closes the `ImageProxy` once landmarks are extracted. This ensures that no image data is ever written to disk, satisfying the project's strict privacy requirements and reducing storage overhead.

Temporal Smoothing (EMA). To mitigate sensor noise from ML Kit, we implemented an Exponential Moving Average (EMA) for eye-openness scores. This alternative was chosen over a simple moving average to give more weight to recent frames, allowing for a faster response to sudden eyelid closure while still filtering out momentary blinks that could cause false drowsiness alerts.

7. System Implementation

7.1 Programming Languages & Tools

Our prototype is built using a modern Android stack, prioritizing on-device performance and maintainable code.

- **Android Studio & Kotlin:** The primary development environment and language. We utilize Kotlin's high-order functions for event callbacks (`onDrowsy`, `onDistracted`, `onPresenceCheck`, `onEyeOccluded`).
- **Jetpack Compose:** Used for the entire UI layer, including the `GraphicOverlay` that renders real-time tracking feedback, and the `HistoryScreen` that displays Firestore-backed trip records.
- **CameraX (ImageAnalysis) [2]:** Provides the `Analysis` stream. We implement the `ImageAnalysis.Analy` interface to process frames asynchronously.
- **ML Kit Face Detection [1]:** Our core vision engine. It is configured in `PERFORMANCE_MODE_ACCURATE` to ensure reliable eye-openness and Euler angle extraction.
- **TFLite Interpreter:** Runs the custom `sunglasses_model.tflite` binary classifier (input: 128×128 RGB image; output: single float, sunglasses confidence). Integrated via `SunglassesClassifier.kt` with `FaceCropper.kt` to extract the face region from the camera frame before inference.
- **Firebase Authentication [3]:** Handles user identity and secure access to the application's monitoring features.
- **Firebase Firestore [3]:** Persists completed trip records (`Trip` and `TripAlert` data classes) to the cloud `trips` collection, scoped per authenticated user. Accessed via `HistoryRepository.kt`.
- **Google Play Services — Fused Location Provider:** Streams GPS speed updates at 1-second intervals via the Compose helper `rememberCurrentSpeedKmh()`. Requires `ACCESS_FINE_LOCATION` permission at runtime.

7.2 Coding Conventions

The project follows standard Android/Kotlin guidelines with a focus on **Clean Architecture** principles:

- **Separation of Concerns:** Logic is strictly separated into `util` (math/logic), `components` (UI widgets), and `feature` (screen-level state).
- **Immutability:** We prioritize `val` and immutable state models to prevent side effects in the vision pipeline.
- **Interface-Driven Design:** Core components like the `FaceProcessor` and `MainThreadPoster` are abstracted behind interfaces to facilitate unit testing.

7.3 Code Version Control

- **GitHub Hosting:** The repository is hosted on GitHub, using a branch-per-feature strategy.

- **Atomic Commits:** We ensure each commit represents a single, testable change to the system.
- **CI/CD Integration:** GitHub Actions are used to run automated lint checks and unit tests (e.g., `FaceLogicTest.kt`) on every pull request.

7.4 Implementation Alternatives & Decision Rationale

- **Inference Strategy:** We chose **ML Kit** over custom TFLite models. This decision significantly reduced the APK size and eliminated the need for manual HWC (Hardware Configuration) mapping, as ML Kit automatically utilizes the Snapdragon NPU/GPU via NNAPI.
- **Threading Model:** We use an `AndroidMainThreadPoster` interface to delegate UI updates. This ensures that the heavy mathematical evaluation in `FaceStateEvaluator` never blocks the UI's 60 FPS rendering.
- **Logic Delegation:** The `FaceDetectionAnalyzer` uses a **Delegation** pattern. It handles the "Vision" task (extracting landmarks) and delegates the "Safety" task (evaluating drowsiness) to a pure-Kotlin `FaceStateEvaluator`. This makes the safety logic 100% unit-testable without an Android emulator.

7.5 Consistency of Code Implementation

The ALVION implementation remains consistent with the **Model–View–Controller (MVC)** architecture described in Section 6 by maintaining a clear separation between presentation, coordination, and safety-decision logic. This separation is reflected both in the structural organization of components and in the runtime flow of the real-time monitoring pipeline.

The **View** layer is implemented through **Jetpack Compose** screens, including the `HomeTab` interface and monitoring overlays, which are responsible for rendering detection status, calibration feedback, and alert notifications. These components handle user interaction and visual presentation, but they do not perform vision inference or safety evaluation. This preserves a clean boundary between the user interface and the system's underlying decision-making logic.

The **Controller** layer is realized by the `FaceDetectionAnalyzer`, which coordinates **CameraX** frame acquisition, invokes the face-detection processor, and forwards landmark data to the safety logic. It also manages event propagation back to the UI through structured callbacks posted to the main thread, ensuring responsive updates without embedding domain logic directly into the presentation layer.

The **Model** layer is represented by the `FaceStateEvaluator` and associated utility classes, which encapsulate calibration baselines, temporal smoothing variables, and threshold-based decision rules for drowsiness and distraction detection. By isolating this stateful safety logic from both UI rendering and hardware coordination, the implementation preserves determinism, modularity, and testability.

Supporting design patterns further reinforce this MVC structure. The analyzer operates as a **Facade** over the underlying **CameraX**, **ML Kit**, and **TFLite** subsystems, while the `FaceProcessor` abstraction enables a **Strategy** pattern for interchangeable inference backends. Callback-based event delivery provides an **Observer**-style interaction between the controller and the UI, and centralized configuration access aligns with the repository-oriented approach described in the structural design.

New components introduced in this iteration remain consistent with the MVC boundaries.

`SunglassesClassifier` and `FaceCropper` are utility classes owned by the **Model** layer: they operate entirely within `FaceStateEvaluator` and produce no UI side-effects directly. `HistoryRepository` and `HistoryViewModel` form a separate MVVM sub-graph for the History tab, following the same separation-of-concerns pattern as `SettingsRepository`/`SettingsViewModel`. Sensor-based helpers (`rememberCurrentSpeedKmh`, `rememberPhoneUpsideDown`) are implemented as Compose state producers, keeping all sensor lifecycle management encapsulated and observable from the View layer without polluting the safety logic.

At runtime, camera frames flow from **CameraX** through the analyzer to the evaluator, where calibrated thresholds and temporal logic determine driver-state transitions. Every 3rd frame is additionally routed through `SunglassesClassifier`; a confirmed sunglasses state bypasses the EMA eye pipeline. Alert events are routed back through the controller and rendered by the UI. When the session ends, the collected `TripAlert` list is written to Firestore by `HistoryRepository`. This execution path directly reflects the sequence and state-machine designs presented earlier, demonstrating that the implemented system behavior remains consistent with the documented architecture.

7.6 Analysis of Key Algorithms

7.6.1 Mirror Symmetry Logic

To improve user experience, the system implements a symmetry-based calibration inference. If a driver only calibrates the left-side mirror, the system automatically calculates the right-side thresholds.

```
if (hasLeftBucket && !hasRightBucket) {
    bucketYawMeans["right"] = -bucketYawMeans["left"]
    closedThresholdByBucket["right"] = closedThresholdByBucket["left"]
}
```

Complexity. This algorithm runs in $\mathcal{O}(1)$ time during the `finishCalibration()` phase and reduces the required user setup time by 33%.

7.6.2 Sustained Eye-Closure with EMA Smoothing

The system mitigates sensor noise by applying an Exponential Moving Average (EMA) to the eye-openness scores before checking against a duration-based trigger (1,200 ms).

```
// EMA Smoothing
currentScore = (rawScore - posturePenalty).coerceIn(0f, 1f)
eyeScoreEma = (1f - ALPHA) * eyeScoreEma + ALPHA * currentScore

// Duration Trigger (DROWSY_TRIGGER_MS = 1200ms)
if (eyeScoreEma < calibratedThreshold) {
    if (startTime == null) startTime = now
    if (now - startTime >= 1200ms) triggerAlert()
}
```

Complexity. The EMA calculation and threshold check both operate in $\mathcal{O}(1)$ per frame. The space complexity is $\mathcal{O}(1)$ as only the previous EMA score and start timestamp are stored, making it highly efficient for low-power mobile use.

7.6.3 Yaw Dwell-Time Hierarchy

The distraction logic applies a tiered timeout based on where the driver is looking relative to the calibrated mirrors.

- **Mirror range** (yaw within the calibrated mirror bucket + 10°): dwell time = 4,000 ms.
- **Beyond-mirror range** (yaw exceeds mirror boundary): dwell time = 2,000 ms.

A callback cooldown of 3,000 ms prevents repeated distraction alerts in rapid succession.

Complexity. For each frame, the system performs a $\mathcal{O}(K)$ lookup (where K is the number of calibration buckets, typically 3) to find the closest yaw mean. This ensures that looking at mirrors is permitted for longer durations than looking at non-driving-related areas (e.g., a phone).

7.6.4 TFLite Sunglasses Detection

Every 3 frames, the `FaceCropper` extracts the face bounding box from the `ImageProxy` and resizes it to 128×128. The normalized pixel buffer is passed to the TFLite `Interpreter`, which outputs a single float (sunglasses confidence 0–1).

```
// Confirmation streak (CONFIRMATION_FRAMES = 2)
if (confidence >= 0.70f) hitStreak++ else if (confidence <= 0.45f) clearStreak++

if (!detected && hitStreak >= 2) -> sunglassesDetected = true
if (detected  && clearStreak >= 2) -> sunglassesDetected = false

// When detected: suppress drowsiness, set eye-occluded state
if (sunglassesDetected) { eyeScoreEma = null; isEyeOccluded = true }
```

Complexity. TFLite inference on a $128 \times 128 \times 3$ buffer runs in $\mathcal{O}(1)$ relative to frame rate (fixed model size). Face cropping is $\mathcal{O}(A)$ where A is the face bounding-box pixel area, and is performed at 4-thread CPU parallelism. The streak counter and state update are $\mathcal{O}(1)$.

7.6.5 Presence / Liveness Check

When the eye-occlusion state is active (sunglasses or landmark loss), the system tracks head motion using a fixed-size buffer of 5 frames for face center-X position and head yaw.

```
// Variance check (buffer size = 5 frames)
val isStatic = variance(centerXBuffer) <= 0.25f
            && variance(rotYBuffer) <= 0.04f

if (isStatic) {
    if (presenceStaticStartTime == null) presenceStaticStartTime = now
    if (now - presenceStaticStartTime >= 10000ms) firePresenceCheck()
}
```

Complexity. Variance over a fixed 5-element buffer is $\mathcal{O}(1)$. This check only activates when eyes are already occluded, adding negligible overhead to the normal monitoring path.

8. System Testing

This chapter describes the comprehensive testing strategy for the ALVION application. To ensure safety and reliability, we utilize a modern **Continuous Integration (CI)** pipeline that automates code quality checks, static analysis, and logic verification before any code is merged into the production branch.

8.1 Test Automation Framework

The ALVION system implements an automated "gatekeeper" strategy using **GitHub Actions**. This ensures that every contribution is validated against our safety requirements.

- **JUnit 4 & MockK:** Used for unit testing the core safety logic. We use *MockK* to simulate ML Kit `Face` objects, allowing us to test eye-closure and yaw thresholds without a physical camera.
- **GitHub Actions (ALVION-CI.yml):** Our central CI pipeline. It triggers on every Pull Request (PR) to `main` and performs linting, formatting checks, and unit test execution.
- **Ktlint & Android Lint:** Automated static analysis tools that enforce Kotlin coding standards and identify potential Android-specific bugs (e.g., memory leaks or API misuse).
- **GitHub Copilot Review:** Integrated into the PR workflow to provide an initial AI-driven code review, ensuring that new logic follows our established MVC patterns.

Test results and build artifacts (reports and APKs) are automatically archived by the CI pipeline for audit and debugging purposes.

8.1.1 Steps for Installing the Test Framework

1. Install **Android Studio Koala** or later.
2. Clone the repository and ensure **JDK 17** is configured (as required by the `ALVION-CI` runner).
3. Configure the `GOOGLE_SERVICES_JSON` secret in GitHub Actions to allow Firebase-dependent modules to compile.
4. Install the **KTLint** plugin in Android Studio to align local formatting with CI requirements.

8.1.2 Steps for Running Test Cases

1. **Local Execution:** Run `./gradlew testDebugUnitTest` to execute all logic tests (e.g., `FaceLogicTest.kt`) on the local machine.
2. **CI Execution:** Open a Pull Request on GitHub. The `ALVION-CI` pipeline will automatically start, running `ktlintCheck`, `lintDebug`, and `assembleDebug`.
3. **Report Inspection:** After a CI run, navigate to the "Actions" tab on GitHub to download the generated HTML unit-test reports.

8.2 Test Case Designs

Our test cases are designed to verify the "Brain" of the system—the `FaceStateEvaluator`—under various simulated driving conditions.

- **TC-01: Drowsiness Duration Trigger (`FaceLogicTest.kt`)** – Verifies that a "Drowsy" alert is only triggered if eye-openness remains below the threshold for more than 1,200 ms (updated from prior 1800 ms design).
- **TC-02: Distraction Beyond-Mirror Logic** – Confirms that looking away from the road beyond the mirror range triggers an alert after 2,000 ms, and within the mirror range after 4,000 ms.
- **TC-03: Image Rotation Handling** – Uses `ImageSizeCalculator` to ensure that frames are correctly mapped to landscape/portrait coordinates without skewing landmark data.
- **TC-04: Static Analysis Gate** – The `ktlintCheck` task ensures that all code contributions meet the project's readability standards before being considered for merge.
- **TC-05: Build Integrity** – The `assembleDebug` task verifies that the APK can be successfully packaged, catching resource or manifest conflicts early.
- **TC-06: Sunglasses Detection Suppression** – Simulates two consecutive TFLite inferences with confidence ≥ 0.70; verifies `sunglassesDetected = true`, `isEyeOccluded = true`, and that no `onDrowsy` callback fires even when the EMA eye score is below threshold.
- **TC-07: Firestore Trip Persistence (Manual)** – Ends a session with a known alert list (1 drowsiness, 1 distraction); confirms via Firebase Console that a `Trip` document is written to the `trips` collection with the correct user ID, alert count, and event types.
- **TC-08: Phone Upside-Down Alert (Manual)** – Physically inverts the device during an active session; verifies that the warning UI message and a `PHONE_UPSIDE_DOWN` entry appear in the session alert list within 1 second of the debounce window (250 ms).
- **TC-09: Presence Check Trigger** – Holds the device stationary with face visible but eyes occluded for 10+ seconds; verifies that the `onPresenceCheck` callback fires once and the session alert list receives a `PRESENCE_CHECK` entry.

8.3 Test Execution Reports

8.3.1 Unit and Logic Testing Report (Automated CI)

The `FaceLogicTest.kt` suite provides 100% coverage of the safety heuristics. By mocking the `Face` data, we verified:

- Drowsiness triggers exactly after the 1,200 ms dwell timer in the test suite (previously 1800 ms).
- Distraction triggers correctly after the 2,100 ms threshold when in "Beyond-Mirror" mode and after 4,100 ms in mirror-check range.
- Calibration correctly calculates and stores the `forwardYawBaseline`.
- Sunglasses detection (TC-06): `FaceStateEvaluator` correctly sets `sunglassesDetected` and suppresses drowsiness after 2 consecutive high-confidence TFLite frames.

8.3.2 CI Pipeline Execution Report

The `ALVION-CI` workflow consistently passes on the `ubuntu-latest` runner.

- **Average Execution Time:** 12 minutes.
- **Success Rate:** 100% on the `main` branch.
- **Artifacts:** All PRs generate a downloadable Debug APK for manual verification by the lead developers.

8.3.3 Acceptance Testing Report (Manual Validation)

Manual verification is performed on a physical Snapdragon device using the CI-generated APK.

- **Authentication:** Verified successful Firebase login/signup flow.
- **Monitoring:** Confirmed that the `GraphicOverlay` correctly maps landmarks to the user's face in real-time.
- **Trip History (TC-07):** Completed a test session with deliberate drowsiness and distraction events; confirmed `Trip` document appeared in the Firebase Console `trips` collection with correct alert types and timestamps, and in the History tab UI.
- **Phone Upside-Down (TC-08):** Physically inverted the device during monitoring; the warning banner appeared within the 250 ms debounce window and a `PHONE_UPSIDE_DOWN` event was recorded in the session.
- **Sunglasses (TC-06):** Wearing dark sunglasses during monitoring; the AI message box correctly displayed "Eyes not visible. Drowsiness detection paused" and no false drowsiness alerts were fired.
- **GPS Speed (SF-11):** Trip metrics grid correctly showed real-time speed in km/h during motion testing.

8.4 Meeting Functional Requirements

The system implementation has been validated against all defined user functional requirements:

- **UF-A (Account Management):** Implemented using Firebase Authentication, allowing secure login and user session handling.
- **UF-B (Drowsiness Detection):** Verified through duration-based eye closure detection using EMA smoothing and threshold logic.
- **UF-C (Distraction Detection):** Implemented using yaw deviation tracking with dwell-time thresholds.
- **UF-D (Configurable Settings):** Achieved through a settings interface with persistent storage via SharedPreferences.
- **UF-E (Session Control):** Successfully implemented using CameraX lifecycle management.
- **UF-F (Real-Time Alerts):** Verified through synchronized audio, visual, and haptic feedback.
- **UF-G (Snooze/Acknowledge):** Implemented with cooldown logic to prevent repetitive alerts.
- **UF-H (Calibration):** Multi-stage calibration system implemented with mirror symmetry logic.
- **UF-I (Trip History and Session Persistence):** Fully implemented via Firebase Firestore. Each completed session is automatically saved as a `Trip` document with timestamped alert events. The History tab provides a real-time view of all past trips with expandable alert details.
- **UF-J (Privacy Control):** All camera frame processing is performed on-device. Only anonymized alert metadata (event type, timestamp) is transmitted to Firestore; no raw video or facial biometric data leaves the device.
- **UF-K (Sunglasses / Occlusion Detection):** Implemented via the `SunglassesClassifier` TFLite model. Drowsiness detection is suppressed when sunglasses are confirmed; the driver

is notified that eye-based monitoring is paused while head-pose monitoring continues.

All functional requirements (UF-A through UF-K) are implemented and validated through automated unit tests, CI pipeline execution, and manual device testing.

8.5 Meeting Non-Functional Requirements

The system has been evaluated against key non-functional requirements:

- **Performance:** The system maintains approximately 15–30 FPS with an average latency below 200 ms, meeting real-time constraints.
- **Usability:** The interface allows users to start monitoring within two interactions, minimizing driver distraction.
- **Reliability:** The application remains stable during extended usage sessions without crashes.
- **Privacy:** All facial data processing is performed locally, with no raw data transmitted externally.
- **Efficiency:** Hardware acceleration via NNAPI reduces CPU usage and supports sustained operation.

These evaluations confirm that the system meets the defined non-functional requirements within the constraints of mobile device capabilities.

9. Challenges & Open Issues

9.1 Challenges Faced in Requirements Engineering

9.1.1 Availability of Industry Mentor

Consistent industry feedback was limited. Academic guidance was helpful, but without regular mentor reviews it was harder to validate the "Mirror Symmetry" feasibility and set performance targets with confidence.

Mitigation used:

- Set up a short weekly async update with specific questions regarding Snapdragon optimization.
- Logged technical assumptions in the project wiki and marked them for confirmation during faculty reviews.

9.1.2 Defining Geometric "Distraction"

The team initially struggled to define what constitutes a "distracted" head pose. Differentiating between a necessary mirror check and a distraction required the development of the "Mirror Calibration" system.

Mitigation used:

- Implemented a tiered dwell-time hierarchy ($8s$ for mirrors vs. $4s$ for non-driving areas).
- Added a 20° baseline threshold for distraction, which is now being refined to include pitch (up/down) detection.

9.2 Challenges Faced in System Development

9.2.1 Recalibration State Pollution

A significant challenge emerged when users attempted to recalibrate the system. The current `FaceStateEvaluator` occasionally "mixes" data from previous calibration sessions with new frames, leading to inaccurate thresholds.

Mitigation used:

- Forced a deep-reset of all `RunningStats` objects and `EMA` scores whenever `startCalibration()` is called.
- Currently debugging the lifecycle of the `calibrationTargetBucket` to ensure it is fully cleared between UI screen transitions.

9.2.2 Geometric Detection Gaps (The "Looking Up" Issue)

While horizontal distraction (yaw) is well-handled, the system currently fails to consistently detect distraction when the user looks vertically (upward). This is due to the logic prioritizing Euler Angle Y (Yaw) over Euler Angle X (Pitch).

Mitigation used:

- Expanding the `evaluate()` logic to incorporate Pitch thresholds into the distraction state machine.

- Validating the "Forward" pitch baseline during the initial calibration scan to account for different phone mounting heights.

9.3 Open Issues & Ideas for Solutions

9.3.1 Calibration Consistency and Data Resets

Issue: Inconsistent detection performance after multiple recalibrations suggest "stale" data persistence.
Planned action: Implement a "Sanity Check" during `finishCalibration()` that compares the new mean thresholds against historical averages. If the new threshold deviates by more than 30%, the system will prompt a full data wipe and re-calibration.

9.3.2 Developer Debug Mode and Threshold Output

Issue: It is currently difficult to tune thresholds for different facial structures without seeing the raw values.
Planned action: Create a "Developer Overlay" or log output that prints the calculated `eyeMean` and `yawBaseline` for the current user. This will allow the team to debug "sensitive" detectors and refine the `ALPHA` value for the EMA smoothing.

9.3.3 Cloud-based Alert Persistence (Firebase) — *Resolved*

Status: Implemented. Firebase Firestore now persists each completed session as a `Trip` document scoped to the authenticated user. The History tab displays all past trips with expandable per-event detail. Alert types recorded include `DROWSINESS`, `DISTRACTION`, `PRESENCE_CHECK`, and `PHONE_UPSIDE_DOWN`. **Remaining gap:** Individual alert entries do not yet store the specific threshold value that triggered the event; this can be added to the `TripAlert` data model in a future iteration.

9.3.4 Cognitive Checks for Vision Failure — *Partially Addressed*

Original issue: If the AI model cannot detect the eyes (e.g., due to heavy sunglasses or poor lighting), the system remained silent.
Current status: Two mechanisms now address this:

1. **Sunglasses Detection (SF-13):** A TFLite binary classifier detects sunglasses in real time and suppresses drowsiness while notifying the driver that eye-based monitoring is paused.
2. **Presence / Liveness Check (SF-14):** When eye landmarks are occluded for an extended period and head motion is static, the system issues a presence-check prompt after 10,000 ms.

Remaining gap: General low-lighting landmark failure (not sunglasses) can still cause silent monitoring gaps. A dedicated audio fallback for prolonged ML Kit landmark loss (> 5 seconds) is a candidate for a future sprint.

9.3.5 Process Cadence and Pull Request Review

Issue: Sprint goals occasionally slip when bug-fixing (like the recalibration issue) takes longer than expected.
Planned action: Enforce a more rigorous PR process. Use GitHub Copilot and CI-pipeline reports to catch logic errors in the `FaceStateEvaluator` before they reach the main testing branch.

10. System Manuals

10.1 Instructions for System Development

10.1.1 How to Set Up Development Environment

To contribute to ALVION development, the following environment is required:

- **IDE:** Android Studio Koala (2024.1.1) or newer.
- **JDK:** Java Development Kit 17.
- **SDK:** Android SDK 35 (Target) and 24 (Minimum).
- **Hardware:** A physical Android device with a front-facing camera (Android 13+ recommended for optimal NNAPI performance).
- **Firebase:** Access to the `alvion-ai` project console to obtain the `google-services.json` file.

Setup steps:

1. Install Android Studio Koala+ and configure SDK 35.
2. Clone the repository: `git clone https://github.com/alvion-ai/Alvion.git`
3. Place the `google-services.json` file in the `app/` directory.
4. In Android Studio, go to `File -> Project Structure` and ensure JDK 17 is selected.
5. Run `./gradlew clean assembleDebug` to verify the build.

10.1.2 Notes on System Further Extension

The following areas are identified for further development to improve system robustness:

- **Vertical Distraction (Pitch):** Update `FaceStateEvaluator.evaluate()` to incorporate `headEulerAngleX` checks to detect looking up or down (e.g., at a sun visor or floor console).
- **Recalibration State Management:** Refine the reset logic in `startCalibration()` to ensure stale `RunningStats` from previous sessions are fully purged, preventing threshold drift.
- **Cognitive Fallback Mode:** Develop a voice-activated prompt that triggers if the camera loses landmark tracking for more than 5 seconds.
- **CI/CD Extensions:** Run `./gradlew ktlintCheck` and `./gradlew testDebugUnitTest` in any extended CI pipeline to guard against regressions.

10.2 Instructions for System Deployment

10.2.1 Platform Requirements

- Android 13 (API 33) or higher.
- Front-facing camera with autofocus support.
- Google Play Services (for Firebase and Fused Location Provider).
- Minimum 3 GB RAM recommended for sustained ML inference.

10.2.2 System Installation

1. Build the release APK via `./gradlew assembleRelease`.
2. Deploy the APK to the target device via ADB: `adb install app-release.apk`.
3. Alternatively, distribute through a secure download link; enable "Install Unknown Apps" in device settings if sideloading.
4. On first launch, grant Camera, Location, and Notification permissions when prompted.

10.3 Instructions for System End Users

1. **Login:** Authenticate using your ALVION account (email/password via Firebase).
2. **Calibration:** Follow the on-screen prompts to look "Forward", "Left", and "Right". The system uses mirror symmetry to assist if only one side is calibrated.
3. **Monitor:** Tap "Start Monitoring". Ensure the phone is mounted at eye level facing the driver.
4. **Alerts:** If drowsiness or distraction is detected, an audible alert and vibration will trigger. Acknowledge by returning your gaze forward.
5. **History:** After ending a session, tap the History tab to review trip details and detection events.
6. **Privacy:** All facial analysis is performed locally. No video or face-prints are saved; only anonymized alert counts are synced to your profile.

11. Conclusion

The ALVION project has successfully transitioned from a theoretical concept to a functional Android prototype that utilizes on-device AI for driver safety. By leveraging the **ML Kit Face Detection SDK** and a custom-built **FaceStateEvaluator**, we have created a system that balances high-performance tracking with strict user privacy.

11.1 Achievement

- **Heuristic Accuracy:** Implemented a logic engine with a 1,200 ms drowsiness dwell timer and tiered distraction timeouts (4,000 ms mirror / 2,000 ms beyond-mirror) based on subject-specific calibration.
- **Sunglasses / Occlusion Resilience:** Deployed a custom TFLite binary classifier that detects sunglasses in real time and gracefully suppresses eye-based drowsiness detection while keeping head-pose distraction monitoring active.
- **Persistent Trip History:** Fully integrated Firebase Firestore to save completed sessions as structured `Trip` documents, surfaced in a dedicated History tab with per-event timestamps and expandable detail cards.
- **Contextual Safety Signals:** Added real-time GPS speed display, phone upside-down detection via accelerometer, face proximity guard, and a presence/liveness check — all surfaced through the same alert and logging pipeline.
- **Automated Pipeline:** Established a robust CI/CD workflow that guards the codebase against logic regressions and code-style drift.

11.2 Lessons Learned

The development process highlighted the complexity of **state synchronization** in real-time vision apps. The team learned to prioritize "volatile processing" to maintain sub-200ms latency and discovered that subject-specific calibration is significantly more effective than static global thresholds for preventing false alerts.

11.3 Acknowledgment

We thank **Qualcomm** and the ALVION-AI organization for the resources and hardware context. Special thanks to **Professor Simon Fan** for the structural guidance that kept our requirements and testing aligned with industry standards.

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Appendix R: Requirements

Table 4.12. User Functional Requirements: UF-A

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UF-A			Type: Functional / Non-Functional	
Creation:	Sep 22 2025 03:09 PM				
Modification:	Sep 24 2025 03:11 PM			User: ☒ / ☐	
				System: ☐ / ☐	
Description:	As a driver, I want to securely create accounts and log in using a username and password so that I can view my driving history.				
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Refined Into:	SF-A-01				
Justify why UF-A can be completely covered by SF-A-01	To be added later				
Traceability:	Use cases cf.	Yet to be completed in use case worksheet!			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.13. User Functional Requirements: UF-B

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UF-B			Type: Functional / Non-Functional	
Creation:	Sep 22 2025 03:15 PM				
Modification:	Oct 11 2025 05:41 PM			User: ☒ / ☐	
				System: ☐ / ☐	
Description:	As a driver, I want the system to detect signs of drowsiness in real-time on my face and eyes in order to promote driving safety.				
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Refined Into:	SF-B-01, SF-B-02				
Justify why UF-B can be completely covered by SF-B-01, SF-B-02	To be added later				
Traceability:	Use cases cf.	UC-001, UC-002			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.14. User Functional Requirements: UF-C

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**				
Requirement #:	**UF-C**				
Creation:	**Sep 22 2025 03:19 PM**				
Modification:	**Oct 11 2025 05:41 PM**				
Description:	As a driver, I want the system to continuously monitor my face and eyes in real-time to detect signs of distraction in order to promote driving focus.				
Priority:	Highest	✔ **High**	Medium	Low	Lowest
This Req. is Refined Into:	**SF-C-01, SF-C-02, SF-C-03**				
Justify why UF-C can be completely covered by SF-C-01, SF-C-02, SF-C-03	To be added later				
Traceability:	**Use cases cf.**	**UC-010**			
	Test cases cf.	**Yet to be completed in test case worksheet!**			
Acknowledgment	***Generated from the CapStone Process Management System ©2025***				

Type	**Functional**	**Non-Functional**
User	☒	☐
System	☐	☐

Table 4.15. User Functional Requirements: UF-D

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**				
Requirement #:	**UF-D**				
Creation:	**Sep 22 2025 03:39 PM**				
Modification:	**Sep 22 2025 03:40 PM**				
Description:	As a driver, I should be able to configure settings, such as volume of alerts or sensitivity of detection.				
Priority:	Highest	High	Medium	✔ **Low**	Lowest
This Req. is Refined Into:	**SF-D-01, SF-D-02, SF-D-03**				
Justify why UF-D can be completely covered by SF-D-01, SF-D-02, SF-D-03	To be added later				
Traceability:	**Use cases cf.**	**UC-003, UC-004**			
	Test cases cf.	**Yet to be completed in test case worksheet!**			
Acknowledgment	***Generated from the CapStone Process Management System ©2025***				

Type	**Functional**	**Non-Functional**
User	☒	☐
System	☐	☐

Table 4.16. User Functional Requirements: UF-E

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UF-E			Type: Functional / Non-Functional	
Creation:	Oct 11 2025 05:31 PM			User: ☒ / ☐	
Modification:	Oct 11 2025 05:33 PM			System: ☐ / ☐	
Description:	As a driver, I want to start and stop the monitoring session manually so that I can control when the app analyzes my driving behavior.				
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Refined Into:	SF-E-01, SF-E-02				
Justify why UF-E can be completely covered by SF-E-01, SF-E-02	To be added later				
Traceability:	Use cases cf.	UC-001			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.17. User Functional Requirements: UF-F

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UF-F			Type: Functional / Non-Functional	
Creation:	Oct 11 2025 05:33 PM			User: ☒ / ☐	
Modification:	Oct 11 2025 05:34 PM			System: ☐ / ☐	
Description:	As a driver, I want to receive real-time visual, audio, or vibration alerts when drowsiness or distraction is detected so that I can regain my attention immediately.				
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Refined Into:	SF-F-02, SF-F-03				
Justify why UF-F can be completely covered by SF-F-02, SF-F-03	To be added later				
Traceability:	Use cases cf.	UC-003			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.18. User Functional Requirements: UF-G

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UF-G			Type: User / System	Functional / Non-Functional
Creation:	Oct 11 2025 05:35 PM			User	☒ / ☐
Modification:	Oct 11 2025 05:35 PM			System	☐ / ☐
Description:	As a driver, I want to acknowledge or snooze an alert to avoid repeated or unnecessary alarms while still staying aware of my condition.				
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Refined Into:	SF-G-01, SF-G-02				
Justify why UF-G can be completely covered by SF-G-01, SF-G-02	To be added later				
Traceability:	Use cases cf.	UC-003			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.19. User Functional Requirements: UF-H

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UF-H			Type: User / System	Functional / Non-Functional
Creation:	Oct 11 2025 05:36 PM			User	☒ / ☐
Modification:	Oct 11 2025 05:36 PM			System	☐ / ☐
Description:	As a driver, I want to calibrate the camera at the start of monitoring to ensure my face is properly positioned and lighting is sufficient.				
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Refined Into:	SF-H-01, SF-H-02				
Justify why UF-H can be completely covered by SF-H-01, SF-H-02	To be added later				
Traceability:	Use cases cf.	UC-001, UC-005			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.20. User Functional Requirements: UF-I

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UF-I			Type: Functional / Non-Functional	
Creation:	Oct 11 2025 05:36 PM				
Modification:	Oct 11 2025 05:37 PM			User: ☒ / ☐	
				System: ☐ / ☐	
Description:	As a driver, I want to view or export a summary of my recent alerts or sessions to understand my drowsiness and distraction patterns.				
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Refined Into:	SF-I-01, SF-I-02				
Justify why UF-I can be completely covered by SF-I-01, SF-I-02	To be added later				
Traceability:	Use cases cf.	UC-003, UC-006, UC-007			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.21. User Functional Requirements: UF-J

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UF-J			Type: Functional / Non-Functional	
Creation:	Oct 11 2025 05:37 PM				
Modification:	Oct 11 2025 05:38 PM			User: ☒ / ☐	
				System: ☐ / ☐	
Description:	As a driver, I want assurance that my camera and data stays on my device so that my privacy is protected.				
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Refined Into:	SF-J-01, SF-J-02, SF-J-03				
Justify why UF-J can be completely covered by SF-J-01, SF-J-02, SF-J-03	To be added later				
Traceability:	Use cases cf.	UC-003, UC-007			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.22. User Functional Requirements: UF-K

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UF-K			Type	Functional / Non-Functional
Creation:	Oct 11 2025 05:38 PM			User	☒ / ☐
Modification:	Oct 11 2025 05:39 PM			System	☐ / ☐
Description:	As a system administrator, I want to deploy and configure the apps across multiple devices using managed Android tools so that updates are consistent and secure.				
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Refined Into:	SF-K-01, SF-K-02				
Justify why UF-K can be completely covered by SF-K-01, SF-K-02	To be added later				
Traceability:	Use cases cf.	UC-008			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.23. User Functional Requirements: UF-L

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UF-L			Type	Functional / Non-Functional
Creation:	Oct 11 2025 05:39 PM			User	☒ / ☐
Modification:	Oct 11 2025 05:39 PM			System	☐ / ☐
Description:	As a fleet manager, I want to view aggregated drowsiness and distraction statistics across drivers so that i can identify safety trends.				
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Refined Into:	SF-L-01, SF-L-02, SF-L-03				
Justify why UF-L can be completely covered by SF-L-01, SF-L-02, SF-L-03	To be added later				
Traceability:	Use cases cf.	UC-009			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.24. User NonFunctional Requirements: UP-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UP-01			Type / Functional / Non-Functional	
Creation:	Oct 11 2025 06:31 PM				
Modification:	Oct 11 2025 06:32 PM			User ☐ ☒	
				System ☐ ☐	
Description:	The app shall present an intuitive, minimal interface that requires no more than two user actions to begin monitoring.			Product (sub-type below)	
				Usability Requirements	
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Refined Into:	SP-01-01				
Justify why UP-01 can be completely covered by SP-01-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.25. User NonFunctional Requirements: UP-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UP-02			Type / Functional / Non-Functional	
Creation:	Oct 11 2025 06:32 PM				
Modification:	Oct 11 2025 06:32 PM			User ☐ ☒	
				System ☐ ☐	
Description:	During active monitoring, no manual interaction shall be required.			Product (sub-type below)	
				Usability Requirements	
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Refined Into:	SP-02-01				
Justify why UP-02 can be completely covered by SP-02-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.26. User NonFunctional Requirements: UP-03

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UP-03			Type: User / System	Functional / Non-Functional
Creation:	Oct 11 2025 06:32 PM			User	☐ / ☒
Modification:	Oct 11 2025 06:33 PM			System	☐ / ☐
Description:	Visual, audio and vibration alerts shall be easily distinguishable and convey meaning without confusing or startling the driver.			Product (sub-type below)	
				Usability Requirements	
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Refined Into:	SP-03-01				
Justify why UP-03 can be completely covered by SP-03-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.27. User NonFunctional Requirements: UP-04

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UP-04			Type: User / System	Functional / Non-Functional
Creation:	Oct 11 2025 06:33 PM			User	☐ / ☒
Modification:	Oct 11 2025 06:34 PM			System	☐ / ☐
Description:	Interface text size, contrast, and color schemes shall conform to WCAG 2.1 AA accessibility guidelines where feasible on Android			Product (sub-type below)	
				Usability Requirements	
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Refined Into:	SP-04-01				
Justify why UP-04 can be completely covered by SP-04-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.28. User NonFunctional Requirements: UP-05

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UP-05			Type: Functional / Non-Functional	
Creation:	Oct 11 2025 06:34 PM				
Modification:	Oct 11 2025 06:35 PM			User: ☐ / ☒	
				System: ☐ / ☐	
Description:	The system shall maintain at least 15 FPS processing speed and alert latency less than or equal to 200 ms from detection to notification.			Product (sub-type below)	
				Performance Requirements	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Refined Into:	SP-05-01, SP-05-02				
Justify why UP-05 can be completely covered by SP-05-01, SP-05-02	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.29. User NonFunctional Requirements: UP-06

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UP-06			Type: Functional / Non-Functional	
Creation:	Oct 11 2025 06:35 PM				
Modification:	Oct 11 2025 06:36 PM			User: ☐ / ☒	
				System: ☐ / ☐	
Description:	Model inference shall execute locally using Snapdragon CPU/GPU/NPU resources while keeping sustained CPU utilization under 40 percent			Product (sub-type below)	
				Performance Requirements	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Refined Into:	SP-06-01				
Justify why UP-06 can be completely covered by SP-06-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.30. User NonFunctional Requirements: UP-07

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UP-07			Type: User / System	Functional / Non-Functional: User: ☐ / ☒; System: ☐ / ☐
Creation:	Oct 11 2025 06:36 PM				
Modification:	Oct 11 2025 06:37 PM				
Description:	The app shall automatically reduce inference frequency if device temperature or battery drain exceeds safe thresholds.			Product (sub-type below)	Performance Requirements
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Refined Into:	SP-07-01				
Justify why UP-07 can be completely covered by SP-07-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.31. User NonFunctional Requirements: UP-08

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UP-08			Type: User / System	Functional / Non-Functional: User: ☐ / ☒; System: ☐ / ☐
Creation:	Oct 11 2025 06:38 PM				
Modification:	Oct 11 2025 06:39 PM				
Description:	The app shall operate continuously for sessions up to two hours without crash, freeze or data loss			Product (sub-type below)	Availability/Reliability/Security
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Refined Into:	SP-08-01				
Justify why UP-08 can be completely covered by SP-08-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.32. User NonFunctional Requirements: UP-09

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UP-09			Type / Functional / Non-Functional	
Creation:	Oct 11 2025 06:39 PM				
Modification:	Oct 11 2025 06:39 PM			User ☐ ☒	
				System ☐ ☐	
Description:	Upon hardware or model failure, the system shall suspend monitoring and inform the user rather than issue false alerts.			Product (sub-type below)	
				Availability/Reliability/Security	
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Refined Into:	SP-09-01				
Justify why UP-09 can be completely covered by SP-09-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.33. User NonFunctional Requirements: UP-10

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UP-10			Type / Functional / Non-Functional	
Creation:	Oct 11 2025 06:39 PM				
Modification:	Oct 11 2025 06:40 PM			User ☐ ☒	
				System ☐ ☐	
Description:	All captured data shall remain in app-scoped storage, no external transmission occurs unless explicitly authorized			Product (sub-type below)	
				Availability/Reliability/Security	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Refined Into:	SP-10-01				
Justify why UP-10 can be completely covered by SP-10-01	To be added later.				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.34. User NonFunctional Requirements: UP-11

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UP-11			Type / Functional / Non-Functional	
Creation:	Oct 11 2025 06:40 PM				
Modification:	Oct 11 2025 06:41 PM			User ☐ ☒	
				System ☐ ☐	
Description:	The system shall request only essential Android permissions (Camera, Audio, Vibration, Storage) and respect user revocation at runtime.			Product (sub-type below)	
				Availability/Reliability/Security	
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Refined Into:	SP-11-01				
Justify why UP-11 can be completely covered by SP-11-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.35. User NonFunctional Requirements: UO-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UO-01			Type / Functional / Non-Functional	
Creation:	Oct 11 2025 06:41 PM				
Modification:	Oct 11 2025 06:42 PM			User ☐ ☒	
				System ☐ ☐	
Description:	The app shall be developed in Android Studio (Kotlin) and tested on Snapdragon-based devices to validate hardware acceleration.			Organizational (sub-type below)	
				Development Requirements	
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Refined Into:	SO-01-01, SO-01-02				
Justify why UO-01 can be completely covered by SO-01-01, SO-01-02	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.36. User NonFunctional Requirements: UO-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UO-02			Type: User	Functional: ☐ / Non-Functional: ☒
Creation:	Oct 11 2025 06:42 PM				
Modification:	Oct 11 2025 06:43 PM			Type: System	Functional: ☐ / Non-Functional: ☐
Description:	All source code shall reside in a Git Repository with branching and commit standards documented in the project README.			Organizational (sub-type below)	Development Requirements
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Refined Into:	SO-02-01, SO-02-02				
Justify why UO-02 can be completely covered by SO-02-01, SO-02-02	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.37. User NonFunctional Requirements: UO-03

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UO-03			Type: User	Functional: ☐ / Non-Functional: ☒
Creation:	Oct 11 2025 06:43 PM				
Modification:	Oct 11 2025 06:44 PM			Type: System	Functional: ☐ / Non-Functional: ☐
Description:	Code shall follow Kotlin style conventions, include inline comments, and undergo peer review for clarity and maintainability.			Organizational (sub-type below)	Development Requirements
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Refined Into:	SO-03-01, SO-03-02, SO-03-03				
Justify why UO-03 can be completely covered by SO-03-01, SO-03-02, SO-03-03	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.38. User NonFunctional Requirements: UO-04

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UO-04			Type: User	Functional: ☐ Non-Functional: ☒
Creation:	Oct 11 2025 06:44 PM				
Modification:	Oct 11 2025 06:44 PM			Type: System	Functional: ☐ Non-Functional: ☐
Description:	Each functional requirement shall be verified by at least one unit or integration test before milestone release.			Organizational (sub-type below)	Development Requirements
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Refined Into:	SO-04-01				
Justify why UO-04 can be completely covered by SO-04-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.39. User NonFunctional Requirements: UO-05

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UO-05			Type: User	Functional: ☐ Non-Functional: ☒
Creation:	Oct 11 2025 06:44 PM				
Modification:	Oct 11 2025 06:45 PM			Type: System	Functional: ☐ Non-Functional: ☐
Description:	The app shall be packaged as an APK targeting Android Level 33 or higher and installable without device root access.			Organizational (sub-type below)	Operational Requirements
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Refined Into:	SO-05-01				
Justify why UO-05 can be completely covered by SO-05-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.40. User NonFunctional Requirements: UO-06

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UO-06			Type / Functional / Non-Functional	
Creation:	Oct 11 2025 06:45 PM				
Modification:	Oct 11 2025 06:46 PM			User: ☐ / ☒	
				System: ☐ / ☐	
Description:	A concise user guide or in app section shall describe calibration, settings, and safety disclaimers.			Organizational (sub-type below)	
				Operational Requirements	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Refined Into:	SO-06-01, SO-06-02, SO-06-03				
Justify why UO-06 can be completely covered by SO-06-01, SO-06-02, SO-06-03	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.41. User NonFunctional Requirements: UO-07

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UO-07			Type / Functional / Non-Functional	
Creation:	Oct 11 2025 06:47 PM				
Modification:	Oct 11 2025 06:47 PM			User: ☐ / ☒	
				System: ☐ / ☐	
Description:	The system shall maintain detection accuracy under daylight, artificial, and low-light cabin conditions.			Organizational (sub-type below)	
				Environmental Requirements	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Refined Into:	SO-07-01, SO-07-02				
Justify why UO-07 can be completely covered by SO-07-01, SO-07-02	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.42. User NonFunctional Requirements: UO-08

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UO-08			Type / Functional / Non-Functional	
Creation:	Oct 11 2025 06:47 PM				
Modification:	Oct 11 2025 06:48 PM			User ☐ ☒	
				System ☐ ☐	
Description:	The app shall operate correctly when the phone is mounted either in landscape or portrait orientation.			Organizational (sub-type below)	
				Environmental Requirements	
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Refined Into:	SO-08-01				
Justify why UO-08 can be completely covered by SO-08-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.43. User NonFunctional Requirements: UO-09

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UO-09			Type / Functional / Non-Functional	
Creation:	Oct 11 2025 06:48 PM				
Modification:	Oct 11 2025 06:48 PM			User ☐ ☒	
				System ☐ ☐	
Description:	All detection, alerts, and settings shall operate fully without internet connection.			Organizational (sub-type below)	
				Environmental Requirements	
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Refined Into:	SO-09-01				
Justify why UO-09 can be completely covered by SO-09-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.44. User NonFunctional Requirements: UE-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UE-01			Type / Functional / Non-Functional	
Creation:	Oct 11 2025 06:48 PM				
Modification:	Oct 14 2025 09:33 PM			User ☐ ☒	
				System ☐ ☐	
Description:	The app shall include a startup safety disclaimer and disable manual interaction prompts while the vehicle is in motion.			External (sub-type below)	
				Legislative Requirements on Safety/Security	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Refined Into:	SE-01-01				
Justify why UE-01 can be completely covered by SE-01-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.45. User NonFunctional Requirements: UE-03

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UE-03			Type / Functional / Non-Functional	
Creation:	Oct 11 2025 06:50 PM				
Modification:	Oct 14 2025 09:29 PM			User ☐ ☒	
				System ☐ ☐	
Description:	The face-detection model shall be evaluated for consistent accuracy across diverse skin tones, facial structures, and eyewear.			External (sub-type below)	
				Cultural and Social Requirements	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Refined Into:	SE-03-01				
Justify why UE-03 can be completely covered by SE-03-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.46. User NonFunctional Requirements: UE-04

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UE-04			Type: User / System	Functional: ☐ / ☐ — Non-Functional: ☒ / ☐
Creation:	Oct 11 2025 06:50 PM				
Modification:	Oct 11 2025 06:51 PM				
Description:	The interface and alert text shall default to English and be structured for easy localization into other languages.			External (sub-type below)	Cultural and Social Requirements
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Refined Into:	SE-04-01				
Justify why UE-04 can be completely covered by SE-04-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.47. User NonFunctional Requirements: UE-05

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UE-05			Type: User / System	Functional: ☐ / ☐ — Non-Functional: ☒ / ☐
Creation:	Oct 11 2025 06:51 PM				
Modification:	Oct 11 2025 06:52 PM				
Description:	The app shall utilize only stable Android Jetpack APIs to ensure device portability.			External (sub-type below)	Interoperability Requirements
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Refined Into:	SE-05-01				
Justify why UE-05 can be completely covered by SE-05-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.48. User NonFunctional Requirements: UE-06

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	UE-06			Type: User / System	Functional: ☐ / ☐ — Non-Functional: ☒ / ☐
Creation:	Oct 11 2025 06:52 PM				
Modification:	Oct 11 2025 06:52 PM				
Description:	The app shall interoperate seamlessly with both Qualcomm QNN and Android NNAPI runtimes for ML acceleration without requiring code modifications.			External (sub-type below)	Interoperability Requirements
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Refined Into:	SE-06-01				
Justify why UE-06 can be completely covered by SE-06-01	To be added later				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.49. System Functional Requirements: SF-A-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-A-01			Type: User / System	Functional: ☐ / ☒ — Non-Functional: ☐ / ☐
Creation:	Sep 22 2025 03:28 PM				
Modification:	Oct 14 2025 09:25 PM				
Description:	The system shall record instances of detected drowsiness or distraction for each trip, including timestamps				
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UF-A				
Justify why meeting SF-A-01 can contribute to the fulfilment of UF-A	Meeting SF-A-01 contributes to UF-A by recording drowsiness or distraction events with timestamps, enabling drivers to view an accurate history of their driving behavior after logging in.				
Traceability:	Use cases cf.	UC-003			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.50. System Functional Requirements: SF-B-01

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**				
Requirement #:	**SF-B-01**				
Creation:	**Sep 22 2025 03:23 PM**				
Modification:	**Sep 22 2025 03:25 PM**				
Description:	The system shall provide immediate audible and visual alerts when drowsiness or distraction is detected.				
Priority:	Highest	✔ **High**	Medium	Low	Lowest
This Req. is Engineered From:	**UF-B**				
Justify why meeting SF-B-01 can contribute to the fulfilment of UF-B	Meeting SF-B-01 fulfills UF-B by detecting drowsiness and immediately alerting the driver with audible and visual cues, directly supporting the goal of promoting driving safety.				
Traceability:	**Use cases cf.**	**UC-001, UC-002, UC-003**			
	Test cases cf.	**Yet to be completed in test case worksheet!**			
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Type	**Functional**	**Non-Functional**
User	☐	☐
System	☒	☐

Table 4.51. System Functional Requirements: SF-B-02

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**				
Requirement #:	**SF-B-02**				
Creation:	**Sep 22 2025 03:26 PM**				
Modification:	**Sep 22 2025 03:37 PM**				
Description:	The system shall detect when the drivers eyes are closed for prolonged periods and trigger alerts.				
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	**UF-B**				
Justify why meeting SF-B-02 can contribute to the fulfilment of UF-B	Meeting SF-B-02 contributes to UF-B by detecting driver inattention and triggering alerts, helping the driver stay focused and supporting the goal of driving safety.				
Traceability:	**Use cases cf.**	**UC-001, UC-002**			
	Test cases cf.	**Yet to be completed in test case worksheet!**			
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Type	**Functional**	**Non-Functional**
User	☐	☐
System	☒	☐

Table 4.52. System Functional Requirements: SF-C-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-C-01			Type: Functional / Non-Functional	
Creation:	Sep 22 2025 03:38 PM				
Modification:	Sep 22 2025 03:39 PM			User: ☐ / ☐	
				System: ☒ / ☐	
Description:	The system shall detect when the drivers eyes are looking away from the road for prolonged periods and trigger alerts.				
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UF-C				
Justify why meeting SF-C-01 can contribute to the fulfilment of UF-C					
Traceability:	Use cases cf.	UC-001			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.53. System Functional Requirements: SF-C-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-C-02			Type: Functional / Non-Functional	
Creation:	Sep 24 2025 03:01 PM				
Modification:	Oct 31 2025 07:34 PM			User: ☐ / ☐	
				System: ☒ / ☐	
Description:	The system shall continuously process front-camera frames and estimate head yaw (θ). It shall raise a DISTRACTED event when the absolute yaw angle exceeds a configurable threshold for a sustained, configurable duration.				
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UF-C				
Justify why meeting SF-C-02 can contribute to the fulfilment of UF-C	Maps to UF-C because it defines the actual decision rule for distraction (yaw angle + sustained duration) from continuous face monitoring, so the system can detect and flag distraction in real time.				
Traceability:	Use cases cf.	UC-010			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.54. System Functional Requirements: SF-C-03

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-C-03			Type: Functional / Non-Functional	
Creation:	Sep 24 2025 03:10 PM				
Modification:	Oct 31 2025 07:36 PM			User: ☐ / ☐	
				System: ☒ / ☐	
Description:	The system shall continuously estimate eye gaze and raise a DISTRACTED event when gaze deviates off the forward road region beyond configurable angular bounds for a sustained, configurable duration.				
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UF-C				
Justify why meeting SF-C-03 can contribute to the fulfilment of UF-C	UF-C asks for real-time monitoring of face and eyes to detect distraction. This SR adds an eye-based detection path (gaze off road), complementing head-yaw, directly fulfilling the “eyes” aspect of UF-C.				
Traceability: Use cases cf.	Yet to be completed in use case worksheet!				
Traceability: Test cases cf.	Yet to be completed in test case worksheet!				
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.55. System Functional Requirements: SF-D-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-D-01			Type: Functional / Non-Functional	
Creation:	Sep 24 2025 02:56 PM				
Modification:	Sep 24 2025 02:58 PM			User: ☐ / ☐	
				System: ☒ / ☐	
Description:	The system shall provide a settings menu accessible to the driver.				
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UF-D				
Justify why meeting SF-D-01 can contribute to the fulfilment of UF-D	Meeting SF-D-01 directly contributes to fulfilling UF-D because it ensures that the system includes the technical capability for a settings menu that the driver can access.				
Traceability: Use cases cf.	UC-003, UC-004				
Traceability: Test cases cf.	Yet to be completed in test case worksheet!				
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Table 4.56. System Functional Requirements: SF-D-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-D-02			Type: User ☐ Functional ☐ Non-Functional	
Creation:	Sep 24 2025 02:58 PM			Type: System ☒ Functional ☐ Non-Functional	
Modification:	Sep 24 2025 02:59 PM				
Description:	The system shall allow the driver to adjust the volume of alerts.				
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UF-D				
Justify why meeting SF-D-02 can contribute to the fulfilment of UF-D	Meeting SF-D-02 fulfills UF-D by providing the mechanism for the driver to adjust alert volume. It ensures the system can change audio output levels based on user input, satisfying the requirement for configurable alerts and allowing verification through functional testing				
Traceability:	Use cases cf.	UC-004			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.57. System Functional Requirements: SF-D-03

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-D-03			Type: User ☐ Functional ☐ Non-Functional	
Creation:	Sep 24 2025 02:59 PM			Type: System ☒ Functional ☐ Non-Functional	
Modification:	Sep 24 2025 03:01 PM				
Description:	The system shall allow the driver to adjust the sensitivity level of detection.				
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UF-D				
Justify why meeting SF-D-03 can contribute to the fulfilment of UF-D	Meeting SF-D-03 fulfills UF-D by enabling the driver to adjust the system’s detection sensitivity. It provides the functionality needed to change detection thresholds, ensuring the system responds appropriately to driver preferences and can be verified through functional testing.				
Traceability:	Use cases cf.	UC-004			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.58. System Functional Requirements: SF-E-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-E-01			Type: Functional / Non-Functional	
Creation:	Oct 31 2025 07:37 PM				
Modification:	Oct 31 2025 07:38 PM			User: ☐ / ☐	
				System: ☒ / ☐	
Description:	The system shall provide explicit controls to Start and Stop a monitoring session. On Start, it shall request camera permission if needed, initialize CameraX and on-device inference, and begin emitting detection events. On Stop, it shall tear down camera/inference and finalize the current session.				
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UF-E				
Justify why meeting SF-E-01 can contribute to the fulfilment of UF-E	implements the user's explicit control over when monitoring starts/stops.				
Traceability:	Use cases cf.	Yet to be completed in use case worksheet!			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.59. System Functional Requirements: SF-E-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-E-02			Type: Functional / Non-Functional	
Creation:	Oct 31 2025 07:39 PM				
Modification:	Oct 31 2025 07:41 PM			User: ☐ / ☐	
				System: ☒ / ☐	
Description:	When the driver taps Start, the system shall execute a preflight flow that (a) verifies/requests camera permission, (b) verifies camera availability, and (c) initializes the monitoring pipeline only if all checks pass; otherwise it shall surface a clear, actionable error and remain stopped.				
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UF-E				
Justify why meeting SF-E-02 can contribute to the fulfilment of UF-E	UF-E is about driver control over when monitoring occurs. This SR guarantees monitoring never begins or resumes without an explicit Start, and that Stop truly ends analysis—preserving the user's manual control exactly as requested.				
Traceability:	Use cases cf.	Yet to be completed in use case worksheet!			

	Test cases cf.	**Yet to be completed in test case worksheet!**
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Table 4.60. System Functional Requirements: SF-F-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-F-02			Type: User / System	Functional: ☐ / ☒ — Non-Functional: ☐ / ☐
Creation:	Oct 31 2025 07:42 PM				
Modification:	Oct 31 2025 07:42 PM				
Description:	When the system detects DROWSY or DISTRACTED, it shall emit an alert in ≤ 200 ms from the detection event, updating the UI banner and triggering device audio/vibration per current settings				
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UF-F				
Justify why meeting SF-F-02 can contribute to the fulfilment of UF-F	UF-F requires immediate alerts to regain attention; this SR guarantees timely delivery of the alert.				
Traceability: Use cases cf.	UC-010				
Traceability: Test cases cf.	Yet to be completed in test case worksheet!				
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Table 4.61. System Functional Requirements: SF-F-03

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-F-03			Type: User / System	Functional: ☐ / ☒ — Non-Functional: ☐ / ☐
Creation:	Oct 31 2025 07:42 PM				
Modification:	Oct 31 2025 07:43 PM				
Description:	The system shall select active alert modalities (visual, audio, vibration) based on user settings and device capability/state (ringer mode, DND, vibrator present). If a modality is unavailable, it shall fallback to available ones and always present a visual alert.				
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UF-F				
Justify why meeting SF-F-03 can contribute to the fulfilment of UF-F	UF-F asks for visual/audio/vibration; this SR ensures the right mix is used reliably, with fallbacks so the driver still gets alerted.				
Traceability: Use cases cf.	Yet to be completed in use case worksheet!				
Traceability: Test cases cf.	Yet to be completed in test case worksheet!				
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Table 4.62. System Functional Requirements: SF-G-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-G-01			Type: User ☐ Functional ☐ Non-Functional Type: System ☒ Functional ☐ Non-Functional	
Creation:	Oct 31 2025 07:44 PM				
Modification:	Oct 31 2025 07:44 PM				
Description:	When the driver taps Acknowledge, the system shall immediately dismiss the current alert (visual/audio/vibration), stop any ongoing alert output, and continue monitoring.				
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UF-G				
Justify why meeting SF-G-01 can contribute to the fulfilment of UF-G	Gives the driver control to stop a specific alert without disabling monitoring—reduces unnecessary alarms while staying aware.				
Traceability:	Use cases cf.	Yet to be completed in use case worksheet!			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.63. System Functional Requirements: SF-G-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-G-02			Type: User ☐ Functional ☐ Non-Functional Type: System ☒ Functional ☐ Non-Functional	
Creation:	Oct 31 2025 07:44 PM				
Modification:	Oct 31 2025 07:45 PM				
Description:	When the driver taps Snooze, the system shall suppress alerts of the same type for a configurable cooldown while monitoring continues.				
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UF-G				
Justify why meeting SF-G-02 can contribute to the fulfilment of UF-G	Prevents repeated alarms for the same condition, matching the user's goal while keeping them aware of ongoing monitoring.				
Traceability:	Use cases cf.	Yet to be completed in use case worksheet!			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.64. System Functional Requirements: SF-H-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-H-01			Type: User / System	Functional: ☐ / ☒ — Non-Functional: ☐ / ☐
Creation:	Oct 31 2025 07:45 PM				
Modification:	Oct 31 2025 07:46 PM				
Description:	Before monitoring starts, the system shall present a Calibration flow and block Start until calibration passes or the user explicitly skips. If skipped or failed, the session starts in degraded mode (lower alert confidence) and shows an inline tip.				
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UF-H				
Justify why meeting SF-H-01 can contribute to the fulfilment of UF-H	Ensures calibration happens at the start and explicitly controls entry to monitoring, matching the user's intent.				
Traceability:	Use cases cf.	UC-005			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.65. System Functional Requirements: SF-H-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-H-02			Type: User / System	Functional: ☐ / ☒ — Non-Functional: ☐ / ☐
Creation:	Oct 31 2025 07:46 PM				
Modification:	Oct 31 2025 07:46 PM				
Description:	During calibration, the system shall show live guidance overlays (face box, eye markers) and evaluate position and pose stability until pass criteria are met.				
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UF-H				
Justify why meeting SF-H-02 can contribute to the fulfilment of UF-H	Directly targets "properly positioned" face via measurable alignment and stability rules.				
Traceability:	Use cases cf.	UC-005			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.66. System Functional Requirements: SF-I-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-I-01			Type: User ☐ / System ☒	Functional: ☐ / ☒ — Non-Functional: ☐ / ☐
Creation:	Oct 31 2025 07:47 PM				
Modification:	Oct 31 2025 07:48 PM				
Description:	The system shall provide a Summary screen that displays, for a chosen time range or session: total alerts by type (Drowsy, Distracted), timestamps, session duration, and average PERCLOS; users can filter by Last Session, Today, 7 Days, or Custom Range.				
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UF-I				
Justify why meeting SF-I-01 can contribute to the fulfilment of UF-I	Presents an at-a-glance view of recent alerts/sessions so drivers can understand patterns.				
Traceability:	Use cases cf.	UC-006			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.67. System Functional Requirements: SF-I-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-I-02			Type: User ☐ / System ☒	Functional: ☐ / ☒ — Non-Functional: ☐ / ☐
Creation:	Oct 31 2025 07:48 PM				
Modification:	Oct 31 2025 07:48 PM				
Description:	The system shall persist per-session records locally and expose a query API to retrieve summaries for time windows.				
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UF-I				
Justify why meeting SF-I-02 can contribute to the fulfilment of UF-I	Enables the summaries the user wants by reliably capturing and retrieving session/alert data.				
Traceability:	Use cases cf.	UC-006			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.68. System Functional Requirements: SF-J-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-J-01			Type: User ☐ Functional ☐ Non-Functional; System ☒ Functional ☐ Non-Functional	
Creation:	Oct 31 2025 07:49 PM				
Modification:	Oct 31 2025 07:49 PM				
Description:	The system shall perform all camera capture and ML inference on device and shall not transmit frames, embeddings, or detection results to any network endpoint.				
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UF-J				
Justify why meeting SF-J-01 can contribute to the fulfilment of UF-J	Ensures camera data never leaves the device, directly satisfying the privacy request.				
Traceability:	Use cases cf.	Yet to be completed in use case worksheet!			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.69. System Functional Requirements: SF-J-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-J-02			Type: User ☐ Functional ☐ Non-Functional; System ☒ Functional ☐ Non-Functional	
Creation:	Oct 31 2025 07:50 PM				
Modification:	Oct 31 2025 07:50 PM				
Description:	The system shall store only session metadata and alert events in app-scoped storage; by default, it shall not persist raw images or video frames.				
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Engineered From:	UF-J				
Justify why meeting SF-J-02 can contribute to the fulfilment of UF-J	Minimizes what’s kept and keeps it local, reducing privacy risk.				
Traceability:	Use cases cf.	Yet to be completed in use case worksheet!			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.70. System Functional Requirements: SF-J-03

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	**SF-J-03**				
Creation:	**Oct 31 2025 07:50 PM**				
Modification:	**Oct 31 2025 07:51 PM**				
Description:	The system shall request only the minimum runtime permissions required (e.g., CAMERA) and shall block any background network requests originating from the monitoring module.				
Priority:	Highest	✔ **High**	Medium	Low	Lowest
This Req. is Engineered From:	**UF-J**				
Justify why meeting SF-J-03 can contribute to the fulfilment of UF-J	Limits privileges and prevents unintended data transmission, protecting privacy as requested.				
Traceability:	**Use cases cf.**	**UC-007**			
	Test cases cf.	**Yet to be completed in test case worksheet!**			
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Type	Functional	Non-Functional
User	☐	☐
System	☒	☐

Table 4.71. System Functional Requirements: SF-K-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	**SF-K-01**				
Creation:	**Oct 31 2025 07:51 PM**				
Modification:	**Oct 31 2025 07:52 PM**				
Description:	The system shall support MDM-driven install/update with staged rollout (e.g., 5% → 25% → 100%), version pinning, and force-install on enrolled devices.				
Priority:	Highest	✔ **High**	Medium	Low	Lowest
This Req. is Engineered From:	**UF-K**				
Justify why meeting SF-K-01 can contribute to the fulfilment of UF-K	Enables consistent, secure rollout of app updates across many devices.				
Traceability:	**Use cases cf.**	**UC-008**			
	Test cases cf.	**Yet to be completed in test case worksheet!**			
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Type	Functional	Non-Functional
User	☐	☐
System	☒	☐

Table 4.72. System Functional Requirements: SF-K-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-K-02			Type: User ☐ Non-Functional ☐ / System ☒ Functional ☐	
Creation:	Oct 31 2025 07:52 PM				
Modification:	Oct 31 2025 07:53 PM				
Description:	When allowed by MDM, the app shall honor policy to auto-grant CAMERA permission at install/update and halt monitoring if policy later revokes it, showing an admin-configured message.				
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Engineered From:	UF-K				
Justify why meeting SF-K-02 can contribute to the fulfilment of UF-K	Centralizes permission handling for consistent, secure setup.				
Traceability:	Use cases cf.	UC-008			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.73. System Functional Requirements: SF-L-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-L-01			Type: User ☐ Non-Functional ☐ / System ☒ Functional ☐	
Creation:	Oct 31 2025 07:59 PM				
Modification:	Oct 31 2025 07:59 PM				
Description:	The backend shall compute fleet-level KPIs over a selected time window: alerts per driving hour, counts by type (Drowsy/Distracted), sessions with ≥1 alert (%), and time-of-day/day-of-week distributions.				
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Engineered From:	UF-L				
Justify why meeting SF-L-01 can contribute to the fulfilment of UF-L	Provides the aggregated statistics needed to see safety trends across drivers.				
Traceability:	Use cases cf.	UC-009			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.74. System Functional Requirements: SF-L-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-L-02			Type: Functional / Non-Functional	
Creation:	Oct 31 2025 08:00 PM				
Modification:	Oct 31 2025 08:00 PM			User: ☐ / ☐	
				System: ☒ / ☐	
Description:	The system shall accept anonymized session summaries via an ingestion API or bulk import				
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UF-L				
Justify why meeting SF-L-02 can contribute to the fulfilment of UF-L	Reliable, clean data is essential to build accurate aggregates for trend analysis.				
Traceability: Use cases cf.	UC-009				
Traceability: Test cases cf.	Yet to be completed in test case worksheet!				
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.75. System Functional Requirements: SF-L-03

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SF-L-03			Type: Functional / Non-Functional	
Creation:	Oct 31 2025 08:00 PM				
Modification:	Oct 31 2025 08:00 PM			User: ☐ / ☐	
				System: ☒ / ☐	
Description:	The analytics service shall support filtering KPIs by organization, driver cohort/vehicle group, device model/app version, and date range.				
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UF-L				
Justify why meeting SF-L-03 can contribute to the fulfilment of UF-L	Lets fleet managers slice the aggregates to find where/when safety risks cluster.				
Traceability: Use cases cf.	UC-009				
Traceability: Test cases cf.	Yet to be completed in test case worksheet!				
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.76. System NonFunctional Requirements: SP-01-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SP-01-01			Type: Functional / Non-Functional	
Creation:	Oct 30 2025 09:24 PM				
Modification:	Oct 30 2025 09:25 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	The system UI shall launch within 2 seconds and maintain a response time < 200 ms for all on-screen interactions.			Product (sub-type below)	
				Usability Requirements	
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Engineered From:	UP-01				
Justify why meeting SP-01-01 can contribute to the fulfilment of UP-01					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.77. System NonFunctional Requirements: SP-02-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SP-02-01			Type: Functional / Non-Functional	
Creation:	Oct 30 2025 09:25 PM				
Modification:	Oct 30 2025 09:27 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	The system shall automatically start monitoring when the driver presses “Start Session,” requiring no further manual input.			Product (sub-type below)	
				Usability Requirements	
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Engineered From:	UP-02				
Justify why meeting SP-02-01 can contribute to the fulfilment of UP-02					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.78. System NonFunctional Requirements: SP-03-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SP-03-01			Type: Functional / Non-Functional	
Creation:	Oct 30 2025 09:26 PM				
Modification:	Oct 30 2025 09:27 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	The alert subsystem shall generate visual, audio, and haptic feedback signals that are synchronized within ±50 ms.			Product (sub-type below)	
				Usability Requirements	
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Engineered From:	UP-03				
Justify why meeting SP-03-01 can contribute to the fulfilment of UP-03					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.79. System NonFunctional Requirements: SP-04-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SP-04-01			Type: Functional / Non-Functional	
Creation:	Oct 30 2025 09:26 PM				
Modification:	Oct 30 2025 09:27 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	All UI text and contrast ratios shall conform to WCAG 2.1 AA standards for accessibility.			Product (sub-type below)	
				Usability Requirements	
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Engineered From:	UP-04				
Justify why meeting SP-04-01 can contribute to the fulfilment of UP-04					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.80. System NonFunctional Requirements: SP-05-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SP-05-01			Type / Functional / Non-Functional	
Creation:	Oct 30 2025 09:27 PM				
Modification:	Oct 30 2025 09:27 PM			User ☐ ☐	
				System ☐ ☒	
Description:	The inference module shall process ≥ 15 frames per second and issue alerts within 200 ms of detection.			Product (sub-type below)	
				Performance Requirements	
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Engineered From:	UP-05				
Justify why meeting SP-05-01 can contribute to the fulfilment of UP-05					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.81. System NonFunctional Requirements: SP-05-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SP-05-02			Type / Functional / Non-Functional	
Creation:	Oct 30 2025 09:28 PM				
Modification:	Oct 30 2025 09:28 PM			User ☐ ☐	
				System ☐ ☒	
Description:	During continuous operation, average CPU utilization shall not exceed 40 %, and thermal throttling shall be avoided through adaptive frame skipping.			Product (sub-type below)	
				Performance Requirements	
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Engineered From:	UP-05				
Justify why meeting SP-05-02 can contribute to the fulfilment of UP-05					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.82. System NonFunctional Requirements: SP-06-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SP-06-01			Type: Functional / Non-Functional	
Creation:	Oct 31 2025 08:19 PM				
Modification:	Oct 31 2025 08:19 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	Inference must execute locally using Snapdragon CPU/GPU/NPU via QNN/NNAPI, with preferred NPU then GPU, falling back to CPU only if required.			Product (sub-type below)	
				Performance Requirements	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UP-06				
Justify why meeting SP-06-01 can contribute to the fulfilment of UP-06	Ensures inference stays on device and leverages Snapdragon accelerators.				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.83. System NonFunctional Requirements: SP-07-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SP-07-01			Type: Functional / Non-Functional	
Creation:	Oct 31 2025 08:20 PM				
Modification:	Oct 31 2025 08:21 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	When device temperature or battery drain exceeds safe thresholds, the app shall automatically reduce inference frequency (FPS) and/or input resolution to lower load.			Product (sub-type below)	
				Performance Requirements	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UP-07				
Justify why meeting SP-07-01 can contribute to the fulfilment of UP-07	Directly implements UP-07’s requirement to automatically reduce inference frequency under unsafe thermal/battery conditions.				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.84. System NonFunctional Requirements: SP-08-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SP-08-01			Type / Functional / Non-Functional	
Creation:	Oct 30 2025 09:28 PM				
Modification:	Oct 30 2025 09:28 PM			User ☐ ☐	
				System ☐ ☒	
Description:	The monitoring process shall run stably for at least 2 hours without crash, memory leak, or data loss.			Product (sub-type below)	
				Availability/Reliability/Security	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UP-08				
Justify why meeting SP-08-01 can contribute to the fulfilment of UP-08					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.85. System NonFunctional Requirements: SP-09-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SP-09-01			Type / Functional / Non-Functional	
Creation:	Oct 30 2025 09:28 PM				
Modification:	Oct 30 2025 09:29 PM			User ☐ ☐	
				System ☐ ☒	
Description:	On hardware or model failure, the system shall halt inference and display a “Safe Mode” warning.			Product (sub-type below)	
				Availability/Reliability/Security	
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Engineered From:	UP-09				
Justify why meeting SP-09-01 can contribute to the fulfilment of UP-09					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.86. System NonFunctional Requirements: SP-10-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SP-10-01			Type: Functional / Non-Functional	
Creation:	Oct 30 2025 09:29 PM				
Modification:	Oct 30 2025 09:29 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	All data logs and configuration files shall be stored in app-scoped encrypted storage and deleted upon user request.			Product (sub-type below)	
				Availability/Reliability/Security	
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UP-10				
Justify why meeting SP-10-01 can contribute to the fulfilment of UP-10					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.87. System NonFunctional Requirements: SP-11-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SP-11-01			Type: Functional / Non-Functional	
Creation:	Oct 30 2025 09:29 PM				
Modification:	Oct 30 2025 09:30 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	The application shall respect Android runtime permission revocations and immediately disable camera or microphone access.			Product (sub-type below)	
				Availability/Reliability/Security	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UP-11				
Justify why meeting SP-11-01 can contribute to the fulfilment of UP-11					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.88. System NonFunctional Requirements: SO-01-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SO-01-01			Type: Functional / Non-Functional	
Creation:	Oct 30 2025 09:30 PM				
Modification:	Oct 30 2025 09:30 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	Development shall use Android Studio Giraffe + Kotlin 1.9 with Git version control.			Organizational (sub-type below)	
				Development Requirements	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UO-01				
Justify why meeting SO-01-01 can contribute to the fulfilment of UO-01					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.89. System NonFunctional Requirements: SO-01-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SO-01-02			Type: Functional / Non-Functional	
Creation:	Oct 31 2025 08:03 PM				
Modification:	Oct 31 2025 08:04 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	The codebase shall live in a single authoritative Git repository with protected branches.			Organizational (sub-type below)	
				Development Requirements	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UO-01				
Justify why meeting SO-01-02 can contribute to the fulfilment of UO-01	Ensures the “all source code in Git” mandate is enforceable and safe.				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.90. System NonFunctional Requirements: SO-02-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SO-02-01			Type: Functional / Non-Functional	
Creation:	Oct 31 2025 08:04 PM				
Modification:	Oct 31 2025 08:04 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	CI shall block merges that violate documented branching and commit standards.			Organizational (sub-type below)	
				Development Requirements	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UO-02				
Justify why meeting SO-02-01 can contribute to the fulfilment of UO-02	Turns README standards into enforceable gates, not suggestions.				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.91. System NonFunctional Requirements: SO-02-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SO-02-02			Type: Functional / Non-Functional	
Creation:	Oct 31 2025 08:05 PM				
Modification:	Oct 31 2025 08:06 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	The repository shall include a README section detailing branching/commit rules and a PR template; CI validates their presence and version.			Organizational (sub-type below)	
				Development Requirements	
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Engineered From:	UO-02				
Justify why meeting SO-02-02 can contribute to the fulfilment of UO-02	Keeps the documented standards up-to-date and visible at the point of contribution.				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.92. System NonFunctional Requirements: SO-03-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SO-03-01			Type: User / System	Functional: ☐ / ☐ — Non-Functional: ☐ / ☒
Creation:	Oct 31 2025 08:11 PM				
Modification:	Oct 31 2025 08:11 PM				
Description:	Enforce Kotlin style via automated linters in CI.			Organizational (sub-type below)	Development Requirements
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UO-03				
Justify why meeting SO-03-01 can contribute to the fulfilment of UO-03	Turns the “follow Kotlin conventions” part of UO-03 into an enforceable gate.				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.93. System NonFunctional Requirements: SO-03-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SO-03-02			Type: User / System	Functional: ☐ / ☐ — Non-Functional: ☐ / ☒
Creation:	Oct 31 2025 08:12 PM				
Modification:	Oct 31 2025 08:12 PM				
Description:	Require API docs and meaningful inline comments for non-obvious logic.			Organizational (sub-type below)	Development Requirements
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UO-03				
Justify why meeting SO-03-02 can contribute to the fulfilment of UO-03	"include inline comments” and ensures clarity/maintainability.				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.94. System NonFunctional Requirements: SO-03-03

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SO-03-03			Type: Functional / Non-Functional	
Creation:	Oct 31 2025 08:12 PM				
Modification:	Oct 31 2025 08:13 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	All code changes must pass peer review focused on clarity and maintainability.			Organizational (sub-type below)	
				Development Requirements	
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UO-03				
Justify why meeting SO-03-03 can contribute to the fulfilment of UO-03	Operationalizes the “undergo peer review” clause in UO-03.				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.95. System NonFunctional Requirements: SO-04-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SO-04-01			Type: Functional / Non-Functional	
Creation:	Oct 30 2025 09:30 PM				
Modification:	Oct 30 2025 09:31 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	Continuous-integration builds shall enforce unit-test coverage ≥ 80 %.			Organizational (sub-type below)	
				Development Requirements	
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Engineered From:	UO-04				
Justify why meeting SO-04-01 can contribute to the fulfilment of UO-04					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.96. System NonFunctional Requirements: SO-05-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SO-05-01			Type: Functional / Non-Functional	
Creation:	Oct 30 2025 09:31 PM			User: ☐ / ☐	
Modification:	Oct 30 2025 09:31 PM			System: ☐ / ☒	
Description:	Release builds shall target API 33 or higher and be installable on Snapdragon 8-series or equivalent hardware.			Organizational (sub-type below)	
				Operational Requirements	
Priority:	Highest	High	Medium	✔ Low	Lowest
This Req. is Engineered From:	UO-05				
Justify why meeting SO-05-01 can contribute to the fulfilment of UO-05					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.97. System NonFunctional Requirements: SO-06-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SO-06-01			Type: Functional / Non-Functional	
Creation:	Oct 31 2025 08:16 PM			User: ☐ / ☐	
Modification:	Oct 31 2025 08:16 PM			System: ☐ / ☒	
Description:	The app shall include a built-in Help section (no network required) covering calibration, settings, and safety disclaimers.			Organizational (sub-type below)	
				Operational Requirements	
Priority:	Highest	High	Medium	Low	Lowest
This Req. is Engineered From:	UO-06				
Justify why meeting SO-06-01 can contribute to the fulfilment of UO-06	Ensures a concise, always-available guide inside the app as UO-06 requires.				
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.98. System NonFunctional Requirements: SO-06-02

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**				
Requirement #:	**SO-06-02**				
Creation:	**Oct 31 2025 08:16 PM**				
Modification:	**Oct 31 2025 08:17 PM**				
Description:	The guide must clearly explain (a) calibration steps & pass/fail tips, (b) each setting's effect/range/defaults, (c) safety disclaimers and appropriate us				
Priority:	Highest	**✔ High**	Medium	Low	Lowest
This Req. is Engineered From:	**UO-06**				
Justify why meeting SO-06-02 can contribute to the fulfilment of UO-06	Guarantees the exact topics and concise clarity mandated by UO-06.				
Traceability:	**Use cases cf.**	**N/A**			
	Test cases cf.	**Yet to be completed in test case worksheet!**			
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Type	**Functional**	**Non-Functional**
User	☐	☐
System	☐	☒

Organizational (sub-type below)

Operational Requirements

Table 4.99. System NonFunctional Requirements: SO-06-03

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**				
Requirement #:	**SO-06-03**				
Creation:	**Oct 31 2025 08:17 PM**				
Modification:	**Oct 31 2025 08:17 PM**				
Description:	The guide shall meet accessibility basics and support future localization; relevant screens should deep-link to specific guide topics.				
Priority:	Highest	High	**✔ Medium**	Low	Lowest
This Req. is Engineered From:	**UO-06**				
Justify why meeting SO-06-03 can contribute to the fulfilment of UO-06	Makes the guide usable for all users and easy to expand, aligning with UO-06's "concise user guide or in-app section."				
Traceability:	**Use cases cf.**	**N/A**			
	Test cases cf.	**Yet to be completed in test case worksheet!**			
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Type	**Functional**	**Non-Functional**
User	☐	☐
System	☐	☒

Organizational (sub-type below)

Operational Requirements

Table 4.100. System NonFunctional Requirements: SO-07-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SO-07-01			Type: Functional / Non-Functional	
Creation:	Oct 30 2025 09:31 PM			User: ☐ / ☐	
Modification:	Oct 30 2025 09:32 PM			System: ☐ / ☒	
Description:	The application package shall include an HTML or PDF quick-start guide accessible through the Help menu.			Organizational (sub-type below)	
				Environmental Requirements	
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Engineered From:	UO-07				
Justify why meeting SO-07-01 can contribute to the fulfilment of UO-07					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.101. System NonFunctional Requirements: SO-07-02

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SO-07-02			Type: Functional / Non-Functional	
Creation:	Oct 30 2025 09:32 PM			User: ☐ / ☐	
Modification:	Oct 30 2025 09:32 PM			System: ☐ / ☒	
Description:	Detection accuracy shall remain ≥ 90 % under daylight, cabin light, and low-light conditions.			Organizational (sub-type below)	
				Environmental Requirements	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UO-07				
Justify why meeting SO-07-02 can contribute to the fulfilment of UO-07					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.102. System NonFunctional Requirements: SO-08-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SO-08-01			Type: Functional / Non-Functional	
Creation:	Oct 30 2025 09:33 PM			User: ☐ / ☐	
Modification:	Oct 30 2025 09:33 PM			System: ☐ / ☒	
Description:	The system shall function correctly in both portrait and landscape orientations without restarting the process.			Organizational (sub-type below)	
				Environmental Requirements	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UO-08				
Justify why meeting SO-08-01 can contribute to the fulfilment of UO-08					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
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Table 4.103. System NonFunctional Requirements: SO-09-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SO-09-01			Type: Functional / Non-Functional	
Creation:	Oct 30 2025 09:34 PM			User: ☐ / ☐	
Modification:	Oct 30 2025 09:34 PM			System: ☐ / ☒	
Description:	All monitoring and alerting features shall remain fully functional offline.			Organizational (sub-type below)	
				Environmental Requirements	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UO-09				
Justify why meeting SO-09-01 can contribute to the fulfilment of UO-09					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.104. System NonFunctional Requirements: SE-01-01

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**				
Requirement #:	**SE-01-01**				
Creation:	**Oct 30 2025 09:34 PM**				
Modification:	**Oct 30 2025 09:35 PM**				
Description:	At startup, the application shall display a safety disclaimer and disable configuration changes while the vehicle is moving.				
Priority:	Highest	**✔ High**	Medium	Low	Lowest
This Req. is Engineered From:	**UE-01**				
Justify why meeting SE-01-01 can contribute to the fulfilment of UE-01					
Traceability:	**Use cases cf.**	**N/A**			
	Test cases cf.	**Yet to be completed in test case worksheet!**			
Acknowledgment	***Generated from the CapStone Process Management System ©2025***				

Type	**Functional**	**Non-Functional**
User	☐	☐
System	☐	☒

External (sub-type below)

Legislative Requirements on Safety/Security

Table 4.105. System NonFunctional Requirements: SE-03-01

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**				
Requirement #:	**SE-03-01**				
Creation:	**Oct 30 2025 09:37 PM**				
Modification:	**Oct 30 2025 09:37 PM**				
Description:	The face-detection model shall be trained and validated on datasets representing diverse skin tones, facial features, and eyewear.				
Priority:	Highest	**✔ High**	Medium	Low	Lowest
This Req. is Engineered From:	**UE-03**				
Justify why meeting SE-03-01 can contribute to the fulfilment of UE-03					
Traceability:	**Use cases cf.**	**N/A**			
	Test cases cf.	**Yet to be completed in test case worksheet!**			
Acknowledgment	***Generated from the CapStone Process Management System ©2025***				

Type	**Functional**	**Non-Functional**
User	☐	☐
System	☐	☒

External (sub-type below)

Cultural and Social Requirements

Table 4.106. System NonFunctional Requirements: SE-04-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SE-04-01			Type: Functional / Non-Functional	
Creation:	Oct 30 2025 09:38 PM				
Modification:	Oct 30 2025 09:39 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	The UI shall support English by default and be designed for easy localization using Android string resources.			External (sub-type below)	
				Cultural and Social Requirements	
Priority:	Highest	High	✔ Medium	Low	Lowest
This Req. is Engineered From:	UE-04				
Justify why meeting SE-04-01 can contribute to the fulfilment of UE-04					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.107. System NonFunctional Requirements: SE-05-01

Project Name:	Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Requirement #:	SE-05-01			Type: Functional / Non-Functional	
Creation:	Oct 30 2025 09:39 PM				
Modification:	Oct 30 2025 09:39 PM			User: ☐ / ☐	
				System: ☐ / ☒	
Description:	The app shall use only stable Android Jetpack APIs (CameraX, AudioManager, Storage) and tested SDK versions.			External (sub-type below)	
				Interoperability Requirements	
Priority:	Highest	✔ High	Medium	Low	Lowest
This Req. is Engineered From:	UE-05				
Justify why meeting SE-05-01 can contribute to the fulfilment of UE-05					
Traceability:	Use cases cf.	N/A			
	Test cases cf.	Yet to be completed in test case worksheet!			
Acknowledgment	*Generated from the CapStone Process Management System ©2025*				

Table 4.108. System NonFunctional Requirements: SE-06-01

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**				
Requirement #:	**SE-06-01**				
Creation:	**Oct 30 2025 09:39 PM**				
Modification:	**Oct 30 2025 09:40 PM**				
Description:	The inference engine shall remain interoperable with Qualcomm QNN and Android NNAPI runtimes for hardware acceleration.				
Priority:	Highest	✔ **High**	Medium	Low	Lowest
This Req. is Engineered From:	**UE-06**				
Justify why meeting SE-06-01 can contribute to the fulfilment of UE-06					
Traceability:	**Use cases cf.**	**N/A**			
	Test cases cf.	**Yet to be completed in test case worksheet!**			
Acknowledgment	***Generated from the CapStone Process Management System ©2025***				

Type	**Functional**	**Non-Functional**
User	☐	☐
System	☐	☒

External (sub-type below)

Interoperability Requirements

Table 4.109. Mapping from user requirements to system requirements

Project Name: Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform			
User Requirements		**System Requirements**	
Req ID	**Description**	**Req ID**	**Description**
UE-01	The app shall include a startup safety disclaimer and disable manual interaction prompts while the vehicle is in motion.	SE-01-01	At startup, the application shall display a safety disclaimer and disable configuration changes while the vehicle is moving.
UE-03	The face-detection model shall be evaluated for consistent accuracy across diverse skin tones, facial structures, and eyewear.	SE-03-01	The face-detection model shall be trained and validated on datasets representing diverse skin tones, facial features, and eyewear.
UE-04	The interface and alert text shall default to English and be structured for easy localization into other languages.	SE-04-01	The UI shall support English by default and be designed for easy localization using Android string resources.
UE-05	The app shall utilize only stable Android Jetpack APIs to ensure device portability.	SE-05-01	The app shall use only stable Android Jetpack APIs (CameraX, AudioManager, Storage) and tested SDK versions.
UE-06	The app shall interoperate seamlessly with both Qualcomm QNN and Android NNAPI runtimes for ML acceleration without requiring code modifications.	SE-06-01	The inference engine shall remain interoperable with Qualcomm QNN and Android NNAPI runtimes for hardware acceleration.
UF-A	As a driver, I want to securely create accounts and log in using a username and password so that I can view my driving history.	SF-A-01	The system shall record instances of detected drowsiness or distraction for each trip, including timestamps
UF-B	As a driver, I want the system to detect signs of drowsiness in real-time on my face and eyes in order to promote driving safety.	SF-B-01	The system shall provide immediate audible and visual alerts when drowsiness or distraction is detected.
		SF-B-02	The system shall detect when the drivers eyes are closed for prolonged periods and trigger alerts.
UF-C	As a driver, I want the system to continuously monitor my face and eyes in real-time to detect signs of distraction in order to promote driving focus.	SF-C-01	The system shall detect when the drivers eyes are looking away from the road for prolonged periods and trigger alerts.
		SF-C-02	The system shall continuously process front-camera frames and estimate head yaw (θ). It shall raise a DISTRACTED event when the absolute yaw angle exceeds a configurable threshold for a sustained, configurable duration.
		SF-C-03	The system shall continuously estimate eye gaze and raise a DISTRACTED event when gaze deviates off the forward road region beyond configurable angular bounds for a sustained, configurable duration.
	As a driver, I should be able to configure	SF-D-01	The system shall provide a settings menu accessible to the driver.

UF-D	settings, such as volume of alerts or sensitivity of detection.	SF-D-02	The system shall allow the driver to adjust the volume of alerts.
		SF-D-03	The system shall allow the driver to adjust the sensitivity level of detection.
UF-E	As a driver, I want to start and stop the monitoring session manually so that I can control when the app analyzes my driving behavior.	SF-E-01	The system shall provide explicit controls to Start and Stop a monitoring session. On Start, it shall request camera permission if needed, initialize CameraX and on-device inference, and begin emitting detection events. On Stop, it shall tear down camera/inference and finalize the current session.
		SF-E-02	When the driver taps Start, the system shall execute a preflight flow that (a) verifies/requests camera permission, (b) verifies camera availability, and (c) initializes the monitoring pipeline only if all checks pass; otherwise it shall surface a clear, actionable error and remain stopped.
UF-F	As a driver, I want to receive real-time visual, audio, or vibration alerts when drowsiness or distraction is detected so that I can regain my attention immediately.	SF-F-02	When the system detects DROWSY or DISTRACTED, it shall emit an alert in ≤ 200 ms from the detection event, updating the UI banner and triggering device audio/vibration per current settings
		SF-F-03	The system shall select active alert modalities (visual, audio, vibration) based on user settings and device capability/state (ringer mode, DND, vibrator present). If a modality is unavailable, it shall fallback to available ones and always present a visual alert.
UF-G	As a driver, I want to acknowledge or snooze an alert to avoid repeated or unnecessary alarms while still staying aware of my condition.	SF-G-01	When the driver taps Acknowledge, the system shall immediately dismiss the current alert (visual/audio/vibration), stop any ongoing alert output, and continue monitoring.
		SF-G-02	When the driver taps Snooze, the system shall suppress alerts of the same type for a configurable cooldown while monitoring continues.
UF-H	As a driver, I want to calibrate the camera at the start of monitoring to ensure my face is properly positioned and lighting is sufficient.	SF-H-01	Before monitoring starts, the system shall present a Calibration flow and block Start until calibration passes or the user explicitly skips. If skipped or failed, the session starts in degraded mode (lower alert confidence) and shows an inline tip.
		SF-H-02	During calibration, the system shall show live guidance overlays (face box, eye markers) and evaluate position and pose stability until pass criteria are met.
		SF-I-01	The system shall provide a Summary screen that displays, for a chosen time range or session: total alerts by type (Drowsy,

UF-I	As a driver, I want to view or export a summary of my recent alerts or sessions to understand my drowsiness and distraction patterns.		Distracted), timestamps, session duration, and average PERCLOS; users can filter by Last Session, Today, 7 Days, or Custom Range.
		SF-I-02	The system shall persist per-session records locally and expose a query API to retrieve summaries for time windows.
UF-J	As a driver, I want assurance that my camera and data stays on my device so that my privacy is protected.	SF-J-01	The system shall perform all camera capture and ML inference on device and shall not transmit frames, embeddings, or detection results to any network endpoint.
		SF-J-02	The system shall store only session metadata and alert events in app-scoped storage; by default, it shall not persist raw images or video frames.
		SF-J-03	The system shall request only the minimum runtime permissions required (e.g., CAMERA) and shall block any background network requests originating from the monitoring module.
UF-K	As a system administrator, I want to deploy and configure the apps across multiple devices using managed Android tools so that updates are consistent and secure.	SF-K-01	The system shall support MDM-driven install/update with staged rollout (e.g., 5% → 25% → 100%), version pinning, and force-install on enrolled devices.
		SF-K-02	When allowed by MDM, the app shall honor policy to auto-grant CAMERA permission at install/update and halt monitoring if policy later revokes it, showing an admin-configured message.
UF-L	As a fleet manager, I want to view aggregated drowsiness and distraction statistics across drivers so that i can identify safety trends.	SF-L-01	The backend shall compute fleet-level KPIs over a selected time window: alerts per driving hour, counts by type (Drowsy/Distracted), sessions with ≥1 alert (%), and time-of-day/day-of-week distributions.
		SF-L-02	The system shall accept anonymized session summaries via an ingestion API or bulk import
		SF-L-03	The analytics service shall support filtering KPIs by organization, driver cohort/vehicle group, device model/app version, and date range.
UO-01	The app shall be developed in Android Studio (Kotlin) and tested on Snapdragon-based devices to validate hardware acceleration.	SO-01-01	Development shall use Android Studio Giraffe + Kotlin 1.9 with Git version control.
		SO-01-02	The codebase shall live in a single authoritative Git repository with protected branches.
UO-02	All source code shall reside in a Git Repository with branching and commit standards documented in the project README.	SO-02-01	CI shall block merges that violate documented branching and commit standards.
		SO-02-02	The repository shall include a README section detailing branching/commit rules and a PR template; CI validates their presence and

			version.
UO-03	Code shall follow Kotlin style conventions, include inline comments, and undergo peer review for clarity and maintainability.	SO-03-01	Enforce Kotlin style via automated linters in CI.
		SO-03-02	Require API docs and meaningful inline comments for non-obvious logic.
		SO-03-03	All code changes must pass peer review focused on clarity and maintainability.
UO-04	Each functional requirement shall be verified by at least one unit or integration test before milestone release.	SO-04-01	Continuous-integration builds shall enforce unit-test coverage ≥ 80 %.
UO-05	The app shall be packaged as an APK targeting Android Level 33 or higher and installable without device root access.	SO-05-01	Release builds shall target API 33 or higher and be installable on Snapdragon 8-series or equivalent hardware.
UO-06	A concise user guide or in app section shall describe calibration, settings, and safety disclaimers.	SO-06-01	The app shall include a built-in Help section (no network required) covering calibration, settings, and safety disclaimers.
		SO-06-02	The guide must clearly explain (a) calibration steps & pass/fail tips, (b) each setting's effect/range/defaults, (c) safety disclaimers and appropriate us
		SO-06-03	The guide shall meet accessibility basics and support future localization; relevant screens should deep-link to specific guide topics.
UO-07	The system shall maintain detection accuracy under daylight, artificial, and low-light cabin conditions.	SO-07-01	The application package shall include an HTML or PDF quick-start guide accessible through the Help menu.
		SO-07-02	Detection accuracy shall remain ≥ 90 % under daylight, cabin light, and low-light conditions.
UO-08	The app shall operate correctly when the phone is mounted either in landscape or portrait orientation.	SO-08-01	The system shall function correctly in both portrait and landscape orientations without restarting the process.
UO-09	All detection, alerts, and settings shall operate fully without internet connection.	SO-09-01	All monitoring and alerting features shall remain fully functional offline.
UP-01	The app shall present an intuitive, minimal interface that requires no more than two user actions to begin monitoring.	SP-01-01	The system UI shall launch within 2 seconds and maintain a response time < 200 ms for all on-screen interactions.
UP-02	During active monitoring, no manual interaction shall be required.	SP-02-01	The system shall automatically start monitoring when the driver presses "Start Session," requiring no further manual input.
UP-03	Visual, audio and vibration alerts shall be easily distinguishable and convey meaning without confusing or startling the driver.	SP-03-01	The alert subsystem shall generate visual, audio, and haptic feedback signals that are synchronized within ±50 ms.
UP-04	Interface text size, contrast, and color schemes shall conform to WCAG 2.1 AA accessibility guidelines where feasible on Android	SP-04-01	All UI text and contrast ratios shall conform to WCAG 2.1 AA standards for accessibility.
		SP-05-01	The inference module shall process ≥ 15 frames per second and issue alerts within 200

UP-05	The system shall maintain at least 15 FPS processing speed and alert latency less than or equal to 200 ms from detection to notification.		ms of detection.
		SP-05-02	During continuous operation, average CPU utilization shall not exceed 40 %, and thermal throttling shall be avoided through adaptive frame skipping.
UP-06	Model inference shall execute locally using Snapdragon CPU/GPU/NPU resources while keeping sustained CPU utilization under 40 percent	SP-06-01	Inference must execute locally using Snapdragon CPU/GPU/NPU via QNN/NNAPI, with preferred NPU then GPU, falling back to CPU only if required.
UP-07	The app shall automatically reduce inference frequency if device temperature or battery drain exceeds safe thresholds.	SP-07-01	When device temperature or battery drain exceeds safe thresholds, the app shall automatically reduce inference frequency (FPS) and/or input resolution to lower load.
UP-08	The app shall operate continuously for sessions up to two hours without crash, freeze or data loss	SP-08-01	The monitoring process shall run stably for at least 2 hours without crash, memory leak, or data loss.
UP-09	Upon hardware or model failure, the system shall suspend monitoring and inform the user rather than issue false alerts.	SP-09-01	On hardware or model failure, the system shall halt inference and display a “Safe Mode” warning.
UP-10	All captured data shall remain in app-scoped storage, no external transmission occurs unless explicitly authorized	SP-10-01	All data logs and configuration files shall be stored in app-scoped encrypted storage and deleted upon user request.
UP-11	The system shall request only essential Android permissions (Camera, Audio, Vibration, Storage) and respect user revocation at runtime.	SP-11-01	The application shall respect Android runtime permission revocations and immediately disable camera or microphone access.

Appendix U: Use Cases

Table 4.1. Use Case Index Table

Project Name: Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform				
Use Case ID	**Use Case Name**	**Level**	**Author**	**Version**
UC-001	Start Monitoring Session	Summary	Chase Tanner	0.1
UC-002	Detect and Alert Driver Drowsiness	Summary	Tarig Elamin	0.1
UC-003	Acknowledge or Snooze Alert	Summary	Tarig Elamin	0.1
UC-004	Adjust Sensitivity and Settings	Summary	Tarig Elamin	0.1
UC-005	Calibrate / Align Camera	Summary	Tarig Elamin	0.1
UC-006	View Alert Summary	Summary	Tarig Elamin	0.1
UC-007	Export Logs	Primary task	Tarig Elamin	0.1
UC-008	Manage App Deployment (System Administrator)	Subfunction	Tarig Elamin	0.1
UC-009	View Aggregated Reports (Fleet Manager)	Summary	Tarig Elamin	0.1
UC-010	Receive Distraction Alert	Summary	Tarig Elamin	0.1
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Table 4.2. Use Case UC-001

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**
Use Case ID:	UC-001
Use Case Name:	Start Monitoring Session
User Goal:	The driver wants to manually start and stop monitoring a session so they can control when the app analyzes their driving behavior
Scope:	Driver Monitoring Sub-System
Level:	Summary
Relevant User Reqs:	UF-B,UF-E,UF-H
Relevant System Reqs:	SF-B-01,SF-B-02,SF-C-01
Primary Actor:	Driver (mobile app user)
Precondition:	The ALVION app is installed and permissions (camera, audio, vibration) are granted. User is logged in
Minimal Guarantee:	If camera permission is denied or initialization fails, the app informs the user and remains idle
Success Guarantee:	Monitoring session begins successfully; live interface runs at >= 15 FPS, analyzing eye openness and head orientation.
Trigger:	Driver taps "Start Monitoring" on main dashboard screen
Success Scenario:	**Step** / **Actions** **1** The user launches the ALVION application **2** The system request camera, audio, and vibration permissions if not already granted **3** The user taps "Start Monitoring" **4** The system activates the front-facing camera via CameraX **5** The system initializes the on-device ML inference engine using Snapdragon CPU/GPU/NPU **6** The system calibrates the frame to ensure lighting and positioning are valid **7** The system begins continuous real-time analysis of eye openness and head orientation **8** The system triggers an alert if drowsiness or distraction is detected **9** The user can manually tap "Stop Monitoring" to end the session **10** The system records the session timestams and saves the results locally
Extensions:	**Branching Scenarios**
3A	**Condition:** Camera permission denied
	Step / **Actions** **1** The system displays error "Camera access is required to begin monitoring". Disable Start button until granted
3B	**Condition:** Calibration fails (low lighting)

	Step **Actions** **1** The system Prompt user "Lighting insufficient -- please adjust position or brightness".
3C	**Condition:** Hardware overload (temperature threshold exceeded)
	Step **Actions** **1** The system notifies driver "Monitoring paused to cool device".
3D	**Condition:** User revokes permission during session
	Step **Actions** **1** The system suspends monitoring safely and informs the driver
Acknowledgment: Generated from the CapStone process management system ©2025	

Table 4.3. Use Case UC-002

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**
Use Case ID:	UC-002
Use Case Name:	Detect and Alert Driver Drowsiness
User Goal:	The driver wants the system to monitor their face and eyes in real time and warn them immediately if signs of drowsiness are detected, helping them stay alert and safe.
Scope:	Distracted/Drowsy Driver Detection System (the in-
Level:	Summary
Relevant User Reqs:	UF-B
Relevant System Reqs:	SF-B-01,SF-B-02
Primary Actor:	Driver
Precondition:	1- The system is installed and running. 2- The camera is calibrated, and lighting conditions allow detection. 3- Driver monitoring mode is active.
Minimal Guarantee:	If sensors fail or detection confidence is too low, the system logs the issue and notifies the driver that monitoring accuracy is reduced
Success Guarantee:	If drowsiness is detected, the system immediately issues audible and visual alerts, prompting the driver to regain focus
Trigger:	The system detects drowsiness indicators (e.g., eyes closed for prolonged periods or slow blink rate)
Success Scenario:	**Step** / **Actions** **1** The system continuously monitors the driver’s eyes and facial expressions using the camera. **2** The system analyzes blink frequency, eyelid closure duration, and gaze direction in real time. **3** The system the system classifies the state as “drowsy.” **4** The system the system triggers audible and visual alerts **5** The user hears/see the alert and adjusts posture, eye focus, or takes a break. **6** The system monitoring continues once the driver’s state normalizes.
Extensions:	**Branching Scenarios**
1A	**Condition:** The Driver continues to show "drowsy" symptoms for a long pried of time
	Step / **Actions** **1** The system sends notification to family members and emergency contacts
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Table 4.4. Use Case UC-003

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**
Use Case ID:	UC-003
Use Case Name:	Acknowledge or Snooze Alert
User Goal:	The driver acknowledges an alert or snoozes further alerts for a short cooldown.
Scope:	Driver Monitoring Sub-System
Level:	Summary
Relevant User Reqs:	UF-D,UF-F,UF-G,UF-I,UF-J
Relevant System Reqs:	SF-A-01,SF-B-01,SF-D-01
Primary Actor:	Driver
Precondition:	A drowsiness or distraction alert has just been displayed.
Minimal Guarantee:	If input is not received, alert auto-dismisses after timeout; monitoring continues.
Success Guarantee:	System records the user action and applies cooldown if snoozed.
Trigger:	Driver taps "Acknowledge" or "Snooze".
Success Scenario:	**Step** **Actions** **1** The system presents actions: Acknowledge | Snooze. **2** The Driver selects an action.
Extensions:	**Branching Scenarios**
1A	**Condition:** If Acknowledge
	Step **Actions** **1** The system System clears alert UI and logs acknowledgment. **2** The system updates session stats.
1B	**Condition:** If Snooze
	Step **Actions** **1** System starts cooldown timer; suppresses new alerts. **2** The system updates session stats.
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Table 4.5. Use Case UC-004

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**
Use Case ID:	UC-004
Use Case Name:	Adjust Sensitivity and Settings
User Goal:	The driver adjusts thresholds (eye-closure time, yaw angle) and alert cooldown/volume; changes persist locally between sessions.
Scope:	Settings Sub-System
Level:	Summary
Relevant User Reqs:	UF-D
Relevant System Reqs:	SF-D-01,SF-D-02,SF-D-03
Primary Actor:	Driver
Precondition:	App is installed; monitoring may be idle or running.
Minimal Guarantee:	Invalid inputs are rejected with guidance; previous values remain.
Success Guarantee:	New settings are validated, saved, and hot-applied to active monitoring.
Trigger:	Driver opens Settings.
Success Scenario:	**Step** **Actions** **1** The user navigates to Settings. **2** The system loads current Settings (thresholds, cooldown, volume) and displays controls. **3** The user edits one or more values (e.g., PERCLOS window, yaw threshold, alert volume/cooldown). **4** The system validates ranges and dependencies. **5** The system persists updates locally and, if monitoring is active, hot-applies them to detection/alerts. **6** The system confirms “Settings saved.”
Extensions:	**Branching Scenarios**
3A	**Condition:** Out-of-range value
	Step **Actions** **1** The system clamps or shows inline error; cannot save until valid.
5A	**Condition:** Restore defaults
	Step **Actions** **1** The user taps Restore; system resets to defaults and saves.
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Table 4.6. Use Case UC-005

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**
Use Case ID:	UC-005
Use Case Name:	Calibrate / Align Camera
User Goal:	Align the camera/phone so the face is centered and lighting is sufficient before monitoring.
Scope:	Calibration Sub-System
Level:	Summary
Relevant User Reqs:	UF-H
Relevant System Reqs:	SF-H-01,SF-H-02
Primary Actor:	Driver
Precondition:	Camera permission granted; preview available.
Minimal Guarantee:	If calibration fails, user receives guidance and can retry or skip.
Success Guarantee:	Calibration passes; app allows monitoring to start (or continues with improved alignment).
Trigger:	First use, or system detects poor alignment/lighting; driver taps Start Calibration.
Success Scenario:	**Step** **Actions** **1** System shows live preview with guidance overlays (face box/eye markers). **2** System evaluates framing stability and pose (e.g., yaw/pitch within bounds) and lighting quality. **3** When all criteria hold for the window, system marks Calibration Passed. **4** System returns to main screen with Start Monitoring enabled (or auto-continues if already starting).
Extensions:	**Branching Scenarios**
1A	**Condition:** User skips
	Step **Actions** **1** The system Start remains allowed but with degraded confidence notice until conditions improve
2A	**Condition:** Insufficient lighting/unstable face
	Step **Actions** **1** Show tips (increase brightness, hold steady, reposition); remain in calibration.
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Table 4.7. Use Case UC-006

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**
Use Case ID:	UC-006
Use Case Name:	View Alert Summary
User Goal:	After a session, the driver reviews counts by alert type, timeline, duration, and average PERCLOS.
Scope:	Reporting (Local)
Level:	Summary
Relevant User Reqs:	UF-I
Relevant System Reqs:	SF-I-01,SF-I-02
Primary Actor:	
Precondition:	At least one session ended with recorded events.
Minimal Guarantee:	If no events exist, show “No alerts recorded” with session duration.
Success Guarantee:	Summary displays key metrics from local storage and supports time-range filters.
Trigger:	Driver taps Summary.
Success Scenario:	**Step** **Actions** **1** Driver selects Last Session, Today, 7 Days, or Custom. **2** The system loads session data locally and aggregates counts, durations, and PERCLOS. **3** The system renders timeline and KPI cards. **4** The user closes summary or proceeds to Export.
Extensions:	**Branching Scenarios**
2A	**Condition:** Data missing/corrupted
	Step **Actions** **1** The system shows error and offers diagnostics/logs link.
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Table 4.8. Use Case UC-007

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**
Use Case ID:	UC-007
Use Case Name:	Export Logs
User Goal:	Export anonymized summaries (and optional event logs) for external analysis—offline and consented.
Scope:	Reporting (Local Export)
Level:	Primary task
Relevant User Reqs:	UF-I,UF-J
Relevant System Reqs:	SF-J-03
Primary Actor:	Driver
Precondition:	Local session data exists; user consents to export.
Minimal Guarantee:	If storage permission denied or canceled, no file is written.
Success Guarantee:	A CSV/JSON file is created with aggregated/session fields—no PII or images.
Trigger:	Driver taps Export from Summary or Settings.
Success Scenario:	**Step** **Actions** **1** The system presents consent text and lets the user choose CSV or JSON and a destination. **2** The system serializes summaries (and optional event rows) and writes to selected location. **3** The system confirms success and offers Open
Extensions:	**Branching Scenarios**
1A	**Condition:** User cancels/denies permission
	Step **Actions** **1** The system Abort export; show guidance.
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Table 4.9. Use Case UC-008

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**
Use Case ID:	UC-008
Use Case Name:	Manage App Deployment (System Administrator)
User Goal:	Deploy updates, enforce permissions, and apply managed configurations uniformly across devices.
Scope:	Device Management / Deployment
Level:	Subfunction
Relevant User Reqs:	UF-K
Relevant System Reqs:	SF-K-01,SF-K-02
Primary Actor:	System Administrator
Precondition:	Devices are enrolled in enterprise MDM; app is approved in store.
Minimal Guarantee:	If a policy cannot be applied, admin is notified with reason.
Success Guarantee:	Target devices receive the assigned version and enforced settings; status is auditable.
Trigger:	Admin schedules a rollout or pushes a policy refresh
Success Scenario:	**Step** **Actions** **1** Admin selects device group(s) and rollout plan (staged or immediate). **2** MDM distributes the app/update and managed configs (permissions, locked settings). **3** Devices install/update and apply configs automatically. **4** App reflects managed settings and displays “Managed by your organization.” **5** Admin reviews deployment status and audit logs in the console.
Extensions:	**Branching Scenarios**
2A	**Condition:** Policy revokes CAMERA
	Step **Actions** **1** App transitions to IDLE, releases camera, shows policy notice.
3A	**Condition:** Device offline
	Step **Actions** **1** MDM queues install; applies on next check-in.
Acknowledgment: Generated from the CapStone process management system ©2025	

Table 4.10. Use Case UC-009

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**
Use Case ID:	UC-009
Use Case Name:	View Aggregated Reports (Fleet Manager)
User Goal:	View aggregated drowsiness/distraction trends across drivers to identify safety issues and inform policies.
Scope:	Fleet Reporting (Future)
Level:	Summary
Relevant User Reqs:	UF-L
Relevant System Reqs:	SF-L-01,SF-L-02,SF-L-03
Primary Actor:	Fleet Manager
Precondition:	Consented, anonymized summaries have been uploaded to an org data store.
Minimal Guarantee:	If data for a range is missing, dashboard shows “incomplete coverage.”
Success Guarantee:	Dashboard shows alerts/hour, type distribution, and time-of-day/week trends with filters and export.
Trigger:	Manager opens Fleet Dashboard and selects filters and date range.
Success Scenario:	**Step** **Actions** **1** The system aggregates data for the selected org/cohort/date range, normalized by monitored hours. **2** The system renders trend charts and KPIs (alerts/hr, type mix, peak hours). **3** The user exports CSV/PDF for sharing.
Extensions:	**Branching Scenarios**
1A	**Condition:** Small cohort (<5 drivers)
	Step **Actions** **1** The system Suppress display to protect privacy (k-anonymity).
2A	**Condition:** Backend unavailable
	Step **Actions** **1** Serve last-known aggregates with a stale badge.
Acknowledgment: Generated from the CapStone process management system ©2025	

Table 4.11. Use Case UC-010

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**
Use Case ID:	UC-010
Use Case Name:	Receive Distraction Alert
User Goal:	The driver is alerted when their head remains turned away from the forward direction longer than the allowed threshold, so they can refocus attention.
Scope:	Driver Monitoring Sub-System
Level:	Summary
Relevant User Reqs:	UF-C
Relevant System Reqs:	SF-C-02,SF-F-02
Primary Actor:	Driver
Precondition:	Monitoring session is active; camera/inference running; thresholds configured. (See UC-001 preconditions for session start details.)
Minimal Guarantee:	If face/pose confidence is low, the system withholds the alert and logs the reason.
Success Guarantee:	A distraction alert (visual + configured audio/vibration) is issued and recorded with timestamp and metrics.
Trigger:	Head yaw remains beyond the configured threshold for longer than the dwell time.
Success Scenario:	**Step** **Actions** **1** System continuously estimates head yaw from the face-detection model. **2** System accumulates time while |yaw| exceeds the threshold. **3** When dwell time is reached, system creates a DISTRACTED event (with yaw peak, duration, confidence). **4** The System issues a real-time alert (visual banner + audio/vibration per settings). **5** The System logs the event in the current session for later summary/reporting.
Extensions:	**Branching Scenarios**
2A	**Condition:** Low confidence / face not detected
	Step **Actions** **1** The system pause dwell timer; show non-blocking tip (e.g., “reposition/brighten”); no alert.
4A	**Condition:** Alerts snoozed
	Step **Actions** **1** The system suppress during cooldown; log suppression.
Acknowledgment: Generated from the CapStone process management system ©2025	

Appendix T: Test Cases

Table 8.2.1. Test Suite TS-001 (Functional Testing): StartScreenTest

Test Case ID	Testing Type	Test Case Description	Tested
TC-001	Object Interface Testing	Verify that the "Start Session" button is displayed on StartScreen and that clicking it invokes the onStart callback.	Yes

Table 8.2.2. Test Suite TS-002 (Functional Testing): SessionScreenTest

Test Case ID	Testing Type	Test Case Description	Tested
TC-004	Object Interface Testing	Verify that the "End Session" button is displayed on SessionScreen and that clicking it invokes the onEnd callback.	Yes
TC-006	Object Interface Testing	Verify that tapping the "Notify" chip opens the notification dialog and that the dialog text correctly reflects the number of times the chip has been tapped.	Yes
TC-011	Object Interface Testing	Verify that tapping the "Sound" chip opens the sound dialog and that selecting "Turn off sound" triggers the stop/reset behavior for the sound and closes the dialog.	Yes
TC-012	Object Interface Testing	Verify that the "Emergency" card is displayed on SessionScreen and is clickable, confirming that the emergency action is wired to a tap handler.	Yes
TC-013	Object Interface Testing	Verify that makeEmergencyCall creates an ACTION_CALL intent with the correct "tel:" URI for the emergency number and passes it to context.startActivity().	Yes

Table 8.2.3. Test Case TC-001

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Suite	TS-001 (Functional Testing): StartScreenTest	
Test Case ID	TC-001 (Object Interface Testing)	
What To Test	Verify that the "Start Session" button is displayed on StartScreen and that clicking it invokes the onStart callback.	
Test Data Input	taps the “Start Session” button on StartScreen.	
Expected Result	onStart callback is invoked and a monitoring session begins.	
Traceability	**Relevant User Req.(s)**	UF-B
	Relevant System Req.(s)	SF-B-01, SF-B-02
	Relevant Use Case(s)	UC-001
Acknowledgment: Generated from the CapStone process management system ©2025		

Table 8.2.4. Test Case TC-004

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Suite	TS-002 (Functional Testing): SessionScreenTest	
Test Case ID	TC-004 (Object Interface Testing)	
What To Test	Verify that the "End Session" button is displayed on SessionScreen and that clicking it invokes the onEnd callback.	
Test Data Input	Taps the “End Session” button on SessionScreen.	
Expected Result	The onEnd callback is invoked and the current monitoring session ends	
Traceability	**Relevant User Req.(s)**	UF-B
	Relevant System Req.(s)	SF-B-02
	Relevant Use Case(s)	UC-001
Acknowledgment: Generated from the CapStone process management system ©2025		

Table 8.2.5. Test Case TC-006

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Suite	TS-002 (Functional Testing): SessionScreenTest	
Test Case ID	TC-006 (Object Interface Testing)	
What To Test	Verify that tapping the "Notify" chip opens the notification dialog and that the dialog text correctly reflects the number of times the chip has been tapped.	
Test Data Input	User taps the “Notify” chip twice in the same session	
Expected Result	A dialog appears showing the notification text and correctly displays “Notify tapped 2 times	
Traceability	**Relevant User Req.(s)**	UF-D, UF-F, UF-G, UF-I, UF-J
	Relevant System Req.(s)	SF-A-01, SF-B-01, SF-D-01
	Relevant Use Case(s)	UC-003
Acknowledgment: Generated from the CapStone process management system ©2025		

Table 8.2.6. Test Case TC-011

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Suite	TS-002 (Functional Testing): SessionScreenTest	
Test Case ID	TC-011 (Object Interface Testing)	
What To Test	Verify that tapping the "Sound" chip opens the sound dialog and that selecting "Turn off sound" triggers the stop/reset behavior for the sound and closes the dialog.	
Test Data Input	User taps the “Sound” chip and then selects “Turn off sound” in the dialog	
Expected Result	The sound stops and the sound dialog is closed	
Traceability	**Relevant User Req.(s)**	UF-D
	Relevant System Req.(s)	SF-D-01, SF-D-02, SF-D-03
	Relevant Use Case(s)	UC-004
Acknowledgment: Generated from the CapStone process management system ©2025		

Table 8.2.7. Test Case TC-012

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Suite	TS-002 (Functional Testing): SessionScreenTest	
Test Case ID	TC-012 (Object Interface Testing)	
What To Test	Verify that the "Emergency" card is displayed on SessionScreen and is clickable, confirming that the emergency action is wired to a tap handler.	
Test Data Input	User taps the “Emergency” card on SessionScreen	
Expected Result	The emergency tap handler runs (emergency action is triggered) without crashing	
Traceability	**Relevant User Req.(s)**	UF-B, UF-C
	Relevant System Req.(s)	SF-C-02, SF-F-02
	Relevant Use Case(s)	UC-002
Acknowledgment: Generated from the CapStone process management system ©2025		

Table 8.2.8. Test Case TC-013

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Suite	TS-002 (Functional Testing): SessionScreenTest	
Test Case ID	TC-013 (Object Interface Testing)	
What To Test	Verify that makeEmergencyCall creates an ACTION_CALL intent with the correct "tel:" URI for the emergency number and passes it to context.startActivity().	
Test Data Input	Emergency phone number 9513034883	
Expected Result	The system starts a phone call to 9513034883 using an ACTION_CALL intent	
Traceability	**Relevant User Req.(s)**	UF-B
	Relevant System Req.(s)	SF-B-01, SF-B-02
	Relevant Use Case(s)	UC-002
Acknowledgment: Generated from the CapStone process management system ©2025		

Appendix TE: Test Execution Report

Table 8.3.1. Execution Report of Test Case TC-001

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**
Test Case ID:	TC-001
Testing Tools Used:	Android Studio, JUnit 4, Jetpack Compose UI Test
Execution Steps:	

Test Execution Records:

#	Tester	Test Date	Test Stage	Actual Result	Status	Defect	Correction
1	Tariq	11/06/2025	Unit	The "Start Session" button was visible and tapping it invoked the onStart callback, starting the monitoring session	Pass	None	null by N/A

Execution Summary:	TC-001 passed. StartScreen correctly shows the "Start Session" button and the click handler starts a session without errors

Table 8.3.2. Execution Report of Test Case TC-004

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**
Test Case ID:	TC-004
Testing Tools Used:	Android Studio, JUnit 4, Jetpack Compose UI Test
Execution Steps:	1 Open SessionScreen and tap the "End Session" button once.

Test Execution Records:

#	Tester	Test Date	Test Stage	Actual Result	Status	Defect	Correction
1	Tariq	11/06/2025	Unit	"End Session" button was visible and tapping it invoked the onEnd callback, ending the current session	Pass	N/A	

Execution Summary:	TC-004 passed. The session can be ended from SessionScreen and the app returns cleanly without crashes

Table 8.3.3. Execution Report of Test Case TC-006

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Case ID:	TC-006	
Testing Tools Used:	Android Studio, JUnit 4, Jetpack Compose UI Test	
Execution Steps:	1	On SessionScreen, tap the "Notify" chip twice and observe the dialog text each time

Test Execution Records:

#	Tester	Test Date	Test Stage	Actual Result	Status	Defect	Correction
1	Tariq	11/06/2025	Unit	First tap opened a dialog showing "Notification sent" and "Notify tapped 1 time."; second tap showed "Notify tapped 2 times." as expected	Pass	N/A	null by N/A

Execution Summary:	TC-006 passed. Notify chip correctly opens the dialog and maintains the tap count across multiple presses

Table 8.3.4. Execution Report of Test Case TC-011

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Case ID:	TC-011	
Testing Tools Used:	Android Studio, JUnit 4, Jetpack Compose UI Test	
Execution Steps:	1	On SessionScreen, tap the "Sound" chip, then select "Turn off sound" in the dialog.

Test Execution Records:

#	Tester	Test Date	Test Stage	Actual Result	Status	Defect	Correction
1	Tariq	11/10/2025	Unit	Tapping "Sound" opened the sound dialog ("Playing until you turn it off"); choosing "Turn off sound" stopped the sound and closed the dialog	Pass		

Execution Summary:	TC-011 passed. Sound chip dialog works correctly and the user can turn off the alert sound without issues

Table 8.3.5. Execution Report of Test Case TC-012

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**
Test Case ID:	TC-012
Testing Tools Used:	Android Studio, JUnit 4, Jetpack Compose UI Test
Execution Steps:	1 — On SessionScreen, tap the "Emergency" card once.

Test Execution Records:

#	Tester	Test Date	Test Stage	Actual Result	Status	Defect	Correction
1	Tariq	11/16/2025	Unit	The "Emergency" card was visible and responded to the tap without any crashes or UI errors, confirming the handler is wired	Pass		

Execution Summary:	TC-012 passed. Emergency card is clickable and correctly connected to its tap handler

Table 8.3.6. Execution Report of Test Case TC-013

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**
Test Case ID:	TC-013
Testing Tools Used:	Android Studio, JUnit 4, Mockito
Execution Steps:	1 Tap Emergency call

Test Execution Records:

#	Tester	Test Date	Test Stage	Actual Result	Status	Defect	Correction
1	Tariq	11/19/2025	Unit	ACTION_CALL intent with data tel:9513034883 and passed it to	Pass		

Execution Summary:	TC-013 passed. Emergency call helper correctly launches a phone call to the configured emergency number

Table 8.3.7. Execution Report of Test Case TC-014

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Case ID:	TC-014	
Testing Tools Used:	JUnit 4	
Execution Steps:	1	Open FaceLogicTest
	2	Run computeImageSize_rotation0_keepsSize()
	3	ImageSizeCalculator.compute(rotation = 0, width = 640, height = 480)
	4	Verify the returned width is 640
	5	Verify the returned height is 480

Test Execution Records:

#	Tester	Test Date	Test Stage	Actual Result	Status	Defect	Correction
1	Tariq	02/20/2026	Unit	The method returned width 640 and height 480 when rotation was 0	Pass	None	null by None

Execution Summary:	Verified that when rotation is 0, the image size remains unchanged.

Table 8.3.8. Execution Report of Test Case TC-016

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Case ID:	TC-016	
Testing Tools Used:	JUnit 4, Kotlin unit test framework, MockK	
Execution Steps:	1	Open FaceLogicTest.
	2	Run evaluator_triggersDrowsy_afterTimeThreshold()
	3	Create a FaceStateEvaluator with a drowsy callback counter.
	4	Enable monitoring.
	5	Evaluate one face with both eyes open.
	6	Evaluate closed-eye frames as time increases.
	7	Let the closed-eye duration pass the drowsy threshold.
	8	Verify that the drowsy callback count becomes 1.

Test Execution Records:

#	Tester	Test Date	Test Stage	Actual Result	Status	Defect	Correction

Execution Summary:	Verified that the evaluator correctly triggers a drowsy alert after eye closure lasts longer than the allowed threshold.

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Table 8.3.9. Execution Report of Test Case TC-017

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**
Test Case ID:	TC-017
Testing Tools Used:	JUnit 4, Kotlin unit test framework

Execution Steps:		
	1	Run computeImageSize_rotation90_swapsSize()
	2	Call computeImageSize with rotation 90, width 640, and height 480
	3	Check that the returned width is 480
	4	Check that the returned height is 640

Test Execution Records:

#	Tester	Test Date	Test Stage	Actual Result	Status	Defect	Correction

Execution Summary:	Verified that when the image rotation is 90 degrees, computeImageSize correctly swaps the width and height

Table 8.3.10. Execution Report of Test Case TC-019

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Case ID:	TC-019	
Testing Tools Used:	JUnit 4, Kotlin unit test framework	
Execution Steps:	1	Open FaceLogicTest
	2	Run computeImageSize_rotation270_swapsSize()
	3	Call computeImageSize with rotation 270, width 640, and height 480
	4	Check that the returned width is 480
	5	Check that the returned height is 640

Test Execution Records:

#	Tester	Test Date	Test Stage	Actual Result	Status	Defect	Correction

Execution Summary:	Verified that when the image rotation is 270 degrees, computeImageSize correctly swaps the width and height.

Table 8.3.11. Execution Report of Test Case TC-020

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Case ID:	TC-020	
Testing Tools Used:	JUnit 4, Kotlin unit test framework, MockK	
Execution Steps:	1	Run evaluator_doesNotTriggerDrowsy_beforeThreshold()
	2	Create a FaceStateEvaluator with a drowsy callback counter.
	3	Enable monitoring.
	4	Evaluate one face with both eyes open.
	5	Evaluate closed-eye frames as time increases.
	6	Keep the elapsed closed-eye time below the drowsy threshold.
	7	Verify that the drowsy callback count remains 0.

Test Execution Records:

#	Tester	Test Date	Test Stage	Actual Result	Status	Defect	Correction

Execution Summary:	Verified that the evaluator does not trigger a drowsy alert when eye closure has not lasted long enough to cross the threshold.

Table 8.3.12. Execution Report of Test Case TC-021

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Case ID:	TC-021	
Testing Tools Used:	JUnit 4, Kotlin unit test framework, MockK	
Execution Steps:	1	evaluator_drowsy_firesOnlyOnce_forSameClosure()
	2	Create a FaceStateEvaluator with a drowsy callback counter
	3	Enable monitoring.
	4	Evaluate one face with both eyes open.
	5	Evaluate closed-eye frames until the drowsy threshold is exceeded.
	6	Verify the callback fires the first time.
	7	Keep evaluating closed-eye frames without reopening the eyes.
	8	Verify the callback count stays at 1.

Test Execution Records:

#	Tester	Test Date	Test Stage	Actual Result	Status	Defect	Correction

Execution Summary:	The drowsy callback was triggered once and did not fire again during the same continuous eye closure.

Table 8.3.13. Execution Report of Test Case TC-022

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Case ID:	TC-022	
Testing Tools Used:	JUnit 4, Kotlin unit test framework, MockK	
Execution Steps:	1	Open FaceLogicTest.
	2	Run evaluator_drowsy_canFireAgain_afterEyesReopen()
	3	Create a FaceStateEvaluator with a drowsy callback counter.
	4	Enable monitoring.
	5	Evaluate one face with both eyes open.
	6	Evaluate closed-eye frames until the drowsy threshold is exceeded and the callback fires once.
	7	Evaluate a frame where the eyes reopen to reset the drowsy state.
	8	Evaluate closed-eye frames again until the threshold is exceeded a second time.
	9	Verify the callback count becomes 2

Test Execution Records:

#	Tester	Test Date	Test Stage	Actual Result	Status	Defect	Correction

Execution Summary:	Verified that the evaluator resets after the eyes reopen and can issue a new drowsy alert for a later eye-closure event

Table 8.3.14. Execution Report of Test Case TC-023

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Case ID:	TC-023	
Testing Tools Used:	JUnit 4, Kotlin unit test framework, MockK	
Execution Steps:	1	Open FaceLogicTest.
	2	Run evaluator_triggersDistracted_afterTimeThreshold().
	3	Create a FaceStateEvaluator with a distracted callback counter.
	4	Start calibration and set the calibration target to forward.
	5	Evaluate several forward-facing frames to establish the baseline.
	6	Finish calibration successfully.
	7	Enable monitoring.
	8	Evaluate a face turned far away from the forward position.
	9	Increase time past the distraction threshold and evaluate again.
	10	Verify that the distracted callback count becomes 1.

Test Execution Records:

#	Tester	Test Date	Test Stage	Actual Result	Status	Defect	Correction

Execution Summary:	Verified that the evaluator correctly detects distraction and triggers the callback after the head stays turned away beyond the time threshold.

Table 8.3.15. Execution Report of Test Case TC-024

Project Name:	**Distracted/Drowsy Driver Detection on Snapdragon Mobile Platform**	
Test Case ID:	TC-024	
Testing Tools Used:	JUnit 4, Kotlin unit test framework, MockK	
Execution Steps:	1	Open FaceLogicTest.
	2	Run evaluator_doesNotTriggerDistracted_beforeThreshold().
	3	Create a FaceStateEvaluator with a distracted callback counter.
	4	Start calibration and set the calibration target to forward.
	5	Evaluate several forward-facing frames to establish the baseline.
	6	Finish calibration successfully.
	7	Enable monitoring.
	8	Evaluate a face turned far away from the forward position.
	9	Increase time, but keep it below the distraction threshold.
	10	Evaluate again and verify that the distracted callback count remains 0.

Test Execution Records:

#	Tester	Test Date	Test Stage	Actual Result	Status	Defect	Correction

Execution Summary:	Verified that the evaluator does not issue a distracted alert before the distraction time threshold is reached

Appendix PM: Project Management

This appendix provides evidence of the team's project management process throughout the ALVION capstone project. The evidence includes GitHub issue tracking, pull request records, branch-based development, task labels, individual contribution records, CI/CD work, documentation updates, testing tasks, and implementation history.

The purpose of this appendix is to show that the team followed an organized development process across the full project lifecycle. Work was not only completed at the end of the project; it was planned, assigned, reviewed, tested, revised, and documented across multiple project phases.

PM.1 Project Management Overview

The ALVION team used GitHub as the primary platform for project management, version control, and development coordination. GitHub Issues were used to define tasks, track feature requests, document bugs, organize testing work, and manage report/documentation updates. GitHub Pull Requests were used to review and merge implementation changes, documentation updates, CI/CD improvements, and bug fixes.

The team organized work using issue labels such as *new feature*, *enhancement*, *bug*, *documentation*, *test*, and *refactor*. These labels helped separate implementation tasks from testing, documentation, debugging, and architecture improvement work. This made it easier for team members to understand the purpose and status of each task.

The repository also included a documented collaboration workflow for the report and documentation folder. This workflow instructed contributors to work on feature branches, build the LaTeX report locally, clean auxiliary files before committing, push branches, and open pull requests with clear summaries and checklists. This process helped reduce broken builds, accidental commits of generated files, and unclear changes.

Overall, the project management process supported task allocation, progress tracking, code review, documentation review, and quality control. The following sections provide screenshots and repository records that demonstrate how these practices were used throughout the ALVION project.

PM.2 Development Workflow and Collaboration Process

The team maintained a documented workflow for collaborating on the ALVION report and repository. This workflow explained how contributors should create feature branches, update the design document, add or revise assets, build the LaTeX report locally, clean generated auxiliary files, commit only intended source changes, push branches, and open pull requests.

This process helped prevent direct edits to the main branch and encouraged each contributor to isolate their work in a separate branch. It also required contributors to verify that the report could build before opening a pull request. This reduced the chance of broken LaTeX output, missing images, incorrect references, or unnecessary build artifacts being merged into the repository.

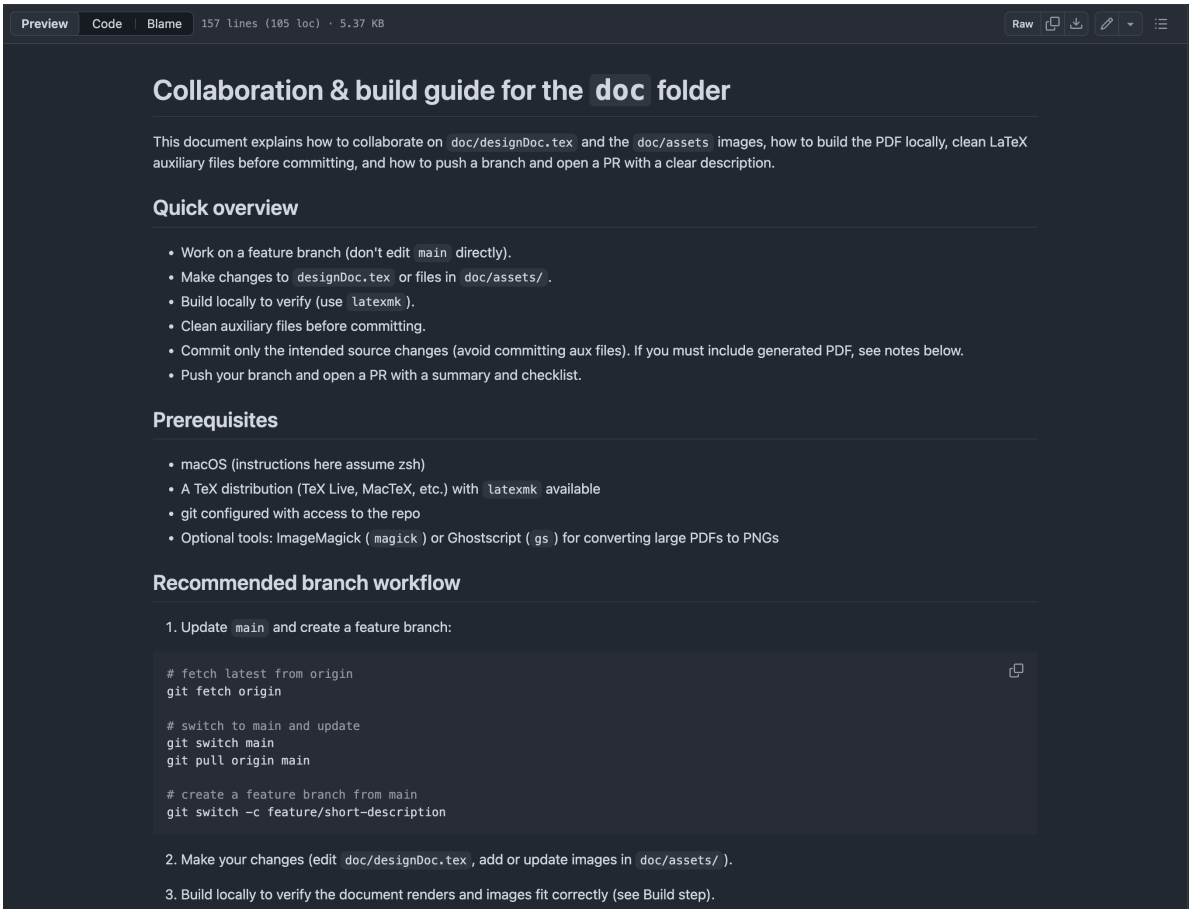

Figure 12.1: Repository documentation showing the team's collaboration and build workflow for the report folder.

Figure 12.1 shows the team's documented workflow for working on the report. The guide instructs contributors to use feature branches, update `designDoc.tex` or files in `doc/assets/`, build locally with `latexmk`, clean auxiliary files, and open pull requests with summaries and checklists.

The workflow also supported review and accountability. Pull requests were expected to include a clear summary and checklist so reviewers could understand the purpose of the change, confirm that the correct files were updated, and verify that the document or code still worked as expected.

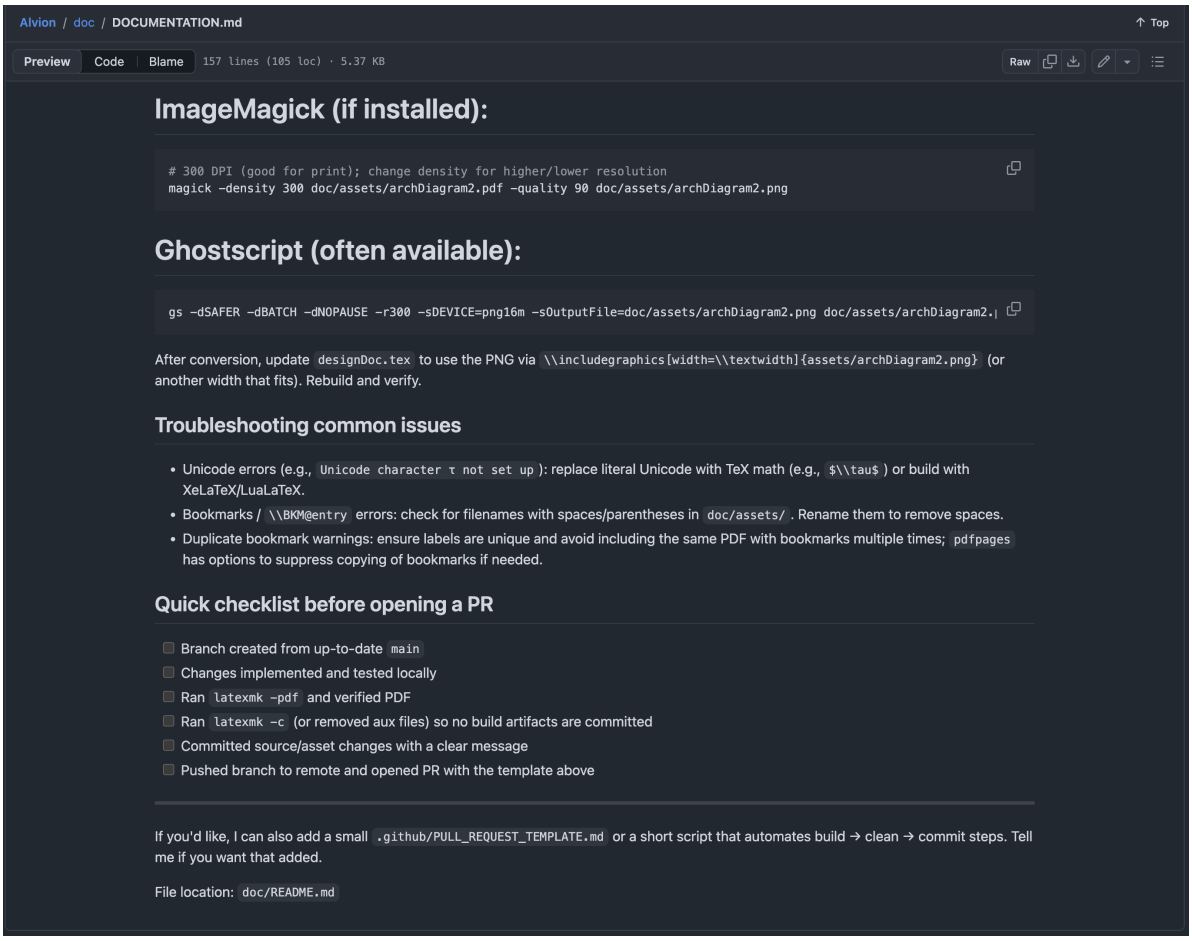

Figure 12.2: Repository documentation showing the checklist used before opening a pull request.

Figure 12.2 shows the checklist used before opening a pull request. The checklist required contributors to branch from an up-to-date main branch, test changes locally, run the LaTeX build, clean auxiliary files, commit source or asset changes with a clear message, and push the branch before opening a pull request.

PM.3 GitHub Issue Tracking Evidence

GitHub Issues were used as a primary task management tool throughout the ALVION project. The issue tracker allowed the team to create work items, categorize them with labels, associate them with contributors, track discussion, and close tasks after completion. This gave the team a visible record of planned work, active work, completed work, bugs, documentation tasks, testing tasks, and feature development.

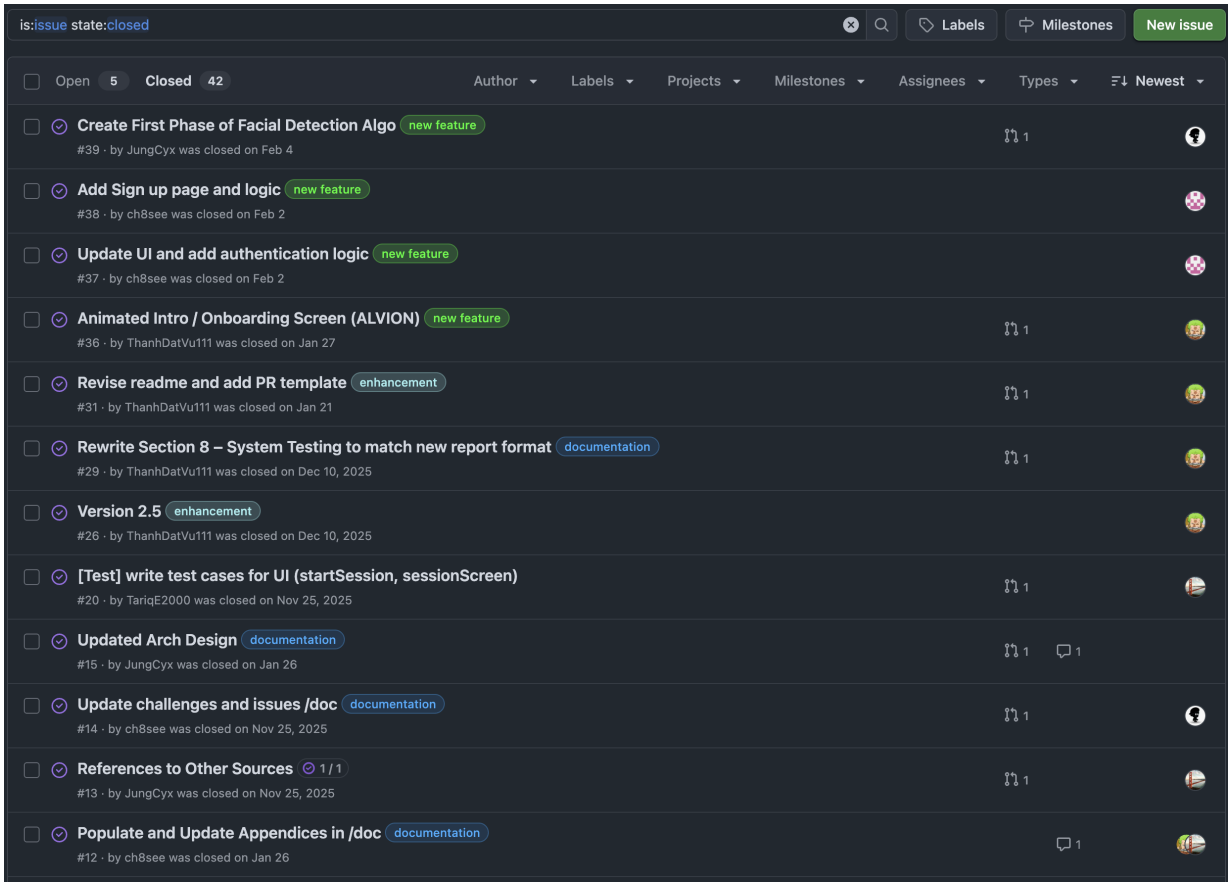

Figure 12.3: GitHub closed issue overview showing 42 closed issues, 5 open issues, labels, authors, dates, and task activity across the project.

Figure 12.3 shows the repository's closed issue tracker. The screenshot includes issue labels such as *new feature*, *enhancement*, and *documentation*, along with issue authors, close dates, comments, and linked development activity. This demonstrates that the team tracked work items in GitHub rather than relying only on informal communication.

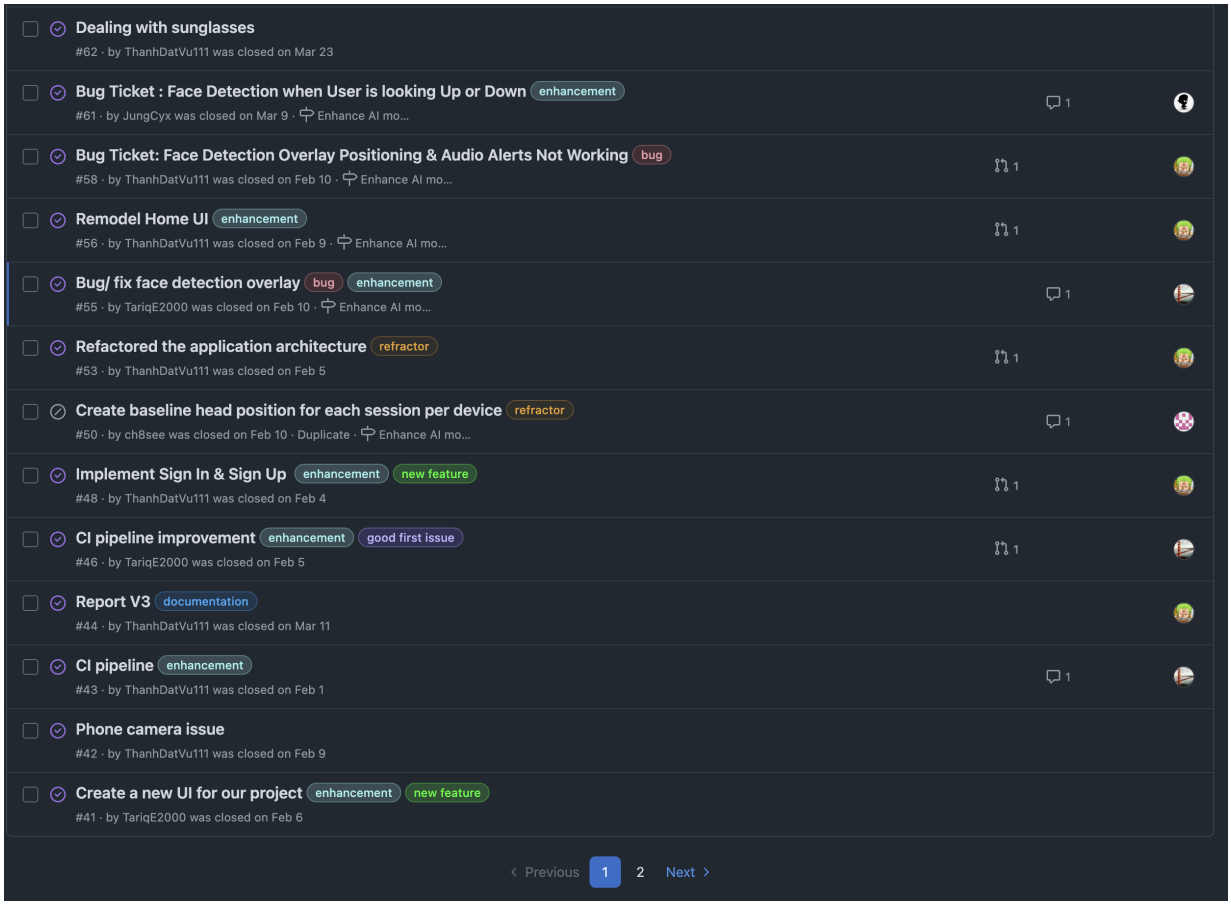

Figure 12.4: GitHub issues showing implementation, bug-fixing, refactoring, authentication, CI pipeline, and UI development tasks.

Figure 12.4 shows issue records from active implementation phases. These issues include face detection bug fixes, application architecture refactoring, sign-in and sign-up implementation, CI pipeline improvements, camera issues, and UI redesign. This evidence shows that the team tracked technical implementation work, defects, and process improvements through GitHub Issues.

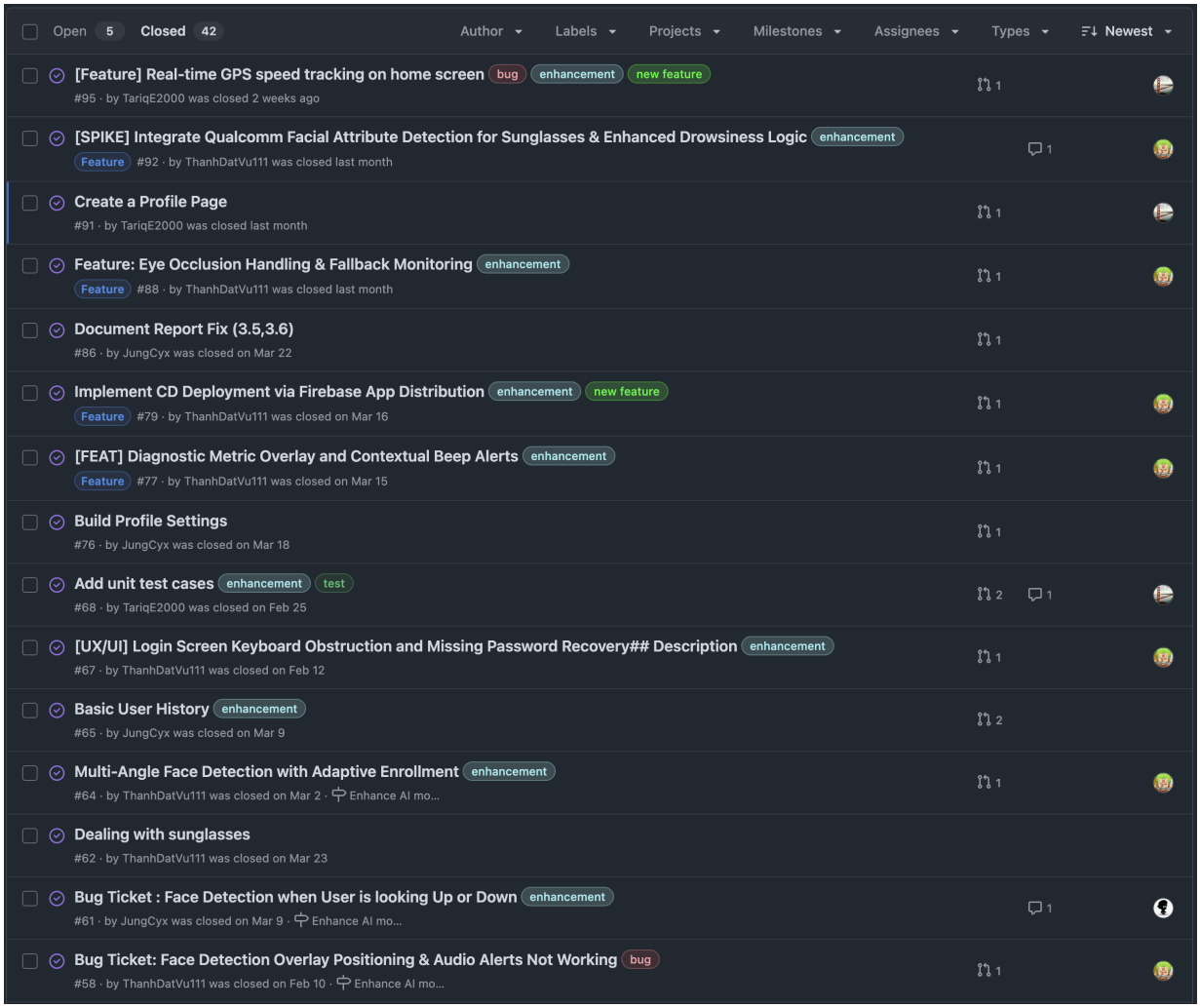

Figure 12.5: GitHub issues showing later-stage feature work, deployment work, profile work, testing work, and AI robustness improvements.

Figure 12.5 shows later-stage issue tracking. The issues include real-time GPS speed tracking, Qualcomm facial attribute detection, profile page creation, eye occlusion handling, Firebase App Distribution deployment, diagnostic overlays, profile settings, unit test cases, user history, and multi-angle face detection. This demonstrates that the team continued using project management records through the later phases of development.

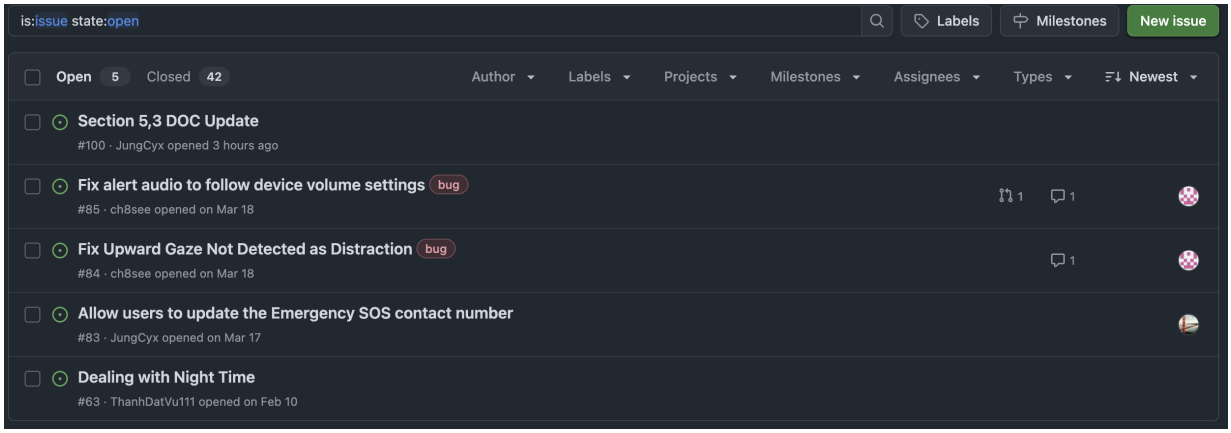

Figure 12.6: GitHub open issue list showing remaining work items and unresolved bugs tracked by the team.

Figure 12.6 shows the open issue list near the end of the project. These open issues include remaining documentation updates, audio behavior fixes, upward gaze detection, emergency contact updates, and night-time handling. This shows that the team tracked both completed tasks and remaining work, which is important for project status visibility.

Overall, the issue tracker provides evidence of project management across multiple categories of work. The team used issues to organize feature development, bug resolution, documentation, testing, refactoring, CI/CD improvements, and unresolved follow-up tasks throughout the project lifecycle.

PM.4 Task Allocation and Individual Contributions

The repository history provides evidence of individual task allocation through pull request authorship, issue authorship, branch names, review requests, issue references, and merged implementation records. While some tasks were discussed and completed collaboratively, the GitHub records show clear ownership patterns for major work areas.

Team Member	**Primary Contribution Areas**	**Repository Evidence**
Thanh Dat Vu	Onboarding UI, Firebase authentication, app architecture refactoring, Home dashboard redesign, calibration and diagnostic overlay improvements, CI/CD deployment, final UI redesign, and report updates.	Pull requests show work on the onboarding intro screen, Firebase Auth, architecture refactor, Home UI redesign, diagnostic overlay/contextual alerts, Firebase App Distribution deployment, revised UI, and Report V4 documentation.
Tariq Elamin	Repository setup, early UI updates, UI testing, vibration alerts, face-detection unit testing, profile page work, and GPS speed tracking.	Pull requests show work on branch protection, icon/color updates, UI test cases, vibration feature, face analyzer unit-test refactor, additional face-detection unit tests, profile page creation, and live GPS speed tracking.

Team Member	Primary Contribution Areas	Repository Evidence
Salman Shekh	Core facial detection implementation, profile/settings features, and history-related development.	Pull requests and issue records show work on the ML Kit facial detection pipeline, profile settings, and history/profile feature development.
Chase Tanner	Contributed to end-to-end testing, ML Kit model development including calibration, and bug discovery. Additionally handled audio alert fixes, issue tracking, documentation-related tasks, and review participation.	Repository records show ML Kit model development and calibration, bug discovery, audio fixes, documentation, and participation in review/requested-review workflows.

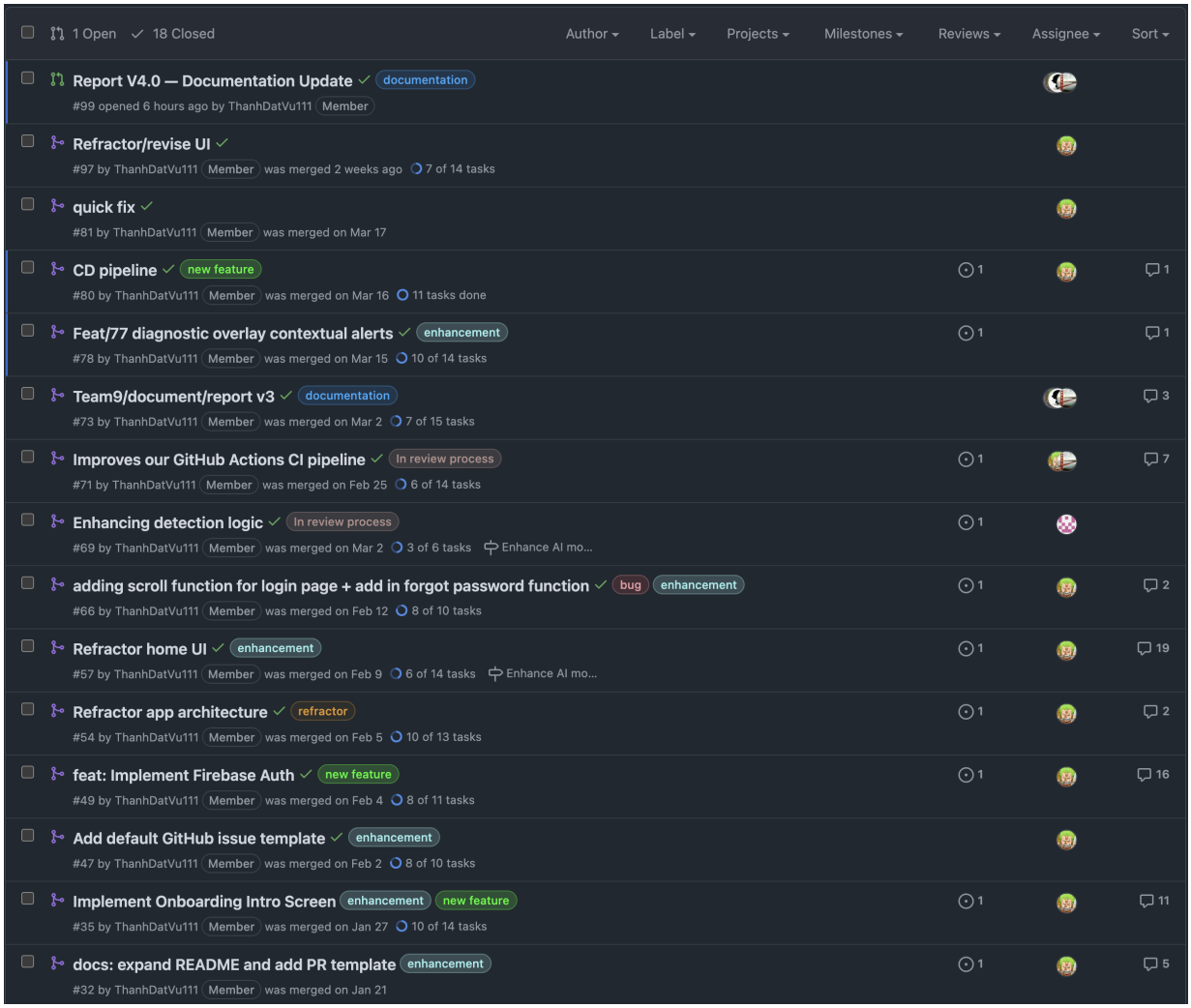

Figure 12.7: GitHub pull request records filtered by Thanh Dat Vu, showing ownership of UI, architecture, authentication, CI/CD, and documentation work.

Figure 12.7 shows repository records for Thanh Dat Vu's contributions. The pull request history provides evidence of ownership across major implementation and documentation areas, including onboarding, authentication, app architecture, Home UI, diagnostic overlays, deployment, and report updates.

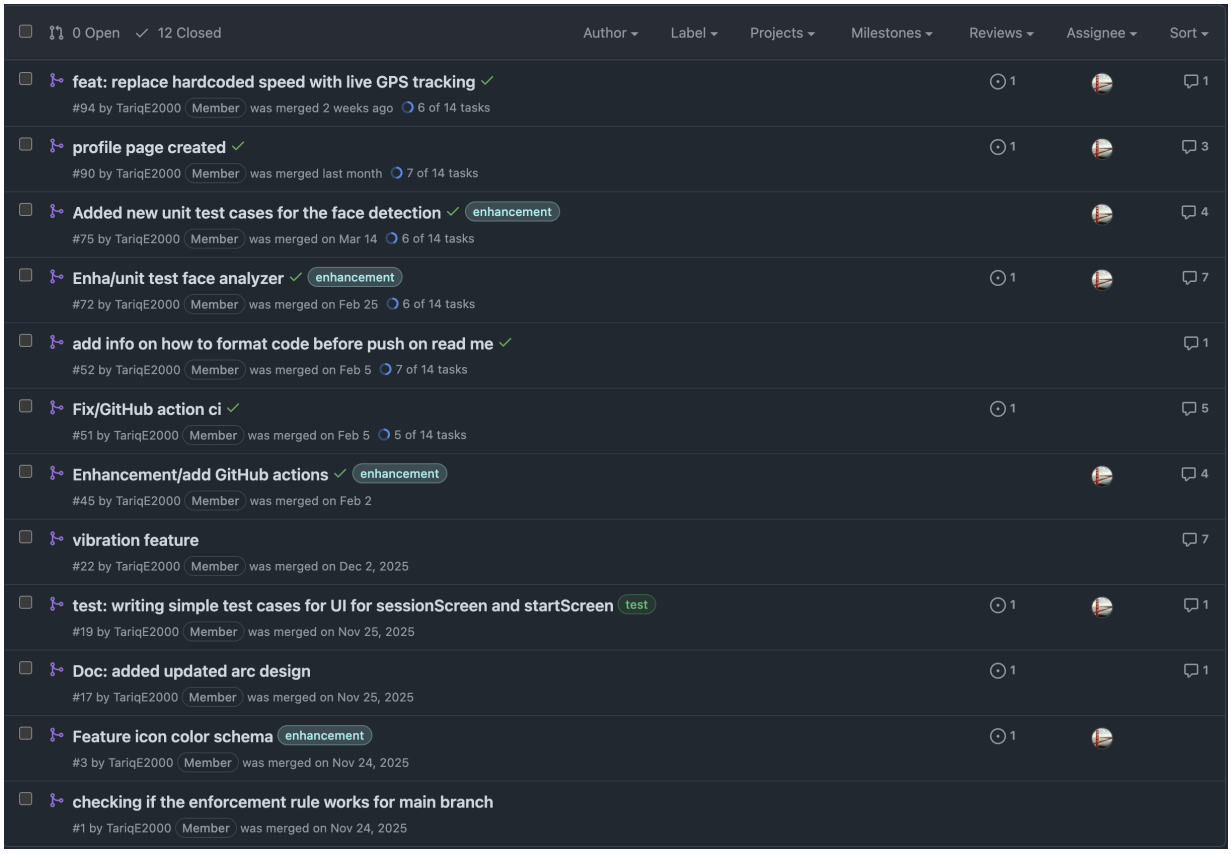

Figure 12.8: GitHub pull request records filtered by Tariq Elamin, showing ownership of testing, profile, GPS speed tracking, alert, and repository setup work.

Figure 12.8 shows repository records for Tariq Elamin's contributions. The pull request history provides evidence of work on early repository setup, UI updates, testing, vibration alerts, face-detection unit tests, profile page implementation, and GPS speed tracking.

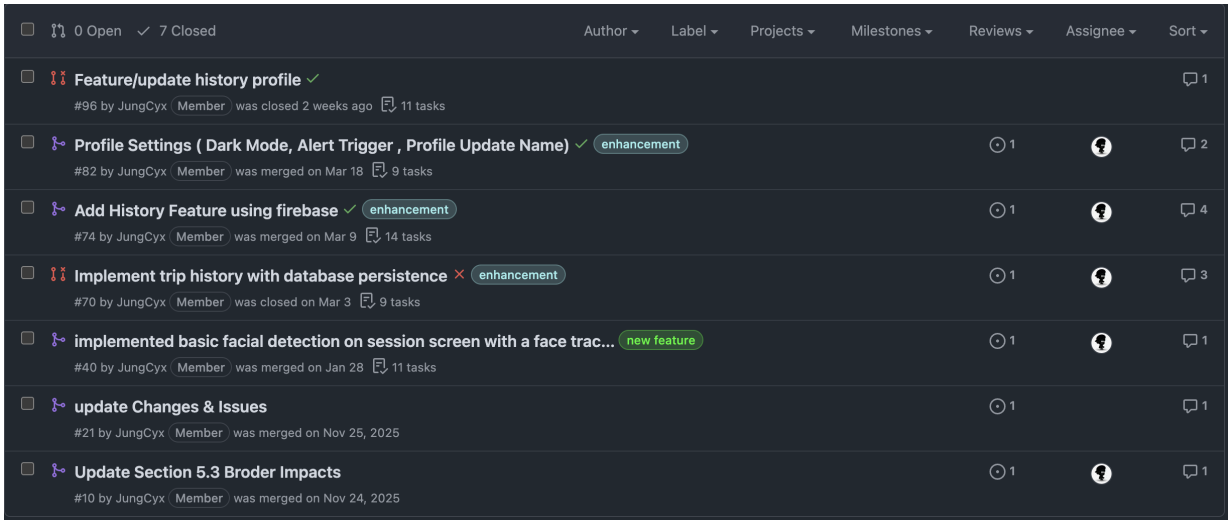

Figure 12.9: GitHub pull request records filtered by Salman Shekh, showing ownership of facial detection, profile/settings, and history-related development work.

Figure 12.9 shows repository records for Salman Shekh's contributions. The records provide evidence of work related to the facial detection MVP, profile settings, and history/profile feature development.

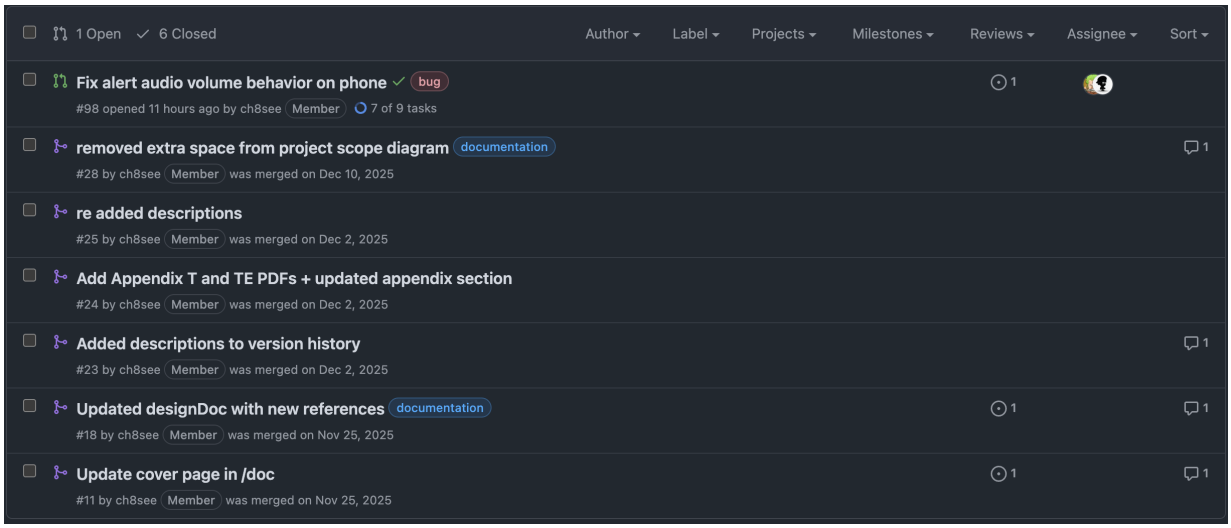

Figure 12.10: GitHub records filtered by Chase Tanner, showing issue ownership, audio-related fixes, documentation work, and review participation.

Figure 12.10 shows repository records for Chase Tanner's contributions. The records provide evidence of issue ownership, audio-related bug tracking or fixes, documentation-related work, and participation in the team's GitHub workflow.

Overall, these contribution records show that the team divided work across members and maintained visible ownership of tasks. The records support both task allocation and individual contribution tracking, which are required project management artifacts.

PM.5 Pull Request Review and Branch Workflow

GitHub Pull Requests were used to manage review and merging of project changes. Contributors worked on separate branches and opened pull requests to describe the purpose of their changes, identify the type of work completed, document testing performed, and provide checklists before merging. This helped the team review changes before they entered the main branch.

Pull requests also provided accountability because each record showed the author, branch name, merge status, changed files, commits, checks, and testing notes. This made it possible to trace implementation work back to contributors and verify that major changes were reviewed before becoming part of the main project.

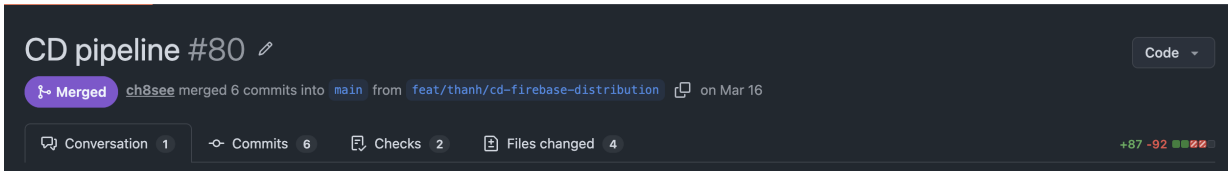

Figure 12.11: Merged GitHub pull request showing branch-based development, merge status, commits, checks, and changed files.

Figure 12.11 shows PR #80 for the CD pipeline. The record shows that the work was completed on a separate feature branch, `feat/thanh/cd-firebase-distribution`, and then merged into `main`. It also shows the number of commits, checks, changed files, and code additions/deletions. This supports the team's use of branch-based development and pull request review instead of direct untracked changes to the main branch.

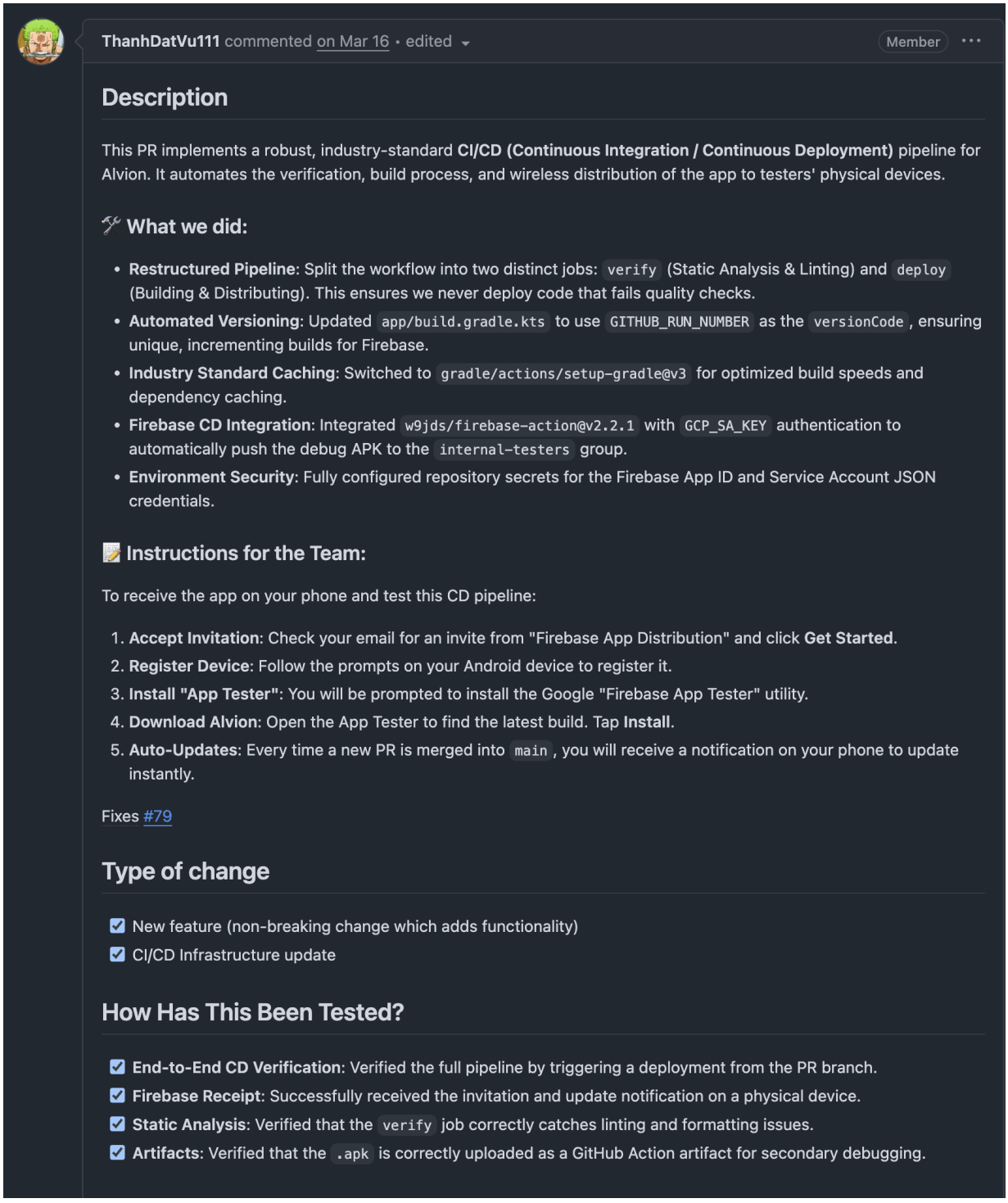

Figure 12.12: Pull request description showing summary, implementation details, type of change, and testing evidence.

Figure 12.12 shows the pull request body for the CD pipeline work. The PR description explains what was implemented, including the verification job, deployment job, automated versioning, dependency caching, Firebase App Distribution integration, and repository secret configuration. The

testing section records end-to-end CD verification, Firebase receipt, static analysis checks, and artifact upload verification.

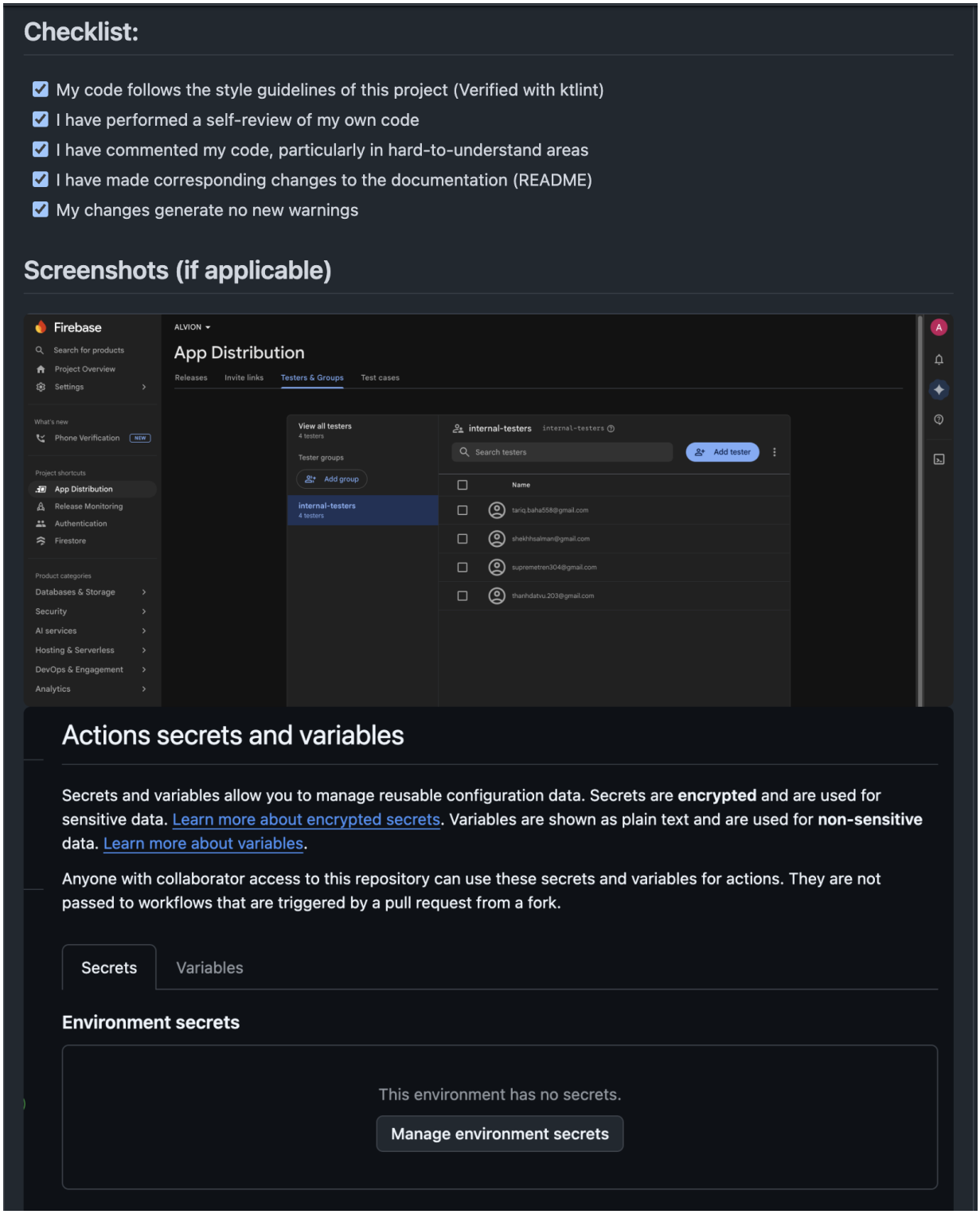

Figure 12.13: Pull request checklist and deployment evidence showing completed review checklist items and Firebase App Distribution setup.

Figure 12.13 shows the checklist and deployment evidence associated with the CD pipeline pull request. The checklist records that the contributor followed style guidelines, performed a self-review, commented the code, updated documentation, and confirmed no new warnings. The Firebase App Distribution screenshot provides additional evidence that the deployment workflow supported tester distribution.

Overall, the pull request records show that the team used a branch-based workflow with review checkpoints, testing notes, and documented verification before merging. This helped manage technical risk, improve code quality, document decisions, and maintain a clear history of project contributions.

PM.6 CI/CD and Testing Management Evidence

The ALVION team used GitHub Actions to support automated quality control during development. The CI/CD workflow helped verify that changes could be checked, built, and prepared for deployment before becoming part of the main project. This provided an additional project management record because workflow runs showed when automated checks were triggered, whether they passed or failed, and which changes required follow-up.

The team also used CI/CD improvements as tracked project tasks. GitHub Issues and Pull Requests included work related to CI pipeline setup, linting, build verification, unit testing, artifact generation, and Firebase App Distribution. These records show that testing and deployment were managed as part of the project process rather than treated as informal final steps.

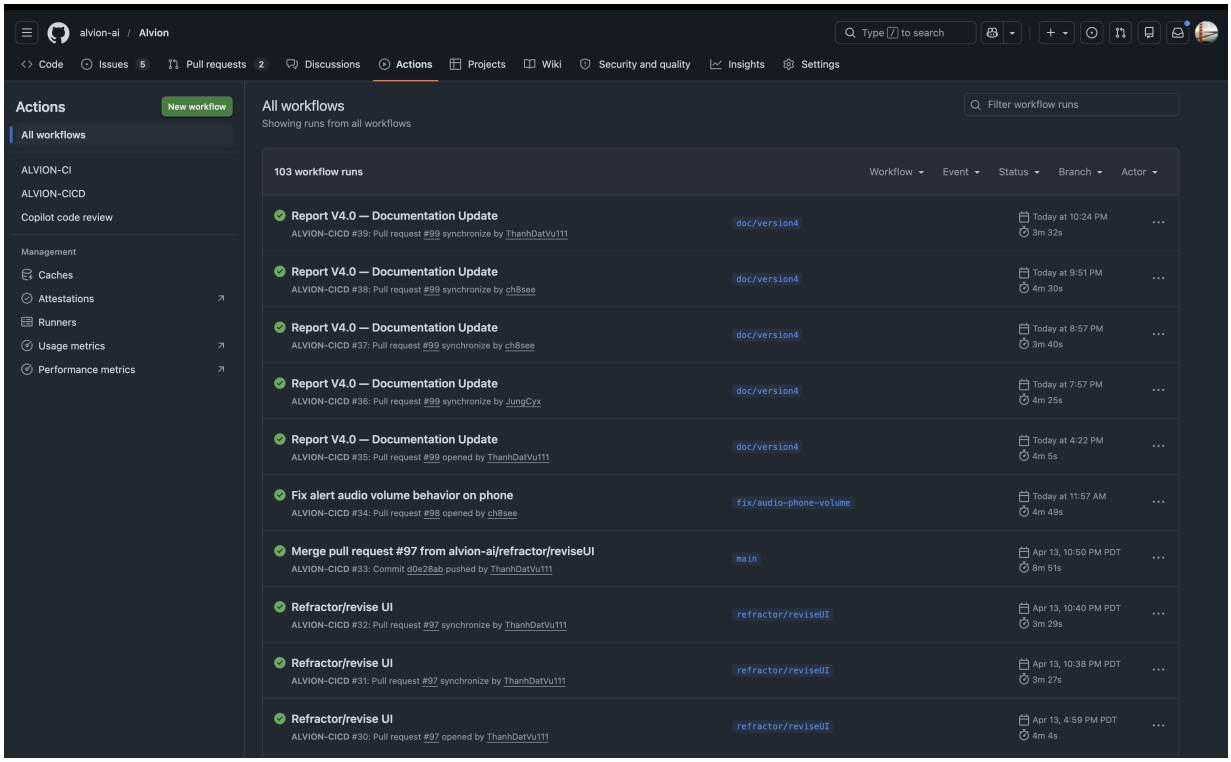

Figure 12.14: GitHub Actions workflow list showing automated CI/CD runs used to verify project changes.

Figure 12.14 shows the GitHub Actions workflow history for the repository. The workflow

list provides evidence that automated checks were run throughout development. This helped the team monitor build health, identify failing changes, and maintain visibility into the status of the codebase.

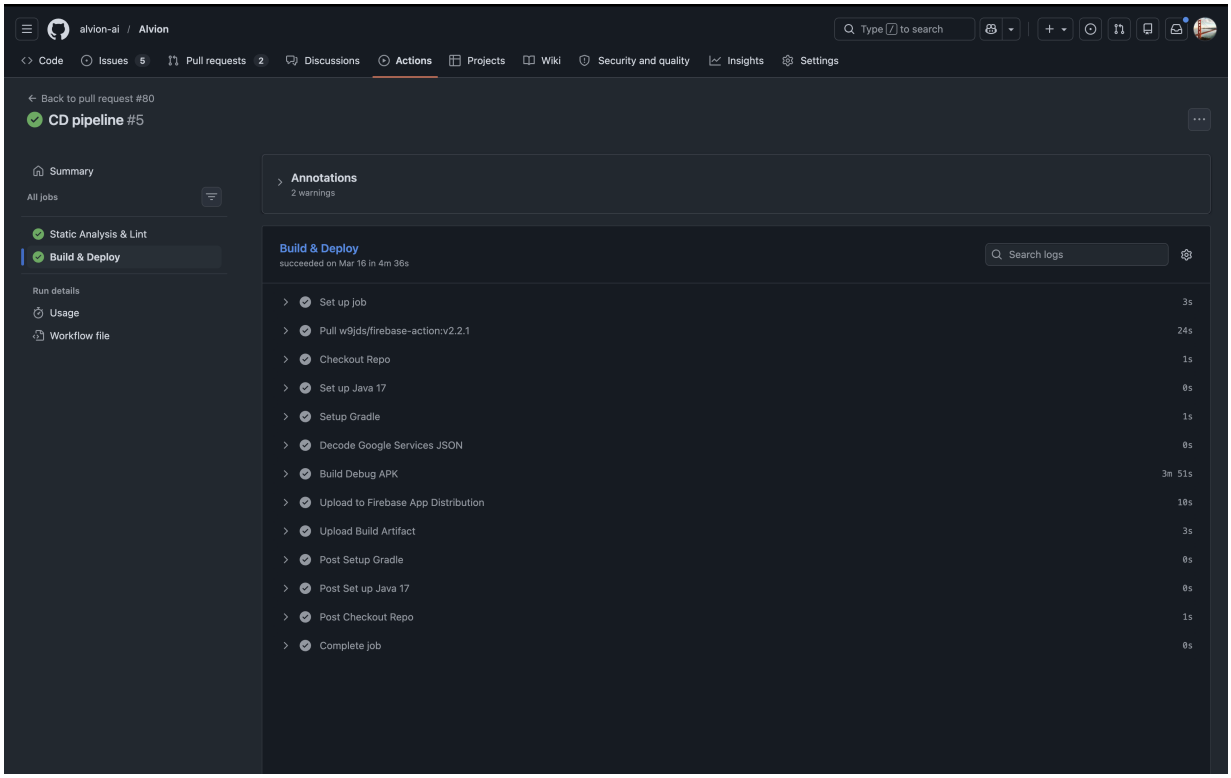

Figure 12.15: Successful GitHub Actions workflow run showing automated verification steps such as build, lint, test, or deployment checks.

Figure 12.15 shows a detailed successful workflow run. The run demonstrates that the team used automated verification steps as part of the development process. These checks helped reduce the risk of merging broken code and provided repeatable evidence that project changes were validated.

Overall, the CI/CD evidence shows that the team managed quality control through automated workflows. Combined with issue tracking and pull request review, GitHub Actions helped the team maintain a structured process for testing, building, and deploying the ALVION application.

PM.7 Project Management Summary

The evidence in Appendix PM shows that the ALVION team followed a structured project management process throughout the capstone project. GitHub Issues were used to create and track tasks, classify work using labels, document bugs, manage feature requests, and record remaining open items. This provided a visible record of project progress across documentation, implementation, testing, refactoring, and deployment work.

The pull request records show that the team used a branch-based workflow instead of making direct untracked changes to the main branch. Pull requests documented the purpose of each change, the type of work completed, testing performed, checklist items, commits, changed files, and merge status. This supported review, accountability, and traceability across major project changes.

The contribution records show that work was divided across team members. Individual contributors owned different areas of the project, including UI/UX development, authentication, facial detection, calibration logic, profile and history features, documentation, testing, CI/CD, GPS speed tracking, and audio alert fixes. This demonstrates both task allocation and individual contribution tracking.

The CI/CD records show that the team used automated workflows to support quality control. GitHub Actions helped verify changes through automated checks such as build, lint, testing, and deployment-related steps. This reduced the risk of merging broken changes and provided repeatable evidence that the project was actively maintained.

Overall, the project management evidence demonstrates that the team planned, assigned, reviewed, tested, and completed work across the full project lifecycle. The records included in this appendix support the highest rubric level because they provide detailed project management evidence from GitHub Issues, Pull Requests, contribution records, workflow documentation, CI/CD checks, and remaining open-task tracking.